IR, NMR

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che-50

⊛ 2018 what are the informations you get from the spectral data of a compound?

⇒ (I) Molecular formula.

(II) Identities of functional group.

(III) carbon connectivity.

(IV) position of substituents and functional groups on the carbon frame work.

(V) steriochemical properties including dynamic & static aspects.

(VI) Types of carbon. & chemical environment.

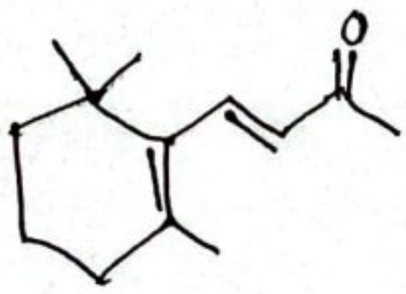
⊛ 2018 State IR value of N-H, C=C, aldehydic C-H, C=O(ester).

⇒ IR value for stretching vibration of some band :-

	$\nu(cm^{-1})$
N-H	3500-3100
C=C	1690-1600 (aliphatic)
	1600-1470 (aromatic)
aldehydic (C-H)	2800-2700

ester (c=o) 1730 - 1730.

2018
Calculate λ_{max} value for the following compounds:-



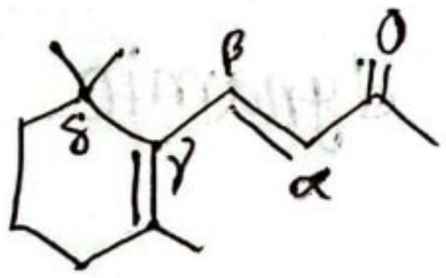
⇒ For :

Base value = 214 nm

2 alkyl substituents = 10 nm

Exocyclic double bond = 5 nm

total = 229 nm

For 

:

Base value = 215 nm

alkyl substituents in γ position = 18 nm

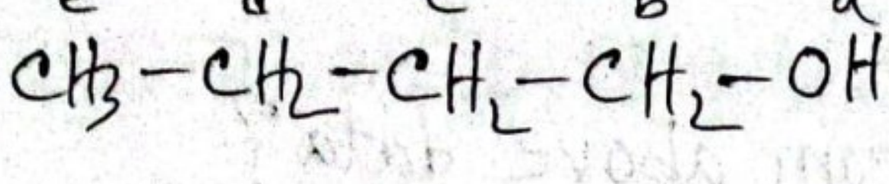
2 alkyl substituent in δ position = 36 nm

extended conjugation = 30 nm

total = 299 nm

2018
Draw the ^{1}H-NMR spectrum of n-butanol. Mention the approximate chemical shift value of each type of protons and explain the spliting pattern indicated your spectrum?

⇒ the structure of n-butanol is



n-butanol

In this compound 5 types of H present,

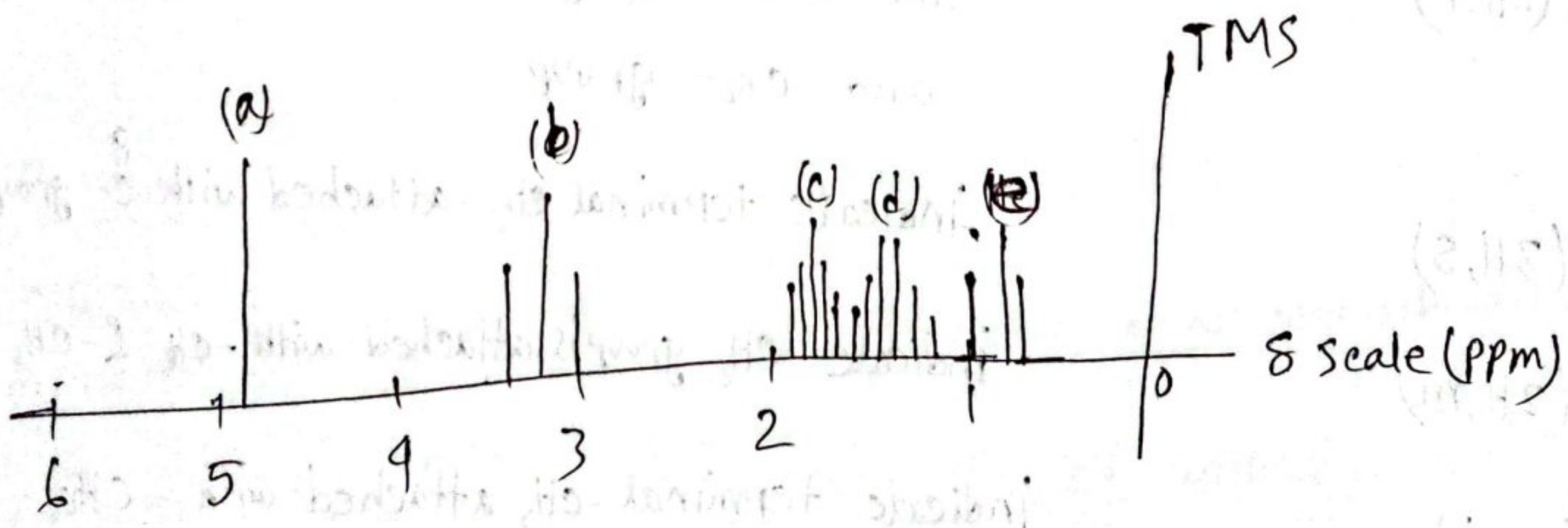
H_a is alcoholic and value is = (1-5) and singlet.

H_b attached with O atom, so δ value is larger = (3.0-3.5)
and multiplicity is triplet

H_c, the δ value is = 1.4 - 1.5 and multiplet

H_d, the δ value is = 1.1 - 1.2 and multiplet

H_e, the δ value is = 0.9 - 1.0 and triplet



2018

An organic compound of formula $C_5H_{10}O$ shows the spectral data:

IR: 2960, 2895, 1718, 1460, 1370 cm^{-1}

^1H NMR: δ 2.3(2H,t), 2.1(3H,s), 1.6(2H,m), 0.9(3H,t)

MS (m/z): 86, 71, 58, 93, 29

Deduce the structure from above data?

⇒

IR value	Type	
2900	$C_{sp^3}-H$	indicates $C_{sp^3}-H$ Str. vibration
2895	$C_{sp^3}-H$	indicates $C_{sp^3}-H$ symmetric Str. vibrations.
1718	$-\overset{O}{\overset{\|\|}{C}}-$	carbonyl $-\overset{O}{\overset{\|\|}{C}}-$ Str. vibration
1460, 1370	$-C-H$ (bending)	indicate $-C-H$ bending vibration

^1H NMR	Inference
2.3(2H,t)	Indicates $-CH_2-$ group is attached with $-\overset{O}{\overset{\|\|}{C}}-$ and $-CH_2-$ group
2.1(3H,S)	indicate terminal CH_3 attached with $-\overset{O}{\overset{\|\|}{C}}-$ group.
1.6(2H,m)	indicate $-CH_2-$ group is attached with $-CH_2-$ & $-CH_3$
0.9(3H,t)	indicate terminal $-CH_3$ attached with $-CH_2-$

So the probable structure is, $CH_3-\overset{O}{\overset{\|\|}{C}}-CH_2-CH_2-CH_3$

2017, 2014

⊗ A compound $C_9H_{10}O_2$ and deduce structure whose

IR: 3100, 2950, 2900, 1719, 1600, 1500, 1000 cm^{-1}

^1H NMR: 1.2 (t, 3H), 4.5 (q, 2H), 7.5–8.5 (m, 5H) ?

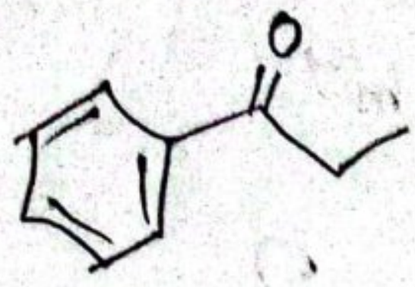
⇒

IR value (cm^{-1})	Type	Indicates
3100	$C_{sp2}-H$	$C_{sp2}-H$ Str. vibration
2950	$C_{sp3}-H$	$C_{sp3}-H$ asymmetric str. vibr.
2900	$C_{sp3}-H$	$C_{sp3}-H$ symmetric str. vib.
1719	$-\overset{O}{\overset{\|\|}{C}}-$	$-\overset{O}{\overset{\|\|}{C}}-$ carbonyl compound
1600	C=C	aromatic C=C
1000	C–O	C–O str. vib.

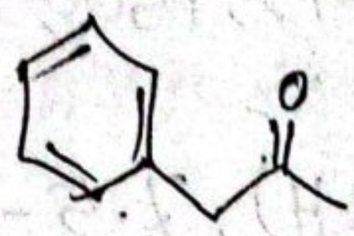
^1H NMR value	Inference
~~1.2 (3H, t)~~ 1.2 (t, 3H)	CH_3 (terminal) attached with $-CH_2-$
4.5 (q, 2H)	$-CH_2-$ attached with $-CH_3$
7.5–8.5 (m, 5H)	mono substituent benzene ring

So, the probable structure is

(✗) 2017 The following two compounds A and B



A



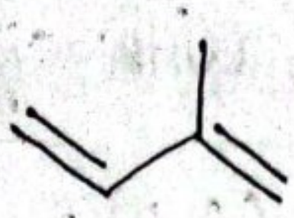
B

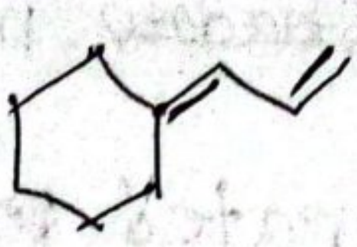
have the C=O absorptions at 1684, 1715 cm^{-1} respectively. Give the C=O absorption frequency for A and B?

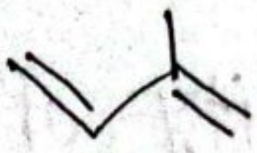
⇒ The absorption for compound A is 1684 cm^{-1}. the absorption for compound B is 1715 cm^{-1}. Both compound contain $-\overset{O}{\overset{\|}{C}}-$ group. But in A compound conjugation occurs & double bond character changes to the single bond & absorption decreases & it is 1684 cm^{-1}.

But no conjugation present in 'B' & absorption is 1715 cm^{-1}.

2017, 2015
Calculate λ_{max} for the following compound:



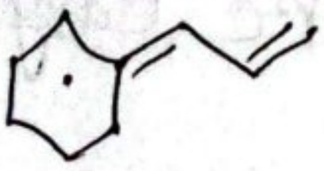


⇒ For  :

Base value = 217 nm

alkyl substituent = 5 nm

total = 222 nm

For  :

Base value = 217 nm

2 alkyl substituent = 10 nm

exocyclic double bond = 5 nm

total = 232 nm

2017
Deuterated (^2H) solvents are used for recording NMR Spectra - why?

⇒ Deuterate solvent are used to make sure that no hydrogent present. If any H present, then it gives signal in the nmr & cannot differentiate which signals gives compound & which signal gives solvent. Using a deuterated solvent minimize the presence of

^1H in the solvent, thus preventing strong solvent peaks that could ~~overshadow~~ interfere with signals from the sample. deuterated ~~[illegible]~~ solvent (^2H) reduces background signals & enhances the visibility of the sample's unique signals. Although deuterated solvent are not completely free of ^1H, the few remaining ^1H atoms produce weaker signals that are easy to distinguish from sample peaks.

⊛ 2017 what is meant by vacuum UV?

⇒ Vacuum UV (VUV) refers to the portion of the UV spectrum with wavelengths btwn 100–200 nm. It is called "vacuum" UV because these λ are strongly absorbed by air, particularly by oxygen.

⊛ 2017 Define blue shift & red shift?

⇒ Blue shift: when light from an object is shifted to shorter λ then it is called blue

Shift. Blue shift are seen in stars, galaxies etc.

Red shift: When light from an object is shifted to longer wavelength is called Red shift. It is commonly seen in distant galaxies moving away from us.

2016
⊛ Define chromophores & auxochromes with example?

⇒ chromophores: are parts of molecules responsible for their color.

The coloured substances show their colour to the presence of one or more unsaturated groups responsible for electronic transition. These groups are called chromophore.

example: C=O, $-NO_2$(nitro), N=N(azo), Benzene ring, C=C.

Auxochromes: are groups attached to chromphores that don't produce colour themselves but enhances/intensify the colour produced by the ~~auxo~~chromophores.

Auxochromes don't absorb light themselves, but they affect the chromophore's ability to absorb light. Auxochromes

generally contain lone pairs of e^-. examples are $-OH$ (hydroxyl), $-NH_2$ (amino), $-OR$ (alkoxy), $-Cl$ (chloro), etc.

2016

An organic compound C_8H_8O. Determine its structur whose

IR : 3080, 2960, 2850, 1690, 1600 cm^{-1}

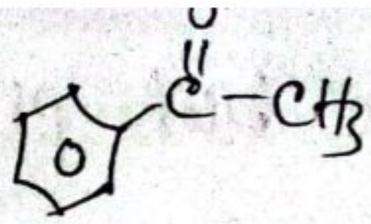
^1H NMR : δ 7.2 - 7.3 (5H, m), 2.2 (3H, s)

MS (m/z) : 120 (M^+), 105, 77, 51 ?

⇒

IR values cm^{-1}	Type	indicates
3080	$C_{sp^2}-H$	aromatic $C_{sp^2}-H$ str. vib.
2960	$C_{sp^3}-H$	aliphatic $C_{sp^3}-H$ asymmetric str. vib.
2850	$C_{sp^3}-H$	aliphatic $C_{sp^3}-H$ symmetric str. vib.
1690	$-\overset{O}{\overset{\|\|}{C}}-$	$-\overset{O}{\overset{\|\|}{C}}-$ group str. vib.
1600	C=C	aromatic C=C double bond

^1HNMR values	Inference/indicate
7.2 - 7.3 (5H, m)	monosubstituted benzene
2.2 (3H, s)	terminal $-CH_3$ attached with $-\overset{O}{\overset{\|\|}{C}}-$

So the probable structure is 

2016
(X) Distinguish btwn propene & propyl Alc. by IR?

⇒ propene ⟶ $CH_3-CH=CH_2$

propyl Alc ⟶ $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-OH$

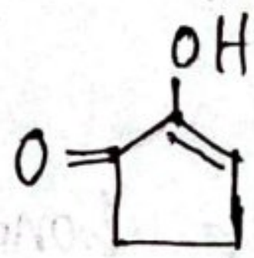
In propene IR value gives C=C bond values this values is around 1680-1620 cm^{-1} also gives $C_{sp^2}-H$ str. vib. which is (3100-3000)cm^{-1}. ~~another valve is $C_{sp^3}-H$ str. and bending~~ There is also a C-H bending vibration (out-of-plane) for alkenes around 910-990 cm^{-1}.

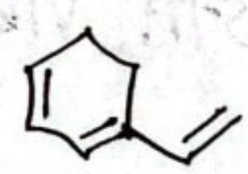
Propyl alcohol contain O-H bond which shows a broad and strong absorption band around 3200-3500 cm^{-1}. due to H-bonding in alcohols. It has C-O stretching vibration around 1050-1150 cm^{-1}.

So, the broad O-H stretch at 3200-3500 cm^{-1} is present in propyl alcohal but absent in propene. so propene will not display 3200-3500 absorption. And propene has the

C=C stretch but propyl Alcohol doesn't have so the propyl Alc will not display absorption around 1640-1680 cm^{-1}. By this process we can distinguish them from each oth

2016
⊛ calculate λ_{max} of the following compound?



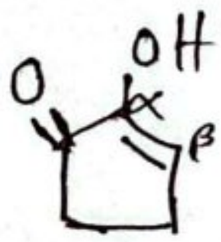
⇒ For  :

Base value = 253 nm

Double bond extended = 30 nm

2 alkyl substituent = 10 nm

total = 293 nm

For 

:

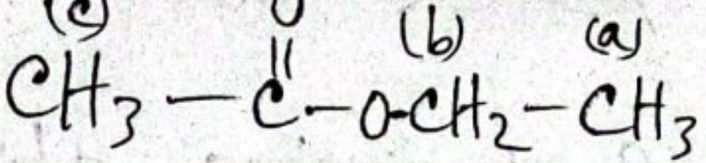
Base value = 215 nm

alkyl substituent in β position = 12 nm

-OH in α position = 35 nm

total = 259 nm

2016
! ⊛ Draw the ^{1}H-NMR spectrum of ethyl acetate? Mention approximate chemical shift value of each type of proton and explain the spliting pattern indicated in spectrum?

⇒ The structure is: 

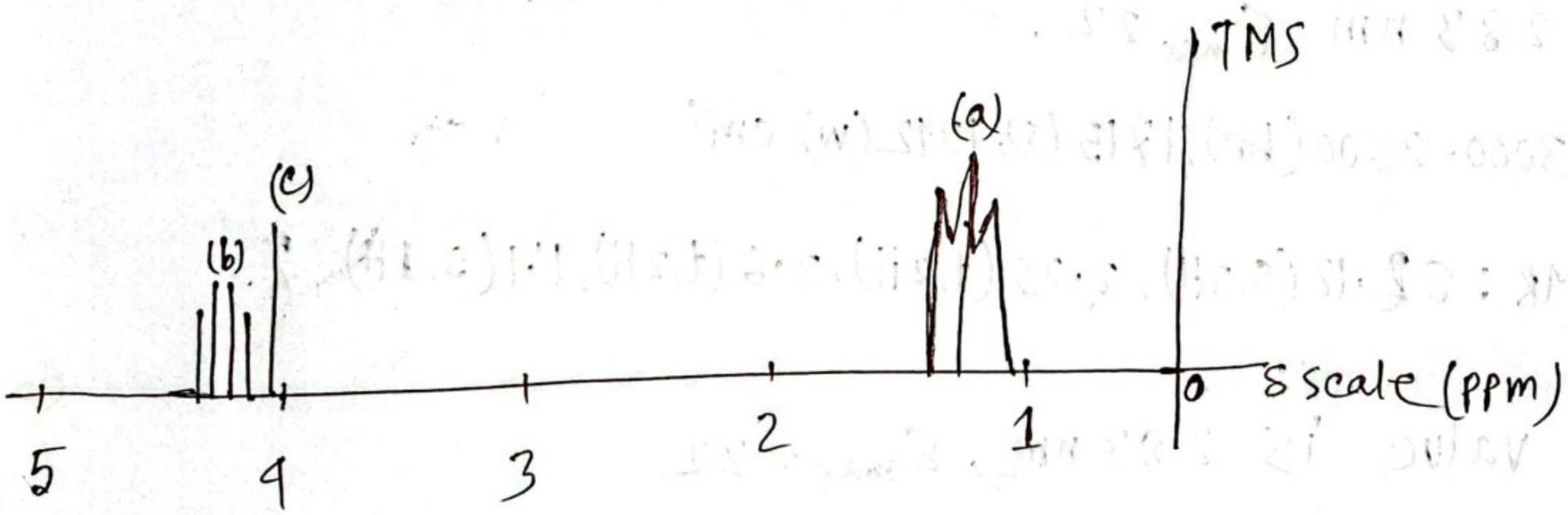
In this compound 3 types of proton is present.

H_a, the δ value is 1.1-1.2 and triplet.

H_b, the δ value is 4.2-4.3 and multiplet.

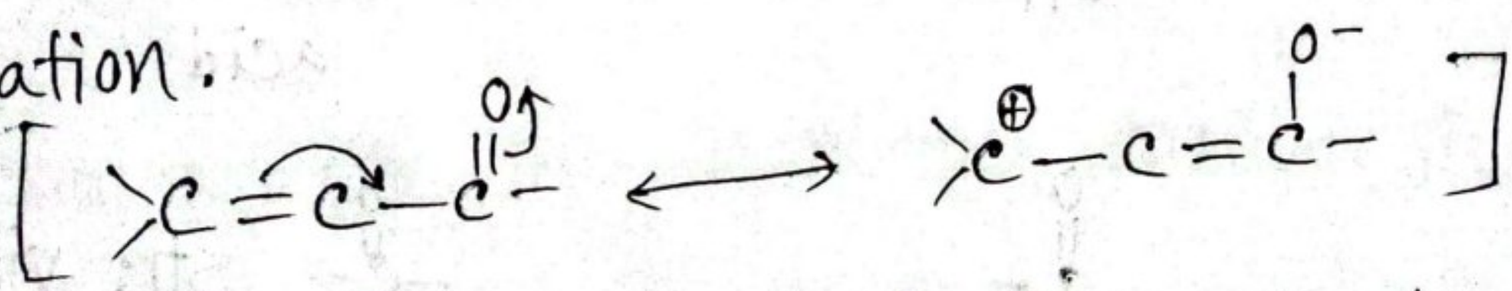
H_c, the δ value is 4.0-4.2 and singlet.

And the ^1H-NMR spectrum's graph is below:



2015
(X) conjugation of carbonyl with a c=c bond shift absorption values to lower frequencies - explain?

⇒ This shift happens due to resonance and electron delocalization.



This are resonance stabilized & single bond character

increased in both >C=C< and >C=O bond. So decreas the force constant & lowering absorption frequencies. Because of resonance, the double bond character of the C=O bond is slightly reduced making it weaker. A weake C=O bond requires less energy to vibrate, shifting the stretching frequency to a lower frequency.

2015

(X) A compound $C_5H_8O_3$ determine this compound whose

(i) UV : 283 nm ε_{max} 22.

(ii) IR : 3000-2500 (br), 1715 (s), 1342 (w) cm^{-1}

(iii) ^1H NMR : δ 2.12 (s, 3H), 2.25 (t, 2H), 2.6 (t, 2H), 11.1 (s, 1H) ?

⇒ UV value is 283 nm, $\varepsilon_{max} = 22$

From UV data it can be said that, the compound contain any chromophoric on auxochromic group.

IR value	type	Inference
3000-2500 (br)	-OH	-OH str. vib. of carboxy acid.
1715 (s)		str. vib.
1342	C-H	aliphatic C-H bond

^1H-NMR (PPM)	Inference
2.12 (3H, S)	terminal $-CH_3$ attached with carbonyl group.
2.25 (2H, t)	$-CH_2-$ attached with $-CH_2-$ group & $-\overset{O}{\overset{\|}{C}}-$.
2.60 (2H, t)	$-CH_2-$ attached with $-CH_2-$ group & $-\overset{O}{\overset{\|}{C}}-$.
11.1 (1H, S)	H in $-COOH$

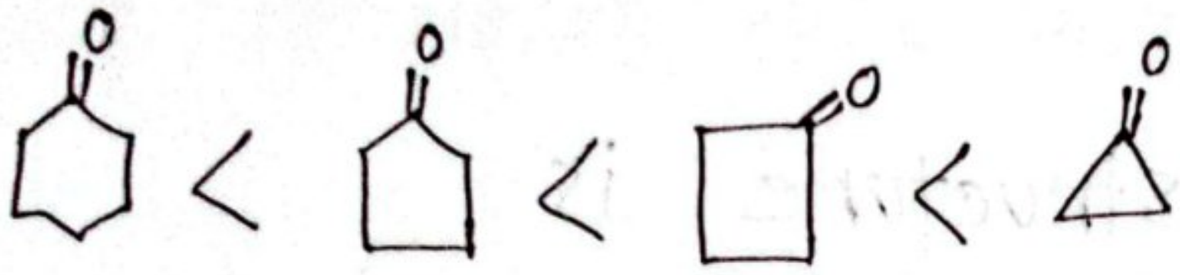
so the probable structure is

$$CH_3-\overset{O}{\overset{\|}{C}}-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-OH$$

2015
⊗ Angel strain raises the carbonyl frequencies?

⇒ The angle strain effect is observed in certain molecular structure, specially when you have an e-withdrawing group attached near a carbonyl group (c=o). When electro-negative atom or group is attached near carbonyl, it withdraws e- density from the c=o. This increases partial positive charge on the carbonyl carbon. The increased positive charge on the carbonyl carbon makes the

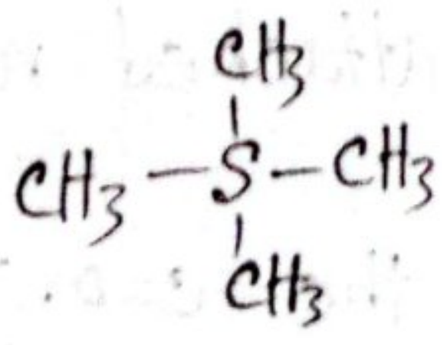
C=O bond more polarized. A more polarized C=O bond has greater double-bond character, since the bond is stronger (more double-bond character), it vibrates at higher frequency, leading to an increase in the IR absorption frequency of the carbonyl group.



Angle strain increases & thus increases carbonyl frequencie

2015
(3) The chemical shift of TMS is set to zero ppm while recording ^{1}H-NMR & ^{13}C-NMR spectra - explain?

⇒ TMS means tetramethyl silane has 12 protons & 4 carbons which are magnetically & chemically equivalent.



that is gives a single, strong signal in the spectrum which turns out to be outside the range of most orgw

compounds. As TMS is considered as standard & chemical shift of TMS is set to zero ppm & all other chemical shift are determined relative to it. Besides TMS is a volatile liquid, inert to most organic reagents & soluble in most organic solvents.

In TMS Silicon is less electronegative than carbon, so the 12 proton is highly shielded. For this reason, the chemical shift found in maximum possible up field region. So it is assumed that the TMS resonate at zero ppm.

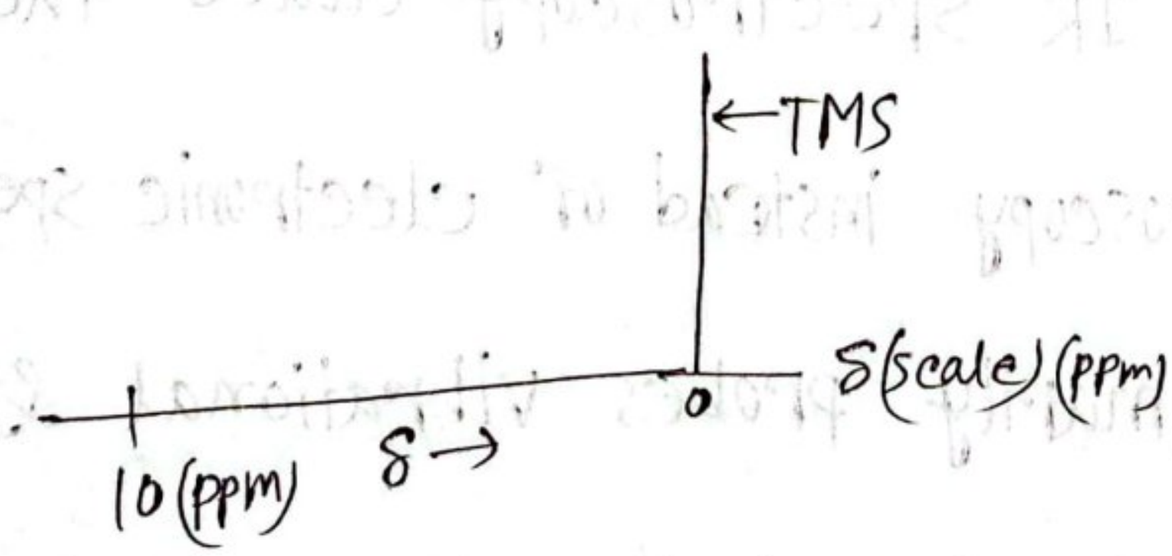
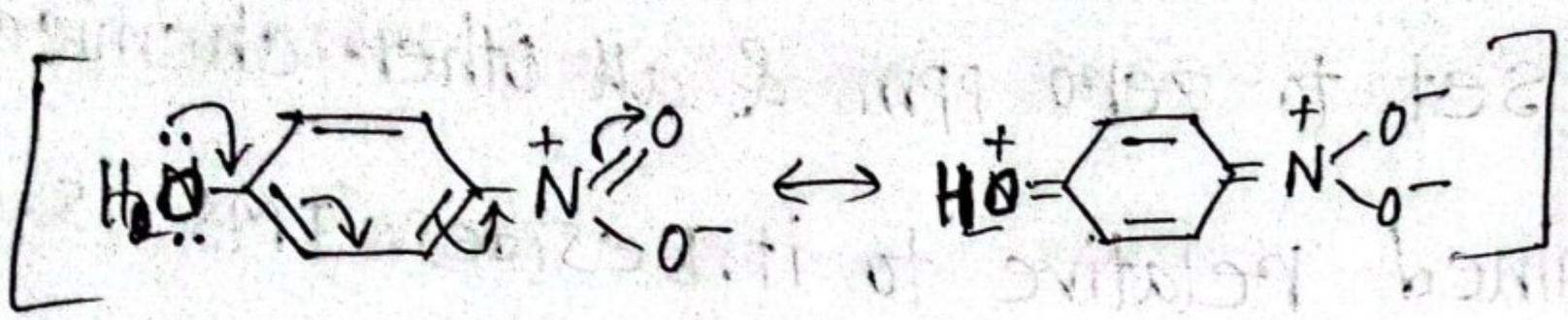


Fig: chemical shift valve in δ value & position of TMS

2015
(X) The λ_{max} valve can be used to differentiate btwn p-nitrophenol & o-nitrophenol - explain?

⇒ In p-nitroaniline direct resonance interaction is

observed btwn. the substituent groups:



P-nitrophenol is resonance stabilized, that's why λ_{max} value is higher. But o-nitrophenol cannot show this reson because both group are so closer steric effect is present & co-planarity inhibit & λ_{max} is smaller.

From value of λ_{max} we can differentiate these compounds...

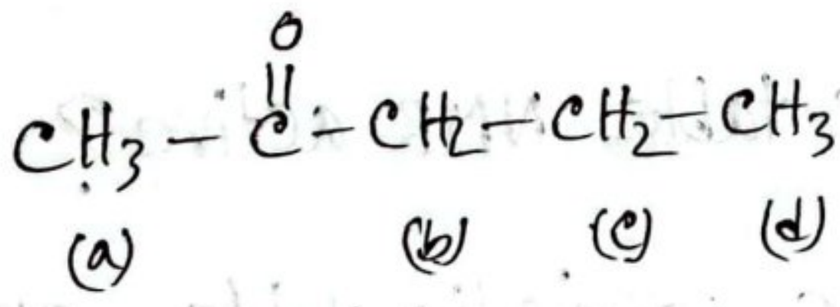
⊗ 2020 Why is the IR spectroscopy called the vibrational rotational spectroscopy instead of electronic spectroscopy?

⇒ Because it primarily probes vibrational & rotational transitions of molecules rather than electronic transition the energy of IR photons matches the energy required to excite molecules from one vibrational state to another. These transitions do not involve changing the electronic state of the molecule, only its

vibrational. (sometimes rotational) state. In gases or other suitable states, molecules undergoing vibrational & rotational transitions. these combined transitions add complexity to the IR spectrum but do not involve changes in electronic energy level. Since IR radiation does not have enough energy to induce electronic transitions, it instead excites vibrational & rotational states, Hence the term vibrational-rotational spectroscopy.

2020
(X) How many types of proton in $CH_3COCH_2CH_2CH_3$. Apply (n+1) rule to predict splitting pattern?

⇒



There are 4 types of proton in this compound. Apply (n+1) rule:

@ For terminal $-CH_3$ group attached with $-\overset{O}{\overset{\|}{C}}-$, $n=0$ ∴ $n+1 = 0+1 = 1$ (singlet)

splitting pattern (3H, S)

(b) For $-CH_2-$ group which attached with $-\overset{O}{\overset{\|}{C}}-$ & $-CH-$,

$n = 2 \therefore n+1 = 2+1 = 3$ (triplet)

Splitting pattern (2H, t)

(c) For $-CH_2-$ group which attached with $-CH-$ & $-CH_3$ group,

$n = 5 \therefore n+1 = 5+1 = 6$ (multiplet)

Splitting pattern (2H, m)

(d) terminal $-CH_3$ which attached with $-CH_2-$,

$n = 2 \therefore n+1 = 2+1 = 3$ (triplet)

Splitting pattern (3H, t).

(X) 2020 Are all element of periodic table NMR active?

⇒ NO, Not all elements in the periodic table are NMR active.

To be NMR active, a nucleus must have a non zero spin quantum number. ($I \neq 0$).

Nuclei with an odd number of proton or neutrons generally have non-zero nuclear spin quantum numb

and making them NMR active. Ex. ^{1}H, ^{13}C, ^{19}F, ^{31}P, ^{15}N, ^{17}O etc.

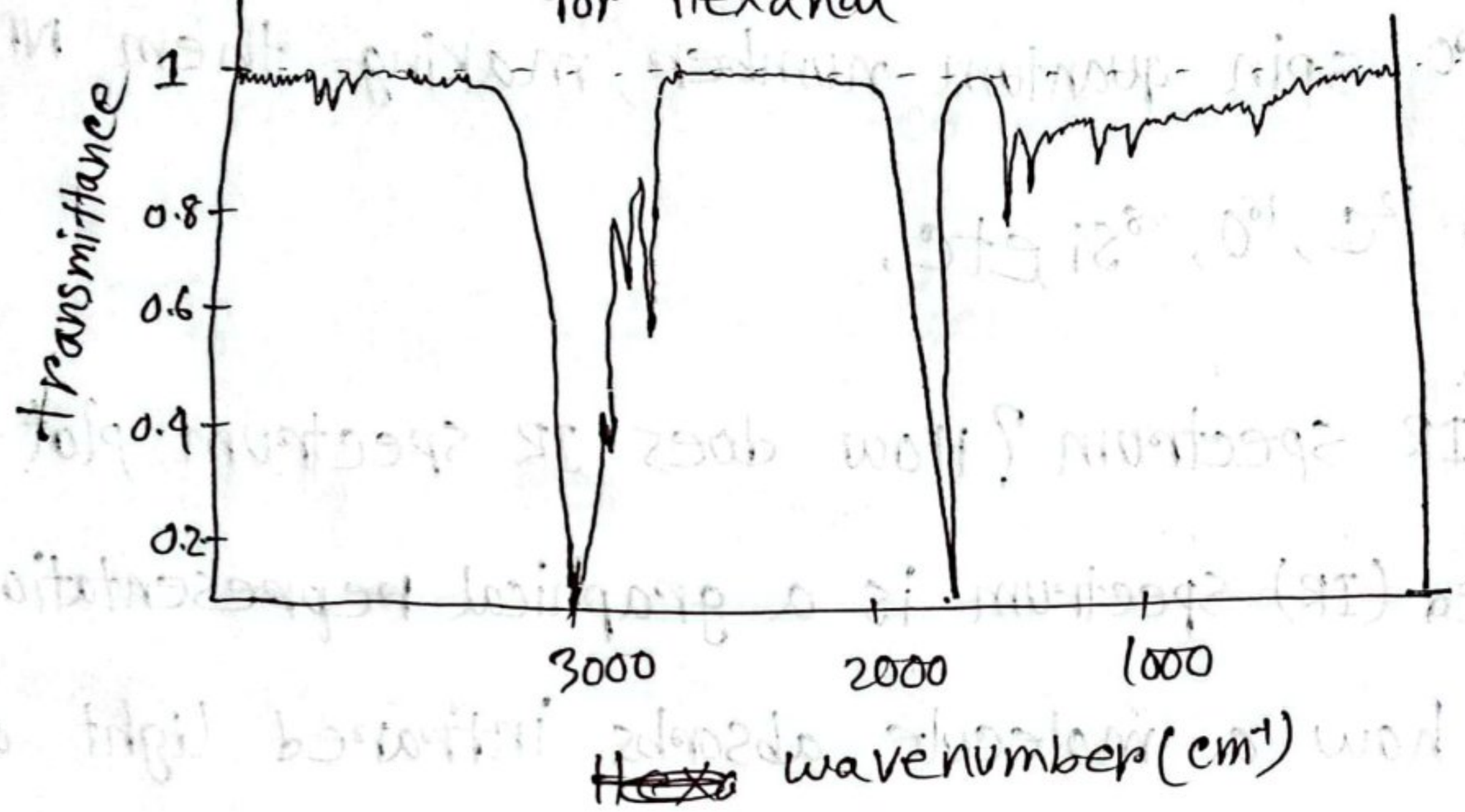
Nuclei with an even number of both protons & neutrons have a zero spin quantum number, making them NMR inactive. EX. ^{12}C, ^{16}O, ^{28}Si etc.

⊗ 2020 what is IR spectrum? How does IR spectrum plot?

⇒ An Infrared (IR) spectrum is a graphical representation that shows how a molecule absorbs infrared light at different wavelengths or frequencies. It is like a fingerprint for a molecule, providing valuable information about its structure & functional group.

⇒ An IR spectrum plot is created by measuring how a sample absorbs infrared (IR) light across a range of wavelength or frequencies. In plotting spectrum the X-axis represent wavenumber (cm^{-1}) & the Y-axis represent transmittance (%) or absorbance. The position of peak on the X-axis (wavenumber) indicates the type of bond that is vibrating. The intensity of a peak in Y-axis indicates the number of

bonds of a particular type in the molecule. By analysing position & intensity of peaks in an IR spectrum, chemists can identify functional group, determine molecular structure.



2020
(*) which region of IR spectrum is extremely useful for the study of organic compound ? what is the absorption range of IR spectroscopy?

⇒ For organic compound ⟶ 1400 - 4000 cm^{-1}

⇒ absorption range ⟶ 400 - 4000 cm^{-1}

2020
(*) why is the UV spectroscopy called electronic spectroscopy

⇒ Because it involves the absorbtion of electro magnet radiation in the ultraviolet and visible regions of the

spectrum, which causes electronic transitions in molecules. the absorption of UV radiation causes e^- to transition from their ground electronic states. These transitions are called electronic transitions.

Since UV spectroscopy directly probes electronic transition so it is called electronic spectroscopy.

2020 what is meant by chemical shift in the NMR spectroscopy? How can you mathematically calculate the chemical shift of a signal?

⇒ A chemical shift is defined as the difference in ppm btwn the resonance frequency of the observed proton & TMS hydrogen.

⇒ chemical shift is a measure of the difference in the resonance frequency of a specific nucleus compared to a standard reference compound. TMS is commonly used as a reference compound, with a chemical shift of 0 ppm. chemical shifts are measured in ppm. It is calculated by using the formula:

$$\delta = \frac{(\text{frequency of nucleus}) - (\text{frequency of standard})}{(\text{frequency of spectrometer})} \times 10^6$$

where frequency of spectrometer is the operating frequency of the NMR instrument (e.g. 300 MHz).

~~(X)~~ 2020 what is mass spectrum? How is it represented?

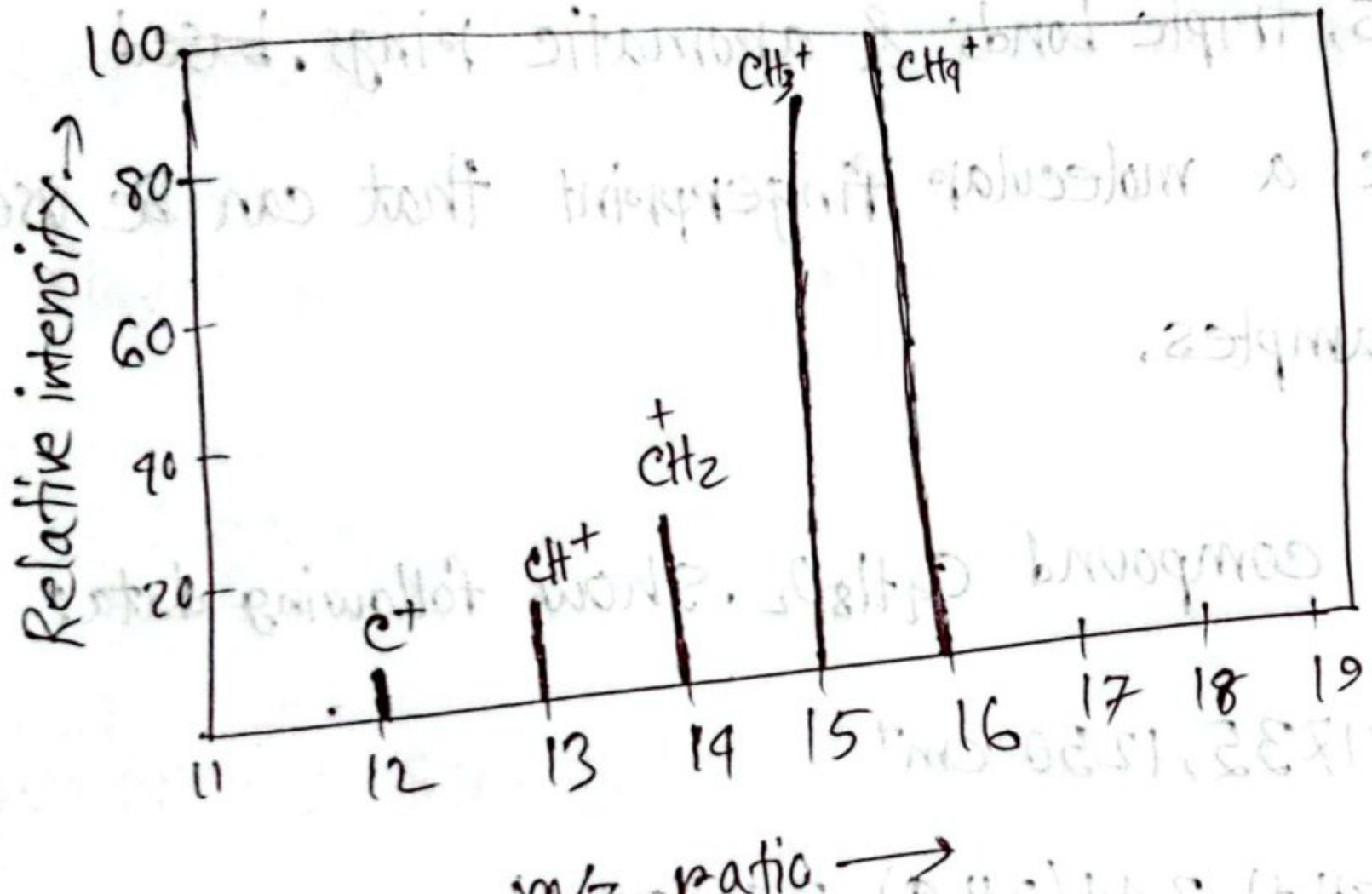
⇒ A mass spectrum is a graph that shows the mass-charge (m/z) ratios of ions in a sample & their relati abundance. In this graph x-axis reppresents the mass-to-charge (m/z) ratio, where m is mass & z is charge. Y-axis shows Relative abundance or intensity of ions. The peak in the spectrum represent the different ions present in the sample. The height of each peak indicate the relative abundance of that ion. The tallest peak in the spectrum is called base peak. It represent the most abundant ion. Mass spectra can be used to identify, confirm or quantify compounds. based on the intensity & m/z ratio of the ions. Mass spectra of CH_4,

its molecular ion is:

$$CH_4^+ \xrightarrow{-H} CH_3^+ \xrightarrow{-H} CH_2^+ \xrightarrow{-H} CH^+ \xrightarrow{-H} C^+$$

$m/z = 16$ $m/z = 15$ $m/z = 14$ $m/z = 13$ $m/z = 12$

The graphical representation is:



2022

⊗ What information do you get from IR spectrum for an organic compound?

⇒ ⊗ provides information about:

(I) the types of bonds present in the molecule.

(II) the functional group it contains.

(III) the presence of impurities.

(IV) It can help to determine the general structure of

a molecule.

(V) It can distinguish btwn isomers with different functional groups.

(VI) By analysing IR Spectrum. you can identify the presence of double bonds, triple bonds & aromatic rings. ~~based~~

(VII) It can provide a molecular fingerprint that can be used to compare samples.

(X) 2022 An organic compound $C_4H_8O_2$. Show following data:

IR: 2960, 2850, 1735, 1250 cm^{-1}

^1H-NMR : δ 1.10(3H, t), 3.44(2H, q), 3.42(3H, s)

determine structure?

⇒

IR value (cm^{-1})	Type	Inference
2960	$C_{sp^3}-H$	$C_{sp^3}-H$ Str. vib.
1735	$-\overset{O}{\overset{\|\|}{C}}-O-$ (ester)	$-\overset{O}{\overset{\|\|}{C}}-O-$ Str. vib.
1250	C-O (ester)	C-O, Str. vib.
2850	C-H (alkane)	C-H Str. Vib.

^{1}H-NMR value, δ (ppm)	Inference
1.1 (3H, t)	terminal $-CH_3$ attached with $-CH_2-$ group.
3.49 (2H, q)	$-CH_2-$ group attached with $-CH_3$ & $-\overset{O}{\overset{\|}{C}}-$.
3.92 (3H, s)	terminal $-CH_3$ attached with $-\overset{O}{\overset{\|}{C}}-$

So the probable structure is

$$CH_3-\overset{O}{\overset{\|}{C}}-O-CH_2-CH_3$$

2021

⊗ Determine structure of C_7H_7ON & which give

IR : 3400, 3350, 3080, 2960, 2850, 1680, 1600 cm^{-1}

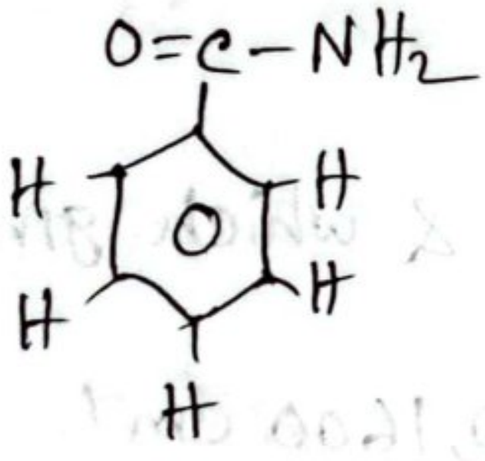
^{1}H-NMR : δ 7.2-7.3 (5H, m), 3.2 (2H, s) ?

⇒ IR value (cm^{-1})	Type	Inference
3400	N-H	N-H asymmetric
3350	N-H	N-H symmetric
3080	C-H	C-H aromatic str.
2960	C-H (alkane)	C-H str. vib.
2850	C-H	C-H str. vib.

1680	C=O	C=O Str. Vib.
1600	C=C	C=C aromatic rings

^{1}H-NMR, δ(ppm)	Inference
7.2-7.3 (5H, m)	mono substituent benzene ring
3.2 (2H, s)	-NH$_2$ group present

So the probable structure is:



2021

(X) Distinguish btwn propanal & propyl alcohol with the help of IR value?

⇒ propanal is an aldehyde and it gives strong, sharp peak due to C=O stretching its absorption band around 1725 cm^{-1} but propyl alcohol lacks of carbonyl group, so this peak is absent in its

IR Spectrum. propyl alcohol has a broad absorption band around 3200-3600 cm^{-1} due to O-H stretch of the Alcohol group. this peak is broad due to H-bonding. But propanal lacks an O-H group; so this broad peak is absent in its IR spectrum. propanal also shows weak to moderate C-H stretch for aldehyde group around 2720-2820 cm^{-1} but propyl alcohol does not have this aldehydic C-H stretch so this peak will be absent. By this process we can distinguish them.

2022
can you differentiate the possible isomers of C_2H_6O by IR data?

⇒ Isomers of C_2H_6O are:

CH_3-CH_2-OH ethanol

CH_3-O-CH_3 dimethyl ether

Now we can differentiate ethanol and dimethyl ether by IR spectroscopy.

Ethanol has an O-H stretch around 3200-3600 cm^{-1}.

which is broad due to H-bonding but dimethyl ether lacks of O-H group, so this broad peak is absent in it's spectrum.

Both have C-O band but the C-O bond in ethanol typically shows a strong absorption band around 1050 cm^{-1}. And C-O bond in dimethyl ether being in a different chemical environment, usually absorbs at a slightly higher frequency around 1100 cm^{-1}.

By this process we can distinguish them.

2021

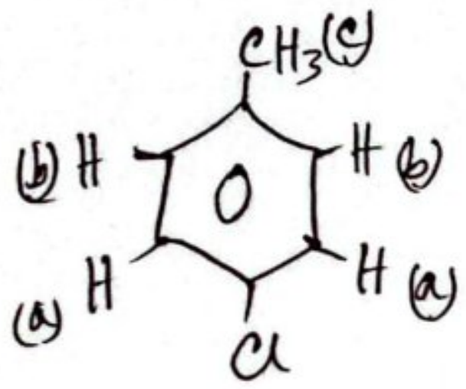
Identification of compounds by spectroscopic metho is a physical technique - explain?

⇒ Because spectroscopic method analyze the interac btwn electromagnetic radiation & matter providing information about the structure, composition & propertie of substance without altering its chemical nature/ composition of the compound. As it is based on observing physical interactions. so they are categorized as physical

technique.

2022

Draw ^{1}H-NMR of p-chlorotoluene. Mention approximate chemical shift valve ~~which~~ of each type of proton & explain the splitting pattern indicated in your spectrum?



There are three types of proton:

H_a, δ value 7.1-7.2, $n=1$ $\therefore n+1=2$ (doublet)

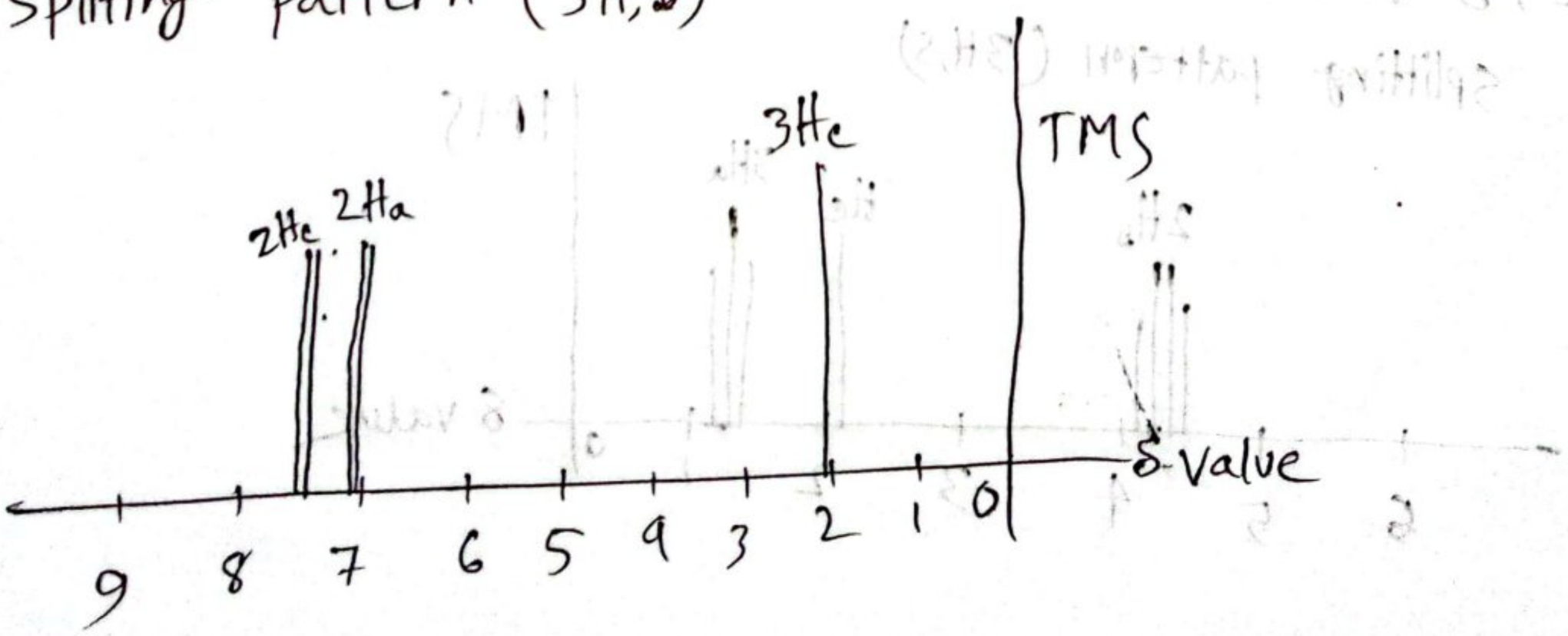
So, spliting pattern (2H, d)

H_b, δ value 7.2-7.3 ppm, $n=1$ $\therefore n+1=2$ (doublet)

So, spliting pattern (2H, d)

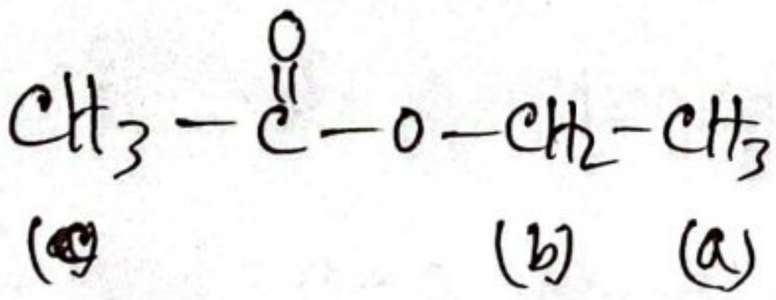
H_c, δ value 2.3-2.5 ppm $n=0$ $\therefore n+1=1$ (singlet)

So spliting pattern (3H, S)



(X) 2021 Draw ^{1}H-NMR spectrum of ethyl acetate. Mention approximate chemical shift value of each type of protons & explain splitting pattern?

⇒ Structure of ethyl acetate:



There are three types of proton.

H_a, δ value 1.26, $n = 2$ $\therefore n+1 = 3$ (triplet)

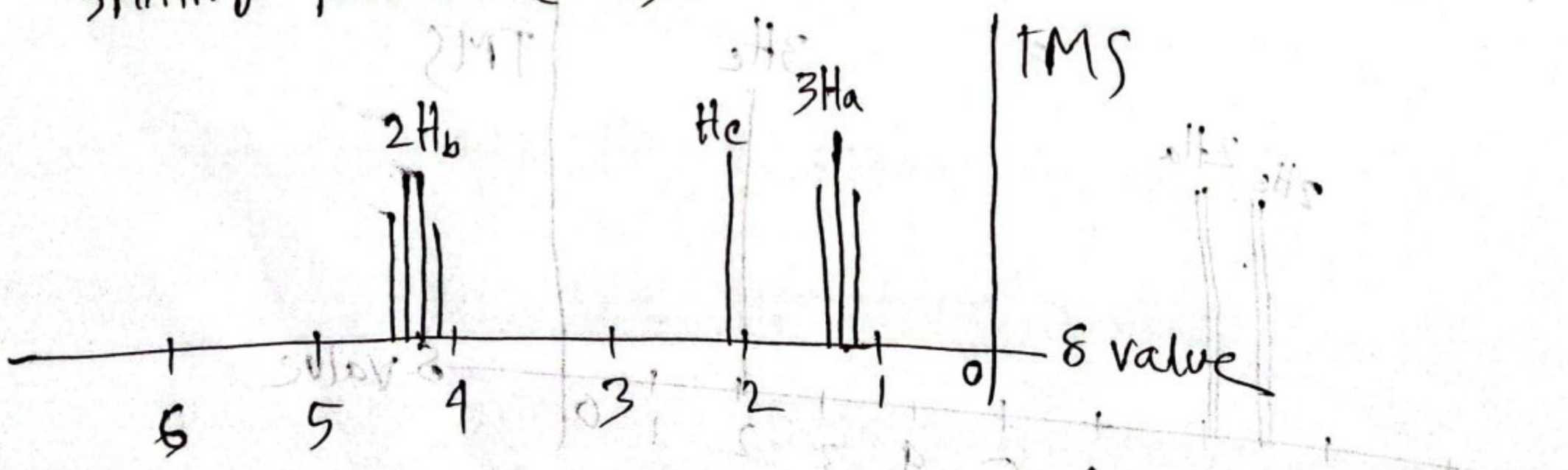
Splitting pattern (3H, t)

H_b, δ value 4.10, $n = 3$ $\therefore n+1 = 4$ (quarted)

Splitting pattern (2H, q)

H_c, δ value 2.09, $n = 0$ $\therefore n+1 = 1$ (singlet)

Splitting pattern (3H, s)

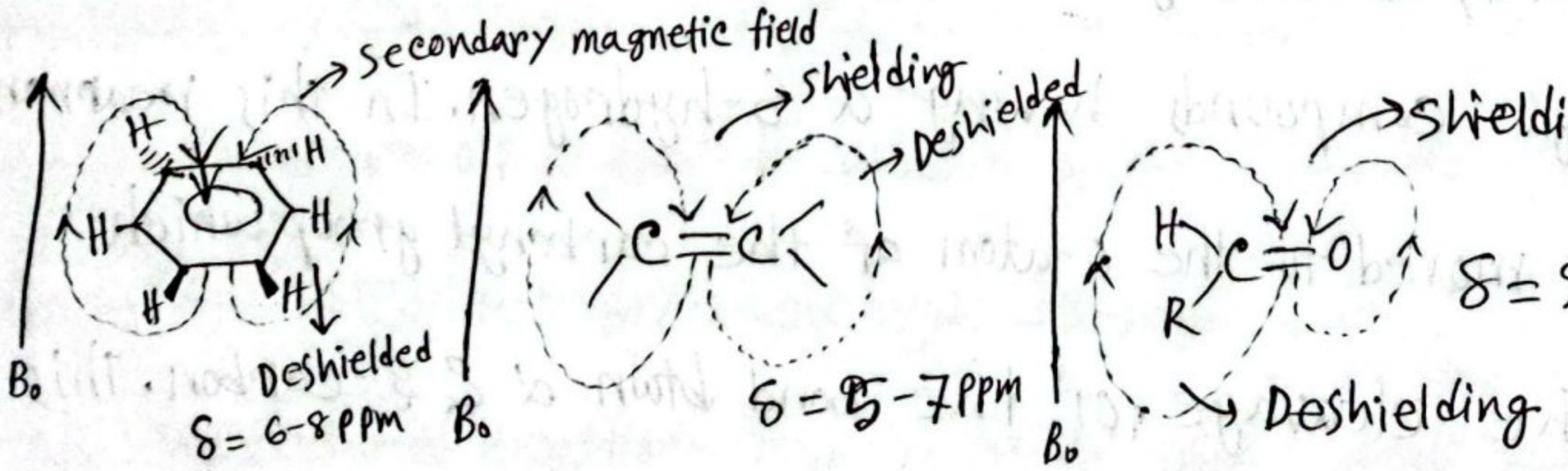


2022, 2021, 2020, 2018, 2016,

⊛ Write short note on "Anisotropic effect on chemical shift value?

⇒ The effects of magnetic field generated by circulating π-e⁻ in double bonds, triple bonds or aromatic rings is called magnetic anisotropic effect.

In aromatic ring, double bond / triple bond, carbonyl groups, the π-e⁻ create an induced magnetic field when an external field is applied. This happens due to the ring current in aromatic systems. The magnetic field is not uniform in all direction around the molecule, leading to regions of shielding & desheilding near the functional groups. In aromatic rings, the circulating π-e⁻ generate current when placed an external magnetic field. the induced magnetic field created deshields the proton of the benzene ring & the absorption is shifted towards downfield at 6-8.5 ppm.



The proton adjacent to the double bond or carbonyl groups are typically in deshielded region, resulting in downfield shifts,

In alkyne, proton attached to sp-hybridized carbons are in the

shielded region which leads to upfield shift. Thus anisotropic effect causes shift in chemical shift values due to the induced magnetic fields from the π-systems.

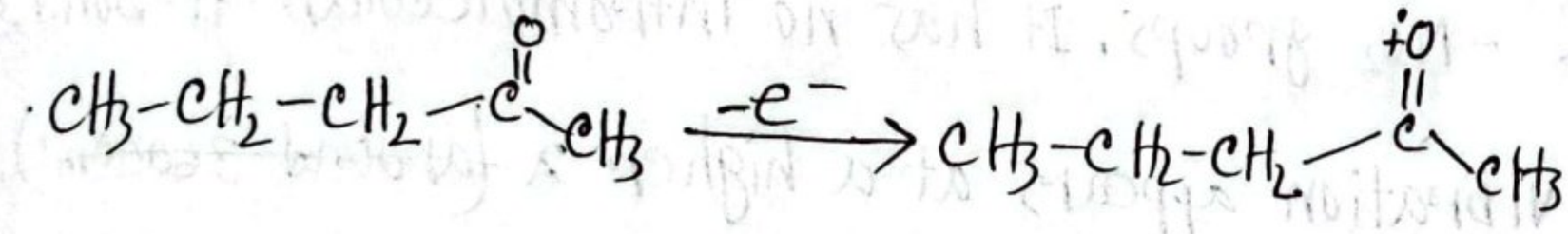
π-e⁻, like sigma e⁻ are induced to circulate in the presence of applied field to generate a magnetic field. the effect of magnetic field generated by circulating π-e⁻. are termed anisotropic effects. because the induced magnetic fields are direction dependent.

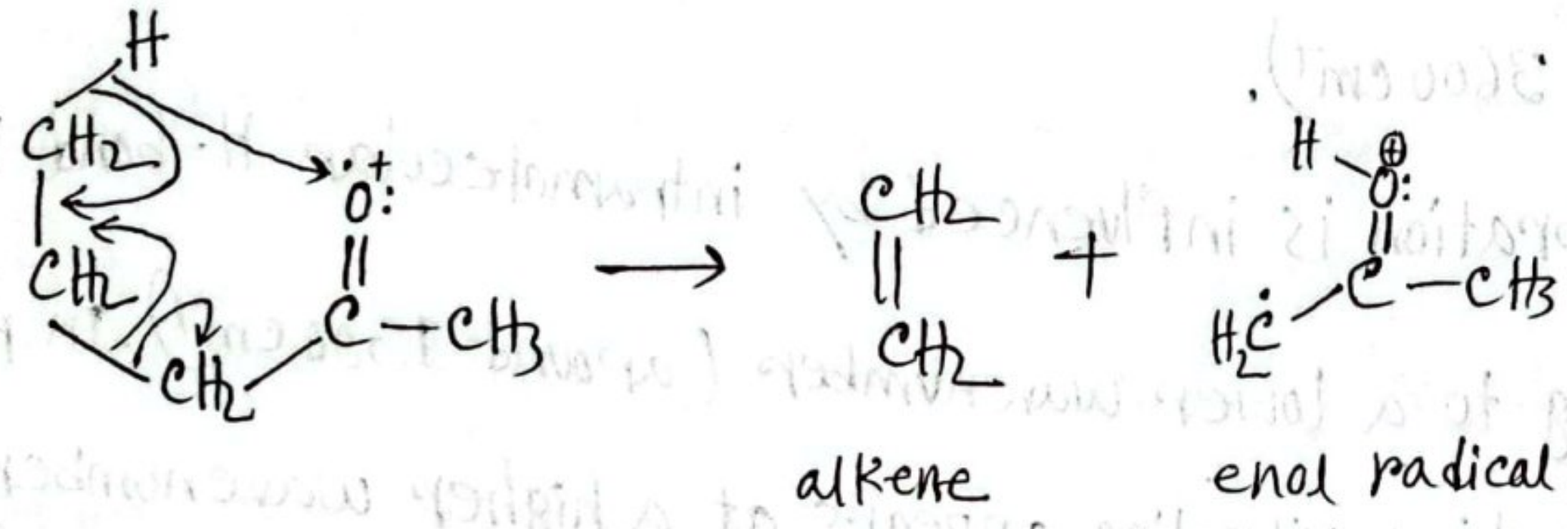
2018
Write short note on Mclafferty ion in mass spectroscopy?

⇒ Mclafferty ion in mass spectroscopy is the product of a specific re-arrangement of organic molecules, especially the carbonyl compounds. The rearrangement is called Mclafferty rearrangement. the Mclafferty rearrangement is a common cleavage that occurs in carbonyl compounds having a δ-hydrogen. In this rearrangement the δ-H is moved to the o-atom of the carbonyl group which leads to the cleavage of the bond btwn α & β-carbon. this results in the formation of an alkene & a charged fragment, known as the Mclafferty ion.

Mclafferty rearrangement is seen in the mass spectroscopy of

2-pentanone : $CH_3-CH_2-CH_2-CO-CH_3$





alkene enol radical cation

Mclafferty ion is particularly useful for the structural elucidation in mass spectroscopy. It help to identify the presence of carbonyl functional groups in a molecule.

2022

(X) How ortho & para nitrophenols can be distinguished by using IR?

⇒ ortho & para nitrophenols distinguished using IR based on differences in their intermolecular interactions & the resulting spectral characteristics. ortho nitrophenol has an intramolecular H-bond btwn -OH & $-NO_2$ because they are adjacent on the benzene ring. This stronger intramolecular H-bond results a shift of the -OH stretching vibration to a lower $\bar{\nu}$ which results weaker & broader peak in the IR spectrum. The -OH stretching vibration appears at a lower wavenumber (around 3200 cm^{-1}).

Para-nitrophenol exhibits intermolecular H-bonding ~~btw~~ due to large distance btwn -OH & $-NO_2$ groups. It has no intramolecular H-bond, so the -OH stretching vibration appears at a higher ~~λ (around 3600cm^{-1})~~. wavenumber (around 3600 cm^{-1}).

The c-o stretching vibration is influenced by intramolecular H-bond in ortho-nitrophenol, leading to a lower wavenumber (around 1300 cm^{-1}). In para nitrophenol, c-o' stretching vibration appears at a higher wavenumber (around 1350 cm^{-1})

2022

⊗ $LiAlH_4$ is better choice for reduction of multifunctional carbonyl compounds than $NaBH_4$-Why?

⇒ DUE to its superior reducing power & broader reactivity.

(ii) $LiAlH_4$ much stronger reducing agent than $NaBH_4$. It can reduce wide range of carbonyl-containing functional groups, including Ald, Ketones, esters, carboxylic acids & amides. BUT $NaBH_4$ is selective & typically reduces only Ald & Ketones under mild conditions.

(iii) $LiAlH_4$ is effective in reducing more reactive & less reactive carbonyl containing functional groups making it versatile for multifunctional compounds. But $NaBH_4$ is ineffective against less reactive groups like esters or carboxylic acids.

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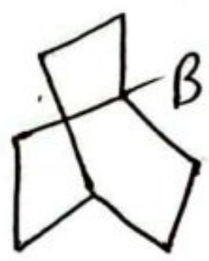
(iv) $LiAlH_4$ possesses highly polar Al-H bond, making the hydride ion (H^-) more nucleophilic & reactive, than $NaBH_4$.

(v) Al is less electronegative than boron leading to more polarized Al-H bond & a stronger reducing agent.

2022, 2018, 2017

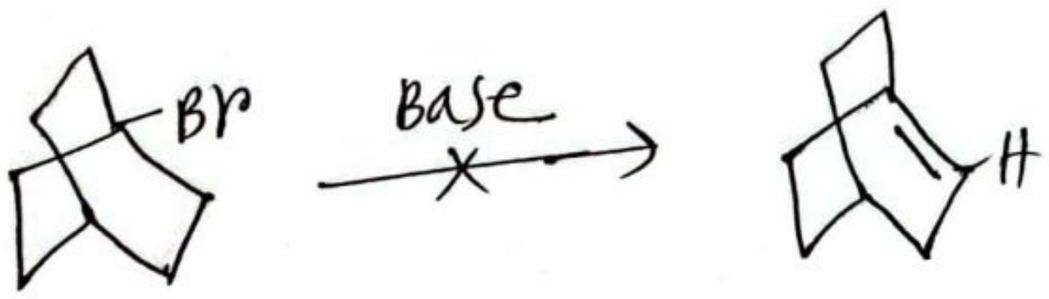
(*) The following compounds is not hydrolyzed even by hot & aqueous sln of KOH.



?

⇒ We know ~~ber~~ bredi's law state that " It is not normally possible to effect an elimination so as to introduce a double bond on a ~~bridging~~ bridge-head carbon atom in a fused ring system".

bredi's rule ~~said~~ In a bicyclic compound with a bridgehead carbon contained small ring is not formed double bond.



Here two p orbitals not form π bond because they are not in the same plane, the angle btwn two p-orbital is 90°. ~~So π orbital~~

So, π-overlap does not take place & reaction is not possible.

⊗ 2017
⊗ Name the method used for inter conversion of cis-trans isomer

⇒ (I) Thermal isomerization: Heat is applied to provide energy for the molecule to overcome the energy barrier & transition btwn cis- & trans form.

(II) Photochemical isomerization: UV or visible ligh is used to excite the molecule to a higher energy state, enabling isomerization.

(III) catalytic isomerization: specific catalysis facilitates conversion.

(IV) Electrochemical isomerization: Electric current is used to alter the electronic structure of the compound, leading to isomerization.

(V) pressure induced isomerization: High pressure can be applied to affect the equilibrium btwn cis & trans forms.

(VI) chemical Reagent isomerization: certain reagent can promote isomerizatio

⊗ 2017
⊗ The λ_{max} valve of $\pi \rightarrow \pi^*$ transition of 1,3-butadiene is larger than that of ethylene - explain?

⇒ In 1,3-butadiene conjugation is present & λ_{max} value is higher because the energy difference btwn $\pi \rightarrow \pi^*$ is lower. on the other hand, in ethylene no conjugation present & energy difference btwn π & π^* is larger.

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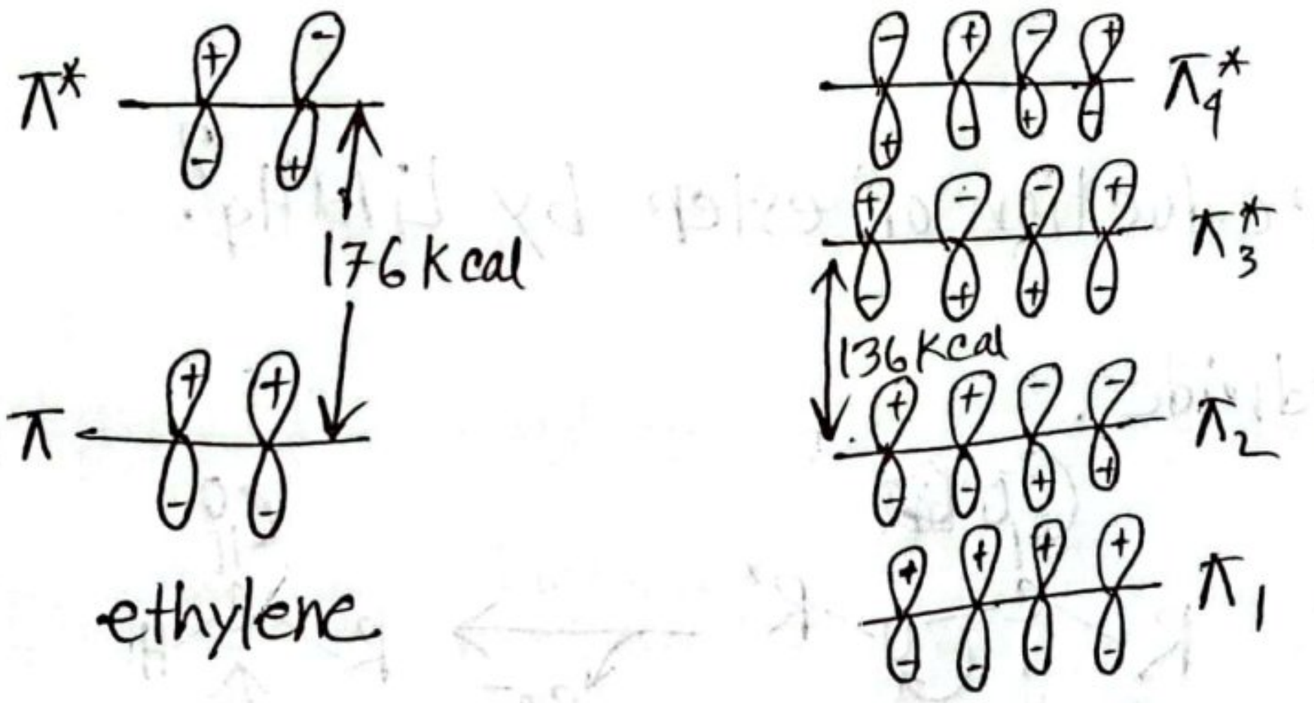
In Butadiene; 4 π orbitals is present two bonding π_1 & π_2 & two antibonding π_3^* & π_4^*. And energy gap is lower. But In ethylene two π-orbital present one is bonding & another antibonding.

In conjugation systems, the π orbitals overlap to form delocalized molecular orbitals. This results lower energy gap btwn Homo & LUMO.

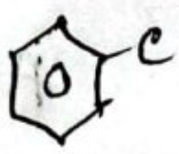
lower energy gap results higher λ as $E = \frac{hc}{\lambda}$.

1,3-bufadiene has conjugate double bonds, leading to delocalized π-e & smaller HOMO-LUMO energy gap. But ethylen has no conjugation.

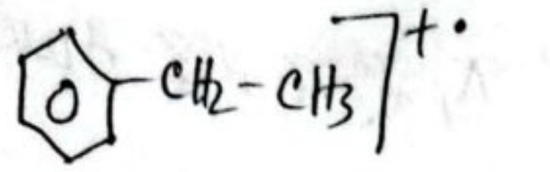


2017

Mass spectrum of ethylbenzene showed a base peak at m/z 91 -explain ?

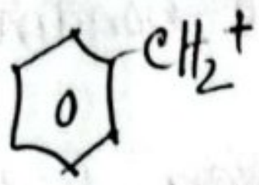
⇒ Structure of ethyl benzene  CH_2-CH_3

The molecular ion peak is found m/z = 106



m/z = 106

(molecular ion)
of ethylbenzene



m/z = 91
more stable

m/z = 77

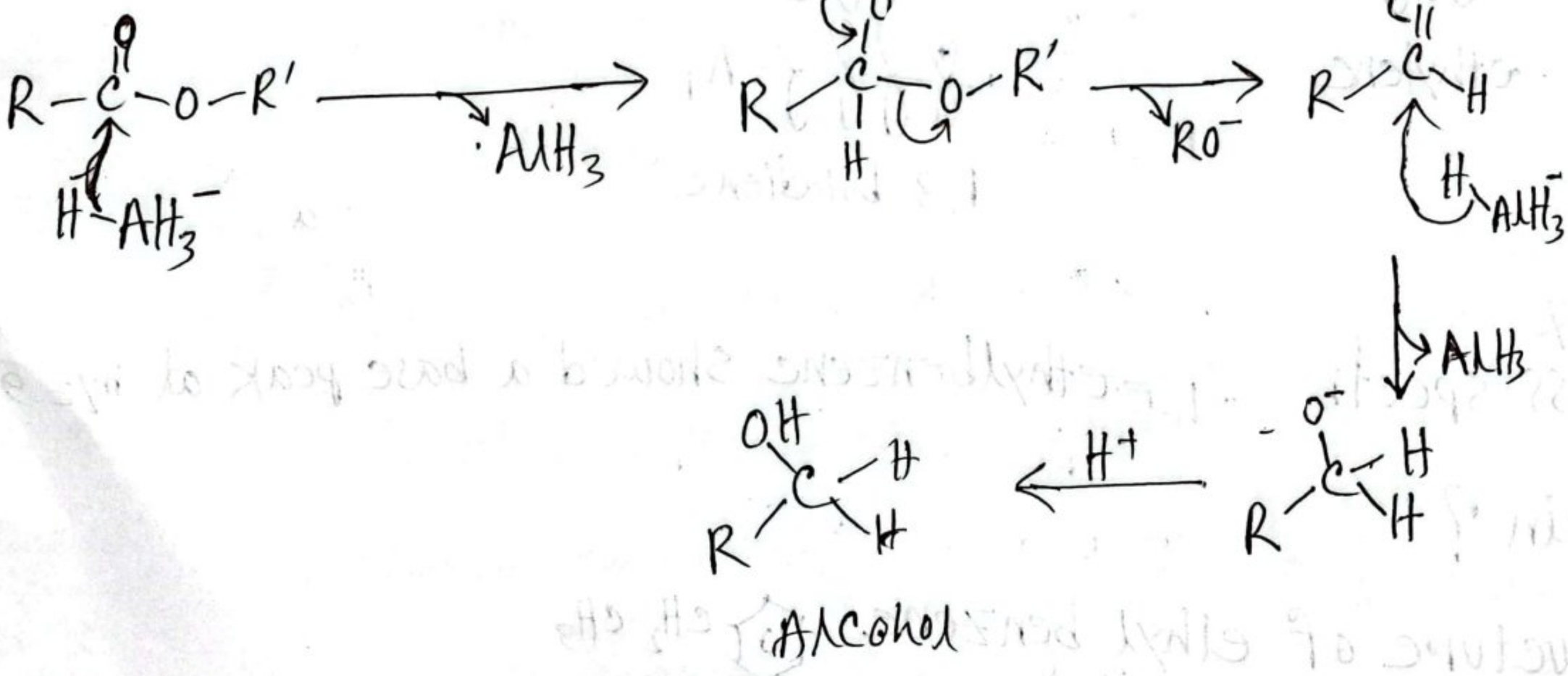
The high stability of tropylium ion makes strong signal at m/z 91. which is appears as base peak

2024
Among $NaBH_4$ & $LiAlH_4$ which one you select to reduce the carbonyl group of an ester?

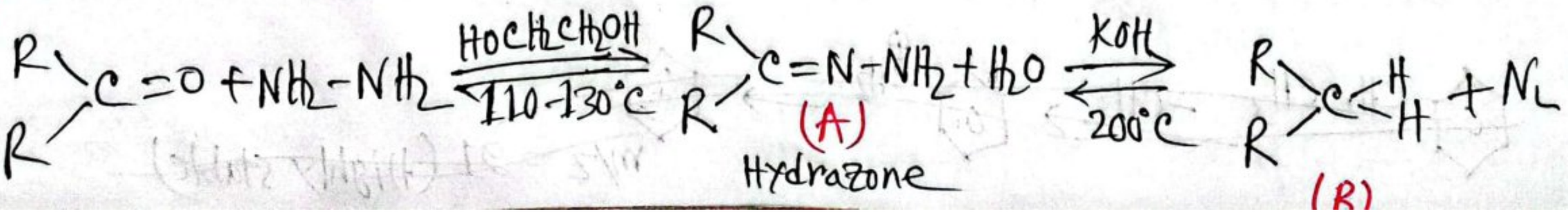
⇒ $LiAlH_4$.

2024
write mechanism of reduction of ester by $LiAlH_4$?

⇒ Step-1 : Addition of hydride.



2024
complete the reaction:



2024
(i) Why this reaction is carried out in two dfrnt temp. (110-130° & 200)?

⇒ To facilitate different steps in the reaction mechanism.

fists step (Hydrazone formation) occurs at a lower Temp. to prevent decomposition of the hydrazine & to allow controlled formation of the hydrazone intermediate. Also for preventing side reaction.

In 2nd step High Temp. (200°C) is required to overcome the Ea for the elimination of N_2 & for formation of $-CH_2$ group. This high temp. in 2nd phase accelerates reduction step & improved yield.

2024
(ii) can you use CH_3CH_2OH as a solvent instead of $HOCH_2CH_2OH$ to carry out above reaction?

⇒ NO. Wolf Kishner reaction typically requires high temp. to proceed effectively. Ethanol's low boiling point (~78°C) makes it unsuitable for this reaction as it will evaporate before reaching the require Temp. But ethylene glycol's high B.P. (~197°C) makes it suitable.

*2024
(iii) write mechanism of 'B' formation from 'A' showing all curved arrows.

⇒ Wolf Kishner reaction mechanism.

(iv) 2021 Write limitations of the reaction (wolf kishner) with example?

⇒ (i) This reaction is unsuitable for substrates with base sensitive groups (epoxides or halides). Ex. $ClCH_2COCH_3$, the chloro group may undergo elimination or substitution under strong basic conditions.

(ii) this reaction uses polar solvents like ethylene glycol, limiting its application for substrates insoluble in these solvents. Ex. Highly non-polar ~~sol~~ Ketones like may have poor solubility & Slower reaction rates.

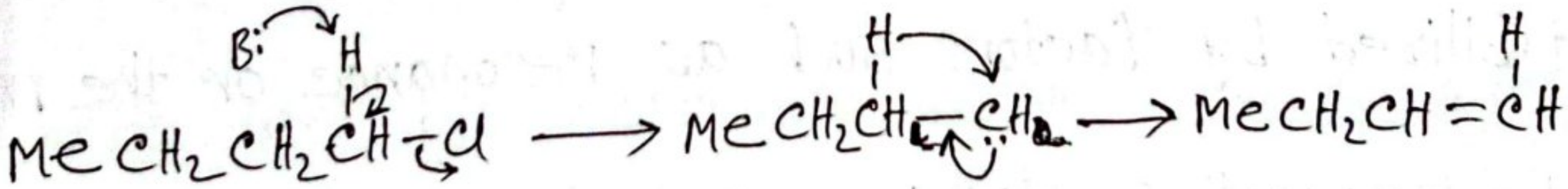
(iii) Initial step involves the formation of hydrazone intermediate, which can be challenging with sterically hindered carbonyl compounds.

(iv) It requires high temp. & strong basic conditions, which can limit its applicability to heat-sensitive substrates.

2017(t)

What do you meant by 1,1-elimination reaction. Give example?

⇒ 1,1-elimination reaction is a type of chemical reaction where two substituents are removed from the same atom, resulting in the formation of a double bond. This reaction typically leads to the formation of an alkene. This is a relatively rare type of elimination reaction, as most elimination reactions are 1,2-elimination, where the substituents are removed from adjacent carbon atom. This 1,2-elimination is often called a "beta-elimination". 1,1-elimination is called α-elimination. Example:



2016, 2021, 2015

Write short note on E_1cB with mechanism & example?

⇒ the E1cB (Elimination, unimolecular conjugate base) mechanism is a type of elimination reaction in organic chemistry. It involves a two-step process

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Where the first step is the formation of a carbanion (a carbon with a negative charge) by removing a proton (H^+) from a carbon adjacent to a leaving group. Secon. Step is the elimination of the leaving group (such as a halide) to form a double bond. This mechanism usually occurs when the substrate has a relatively acidic H atom adjacent to the leaving group & when a strong base is used.

Mechanism: ⓐ A. Strong base removes a proton from the carbon adjacent to the leaving group, forming a carbanion. This step is usually reversible. This carbanion is stabilized by factors such as resonance or the presen of electronegative atoms nearby.

ⓑ The carbanion then expels the leaving group to form a double bond & producing an alkene. This step is irreversible.

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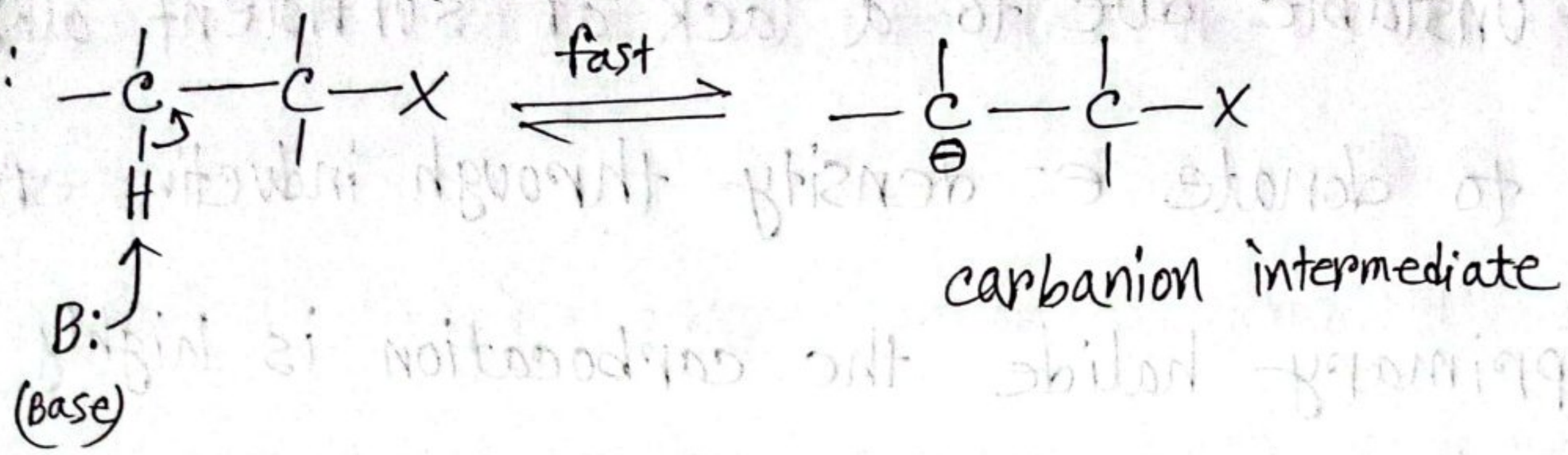
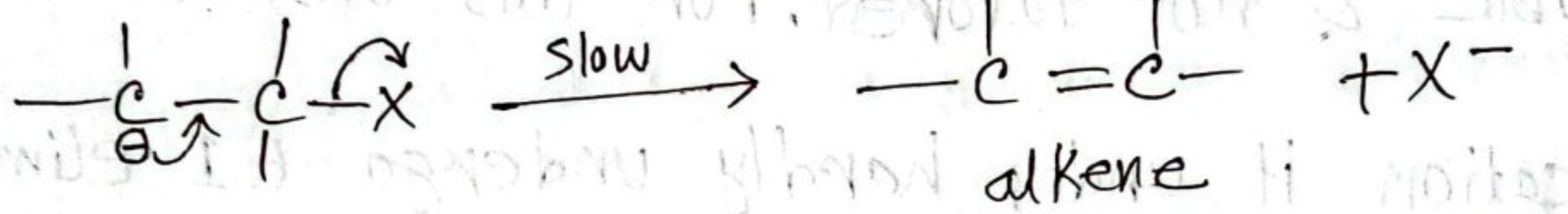
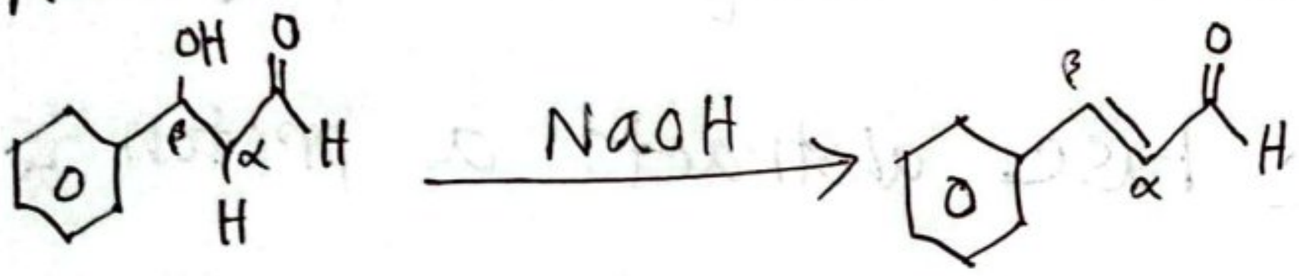
step-1:



step-2:



example:



2018, 2016

(X) primary halides hardly undergo E1 elimination? explain?

⇒ primary halides hardly undergo E1 elimination because the E1 mechanism involves the formation of a carbocation intermediate: In a primary halide, the carbon bearing the leaving group is a primary carbon, meaning it is only attached to one other carbon atom. This makes the resulting carbocation

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highly unstable due to a lack of sufficient alkyl groups to donate e⁻ density through inductive effect. For primary halide the carbocation is highly unstable & not favored. For this unstable nature of carbocation it ~~under~~ hardly undergo E1 elimination. primary halides are more likely to undergo E2 elimation, where the base abstracts a proton & a leaving group in one step. If strong base is present, E2 elimination can occur.

⊗ 2018. Write effect of e⁻ withdrawing groups in the elimination reaction?

⇒ e⁻ withdrawing group (EWG) withdraw e⁻ density from the carbon atom to which they are attached. This makes the adjacent carbon (β carbon) more e⁻ deficient & there H more acidic.

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(II) In elimination reaction, EWG makes the proton removal easier & faster.

(III) EWG pull e density away from the carbon atom bearing the leaving group (LG). This weakens the bond btwn the carbon & the LG, making the LG more prone to depart.

(IV) In E1cB, when strong EWG are present, the formation of a carbanion intermediate becomes more favorable.

(V) EWG favor the E1cB mechanism over E1 & E2.

(VI) EWG can also influence the ragioselectivity of elimination reaction

(VII) In E1, EWG can stabilize the intermediate carbocation by delocalizing the positive charge. EWG increase the rate of E1 reaction by stabilizing the carbocation.

(VIII) In E2 reaction, EWG can stabilize the transition state. & Enhance the reaction rate by making H more acidic.

Such EWG are $-NO_2$, $-CN$, $>C=O$, $-CHO$ etc.

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(A) 2017 Elimination is promoted by e⁻ withdrawing substitution. Explain with example?

⇒ This statement refers to the effect of EWG on elimination reaction. In E1 reaction, the EWG helps to stabilize the carbocation through inductive effect & promoting the elimination.

In E2 reaction, the presence of EWG near the leaving groups helps to stabilize the ~~carbanion~~ negative charge that develops on the adjacent carbon during the transition state & promotes the elimination reaction.

EWG pull e⁻ away from the carbon they are attached to. By pulling e⁻ away from the carbon EWG developing negative charge, EWG help stabilize the transition state this makes the elimination reaction more favorable.

EWG stabilize the carbanion intermediate formed in the first step of E1cB reactions & promoting E1cB reactions

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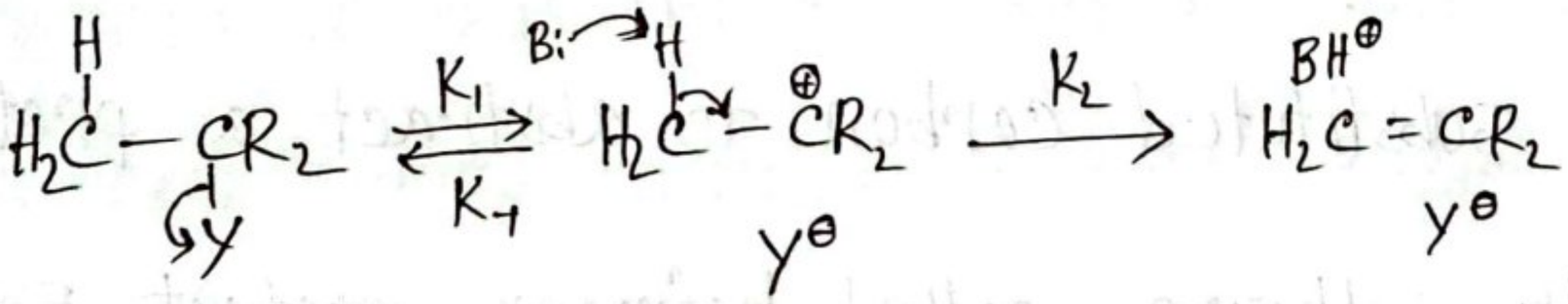
2017, 2017 (t)

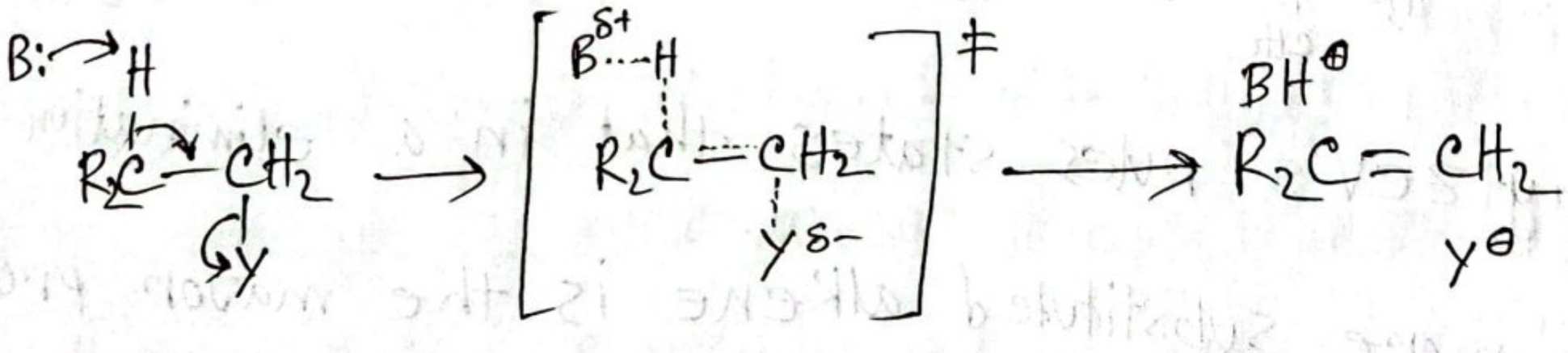
⊗ show three dfrnt schematic representation for 1,2-elimination?

⇒ Three different schematic representations of the 1,2-elimination : E1, E2 & E1cB. these 1,2-elimination reaction involves the removal of two substituents (such as proton & leaving group) from adjacent carbon atoms, resulting in the formation of a double bond.

If the C–Y bond ~~is broken first~~ & H–C bonds can be broken seperately in two-step process. If the C–Y is broken first & carbocation is formed, this is E1.



If H–C & C–Y breaking in one step process, passing through a single T.S. This is E2 mechanism.



If the H–C bond broken first & involving a carbanion intermediate, this mechanism is E1cB.

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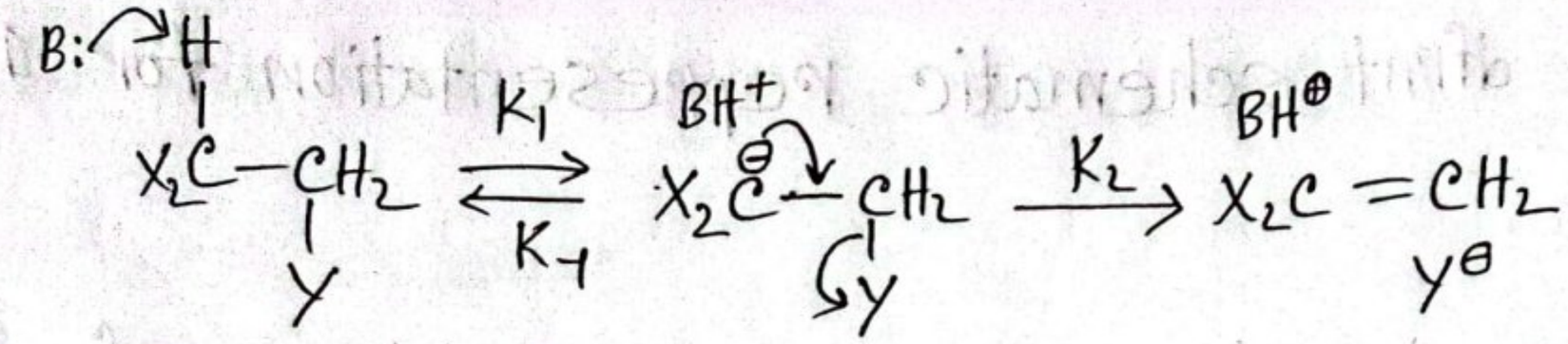
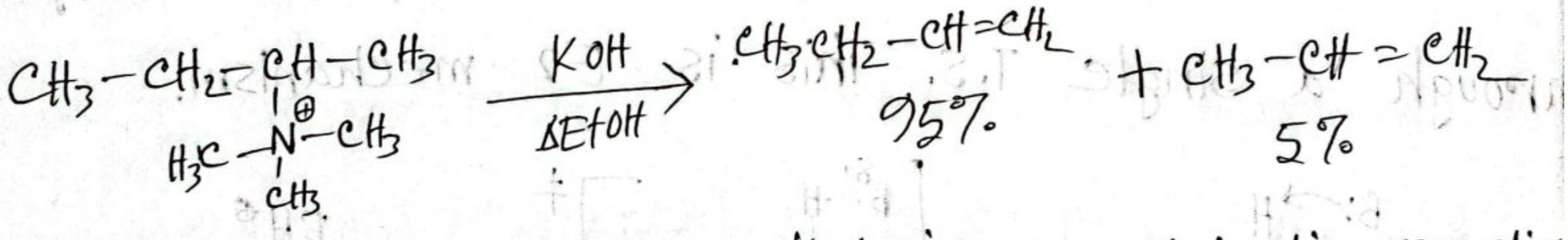


2016
⊛ what are Hofman & saytzev rules of elimination? Explain the rules using appropriate examples?

⇒ Hofmann elimination is an elimination reaction of an amine to form amines. It state's that if you use a bulky (hindered) base in an elimination reaction, the less substituted alkene will be the major product. This is because the bulky base has difficulty accessing the more substituted carbon to abstract a proton. And least stable alkene called hofmann product. Example



⇒ saytzev's rules states that in a elimination reaction the more substituted alkene is the major product. because the more substituted alkene is generally more stable. Example:

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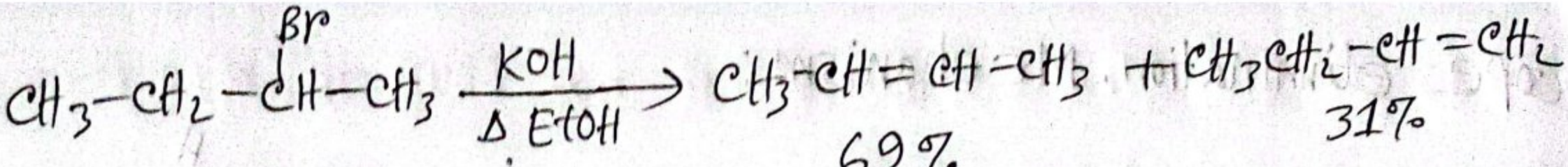

Hofmann's rule is the opposite of saytzeff's rule. It applies when there is significant steric hindrance, especially with large leaving groups like quarternary ammonium salts. generally saytzeff's rule is applies when there is no steric hindrance & Hofmann's rule is often observed when bulky groups are involved, such as in reactions with quarternary ammonium salts.

2016, 2017(t)

Ⓧ Give the mechanism of pyrolytic SYN elimination reaction considering tertiary amine oxide & xanthates as examples?

⇒ Both tertiary amine oxides and xanthates undergo pyrolytic syn elimination reactions, which involve the elimination of groups from the same side of the molecule through a mechanism. These reactions proceed through cyclic transition states and forming alkenes. pyrolytic syn elimination in tertiary amine oxides is called

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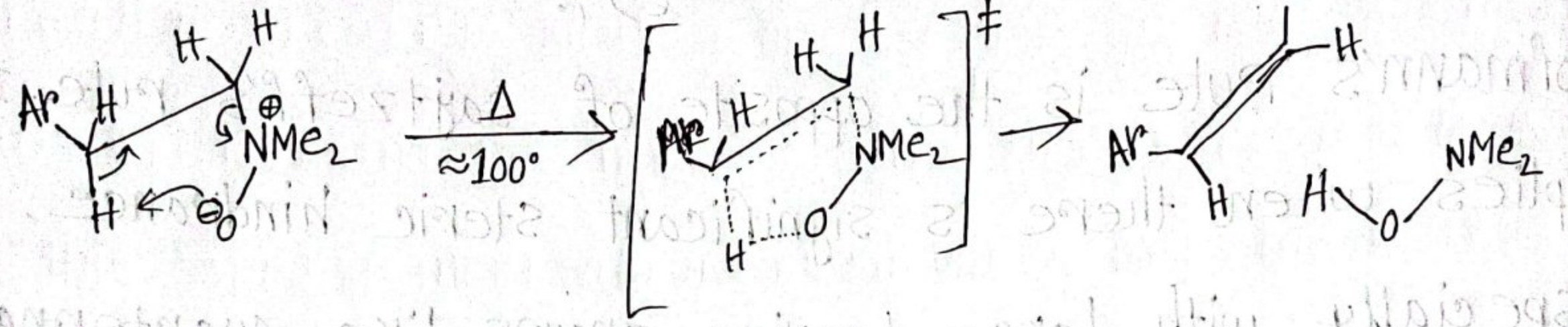
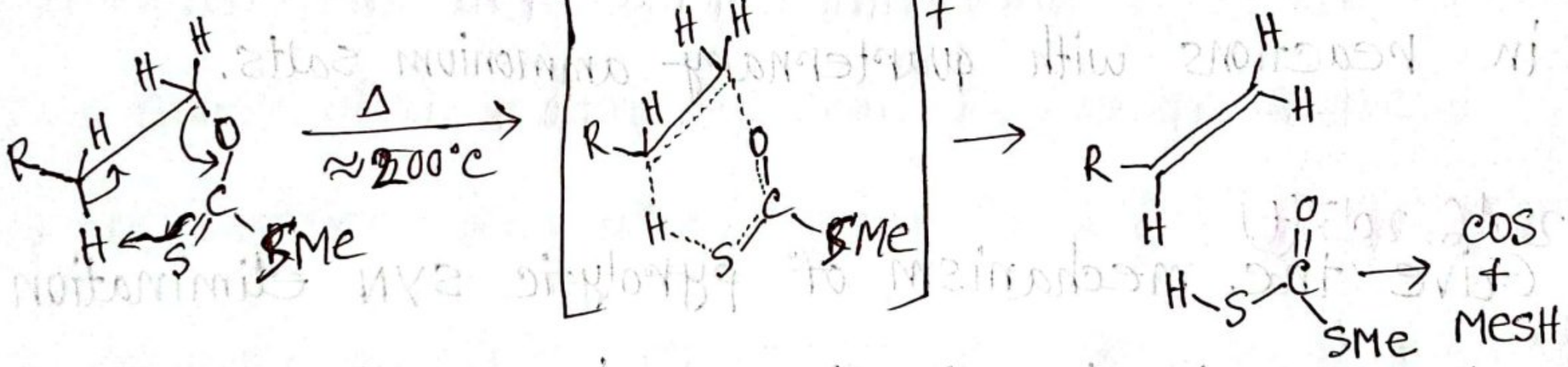
Cope elimination. Mechanism:



The pyrolysis of xanthates is called chugaev reaction.

In the transition state, six-membered ring is formed.

Mechanism:



pyrolytic syn elimination also known as Ei eliminations (elimination intra-molecular).

2017(t)
⊛ under comparable conditions, CH_3CH_2Br yielded ≈1% alkene while $PhCH_2CH_2Br$ yielded ≈99% alkene - explain?

⇒ the primary carbocation formed from CH_3CH_2Br is very unstable. Hence, the elimination is not favoured

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and very little alkene is produced which is 1%. But the benzylic carbocation formed from $PhCH_2CH_2Br$ is very stable due to resonance with the phenyl ring. The positive charge can be delocalised onto the phenyl ring, making the carbocation much more stable than a simple primary carbocation. The greater stability of the benzylic carbocation makes the elimination reaction much more favourable, resulting in 99% alkene production.

2016
⊗ What are the possible grounds which can be stated for favouring E2 elimination of acyclic molecules from the anti-periplanar conformation?

⇒ The E2 elimination mechanism involves the simultaneous removal of a proton (H^+) from a β-carbon & the departure of a leaving group from the α-carbon. This process is highly depend on the geometry of the molecule, specifically the antiperiplanar conformation, which is the most favourable for E2 elimination.

① In an antiperiplanar arrangement, the C-H and C-LG

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bonds are in the same plane but point in opposite directions; creating ideal geometry for interaction of the orbitals, and making E2 elimination more favourable.

2. This arrangement allows maximum overlap btwn the orbitals of the leaving group, the hydrogen, and the developing π system of the forming double bond in the alkene.

3. The antiperiplanar conformation facilitates a smooth, low-energy transition state, leading to a faster r. rate.

4. Antiperiplanar arrangement has a lower activation energy for the reaction compared to other conformation.

5. As the C-H and C-LG bonds aligned in antiperiplanar orientation so deprotonation and the departure of the leaving group can occur simultaneously.

6. This conformation reduces steric hinderance which allows the base to more easily abstract the β-hydrogen, making the reaction more efficient.

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2021
(iv) Write limitations of the reaction (wolf kishner) with example?

⇒ (i) This reaction is unsuitable for substrates with base sensitive groups (epoxides or halides). Ex. $ClCH_2COCH_3$, the chloro group may undergo elimination or substitution under strong basic conditions.

(ii) This reaction uses polar solvents like ethylene glycol, limiting its application for substrates insoluble in these solvents. Ex. Highly non-polar ~~sol~~ Ketones like may have poor solubility & slower reaction rates.

(iii) Initial step involves the formation of hydrazone intermediate, which can be challenging with sterically hindered carbonyl compounds.

(iv) It requires high temp. & strong basic conditions, which can limit its applicability to heat-sensitive substrates.

2022, 2020, 2016, 2015, TU
(x) How does trapping of intermediate help to investigate r. mechanism?

⇒ Trapping of an intermediates is a technique that can help to investigate a reaction mechanism by allowing the identification of short lived & unstable intermediates.

During a reaction, a short-lived intermediate species forms, then trapping agent added to the reaction mixture. Trapping agent

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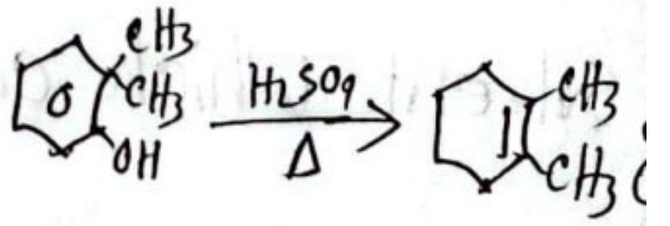
react with intermediate. the trapped product is isolated & analyzed using spectroscopic technique (IR, NMR, mass spectroscopy) to determine its structure & properties. EX. Benzyne intermediate.

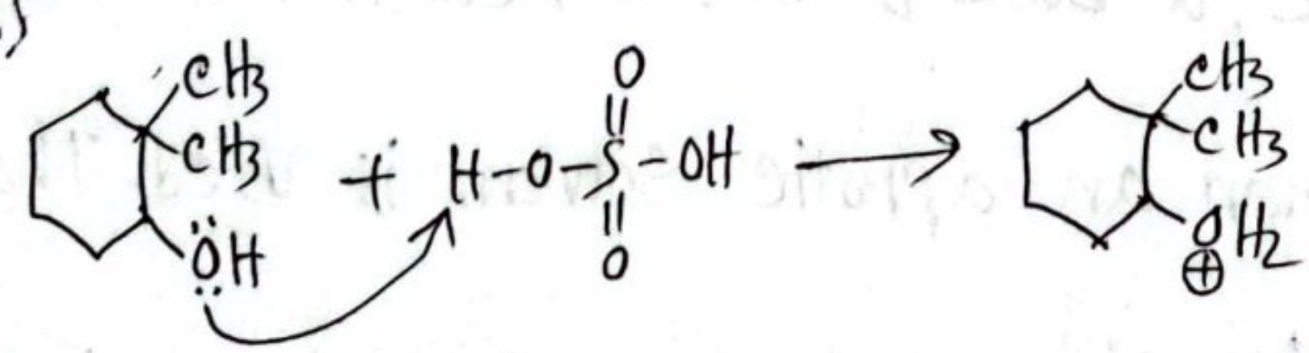
Benzyne is highly reactive species that can be generated through various methods, including the elimination of a halide ion from an aryl halide. To trap benzyne, a diene such as furan can be added to the reaction mixture. The diene undergoes a diels-alder reaction with benzyne, forming an adduct. By isolating this adduct & analyzing it we gain information about its reactivity & geometry. Trapping of intermediates proposed mechanisms by providing experimental evidence.

2017, 2015

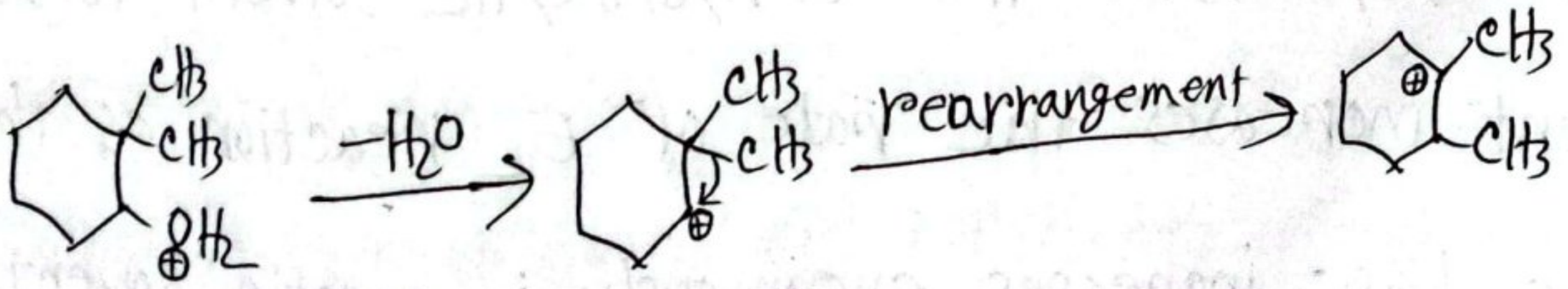
(X) Propose a detailed, stepwise mechanism for:



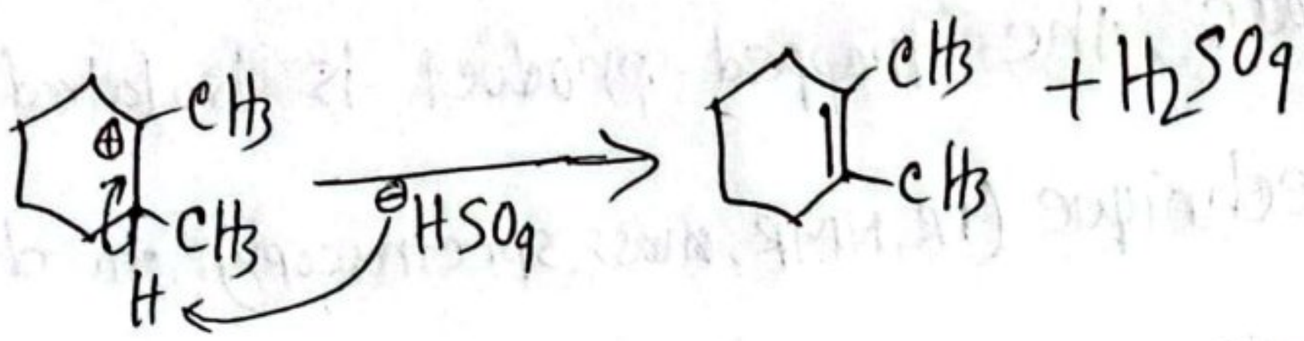
⇒ step-1:



Step-2:



Step-3:



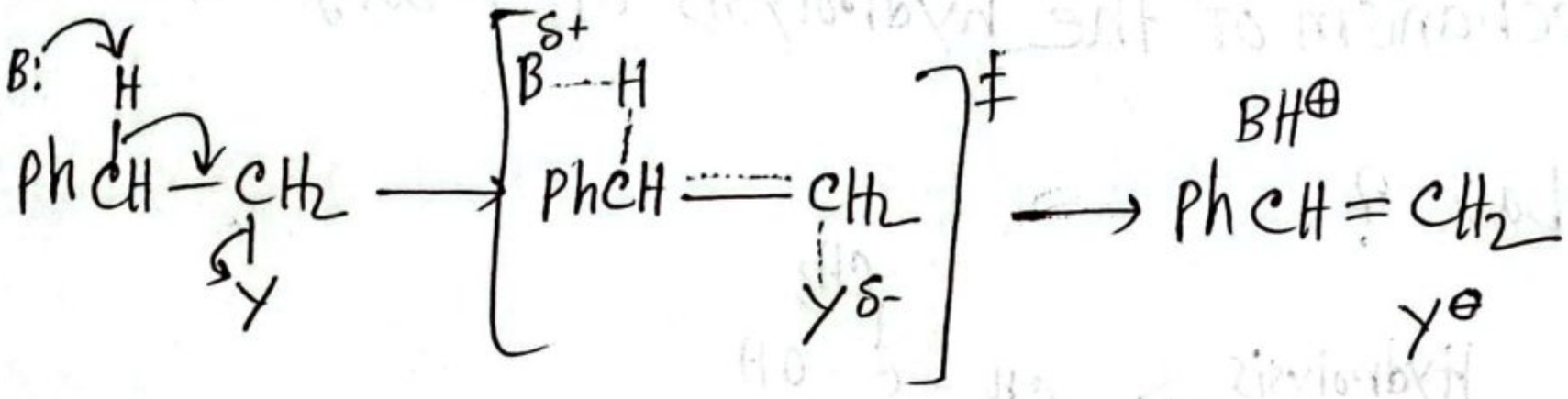
(*) 2017 with a given base, transfer from a hydroxylic solven like H_2O/EtOH to a polar aprotic one like DMF/DMSO can have Profound effect on the rate of E_2 reactions - explain?

⇒ In E_2 reaction, as the strength of the base increases, the rate of reaction increases too. In polar protic solvents or hydroxylic solvents such as H_2O/EtOH, the base is solvated in the solvent due to H-bonding. This solvation hinders the base's ability to deprotonated a substrate effectively. But in polar aprotic solvents, the base do not form H-bond & do not get solvated. Hence, the base acts as a stronger base in aprotic solvent which can ~~act~~ now abstract a proton from the substract easily. Therefore, a base is more reactive towards the substrate for E_2 reaction when an aprotic solvent is used. That is, with a given base, transfer from a hydroxylic solvent to an aprotic solvent increases the rate of E_2 reaction as the strength of the base increases enormously in aprotic solvents.

2017
⊛ Explain the relative rate sequence observed for $PhCH_2CH_2Hal$ with $\bar{O}Et/EtOH$:

	$PhCH_2CH_2F$	$PhCH_2CH_2Cl$	$PhCH_2CH_2Br$	$PhCH_2CH_2I$
R. rate	1	70	4.2×10^3	2.7×10^9 ?

⇒ From the Mechanism of E_2 elimination reaction we see:



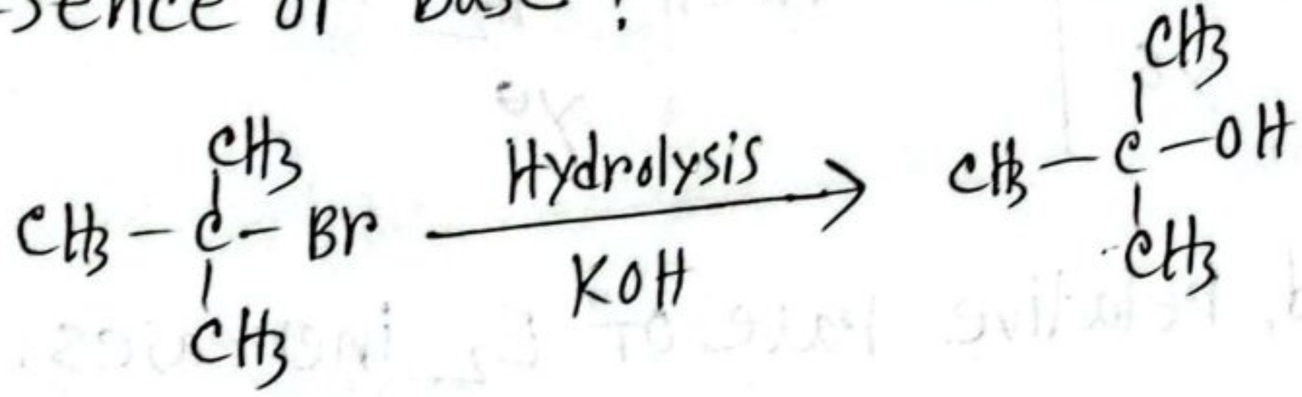
the weaker C-Y bond, relative rate of E_2 increases. As the size of halogen I > Br > Cl > F so, bond strength sequence is C-F > C-Cl > C-Br > C-I So the leaving group F is good than other. So the R. rate among them like above.

2017
⊛ which reagent in each of the following pairs is more nucleophile & why: ① $(CH_3)_3B$ & $(CH_3)_3N$ ② H_2O & H_2S ?

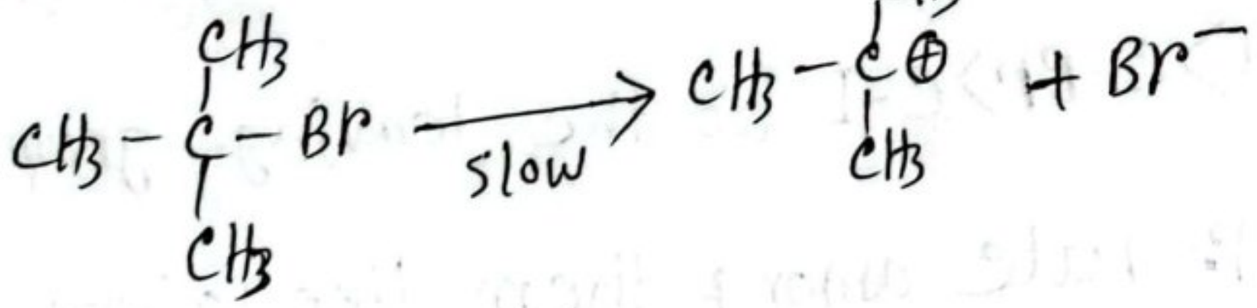
⇒ ① $(CH_3)_3N$: is more nucleophile than $(CH_3)_3B$. because N atom has a lon pain & methyl group donate e^- to the N atom so it can attack any positive charge part easily. But B has incomplete octate, it ~~contain only six elect~~ has lacks of lone pair. It has an empty p-orbital, making it e^--deficient. it tends to accept e^- rather than donate ~~e~~ e^-, so $(CH_3)_3N$ is more nucleophile.

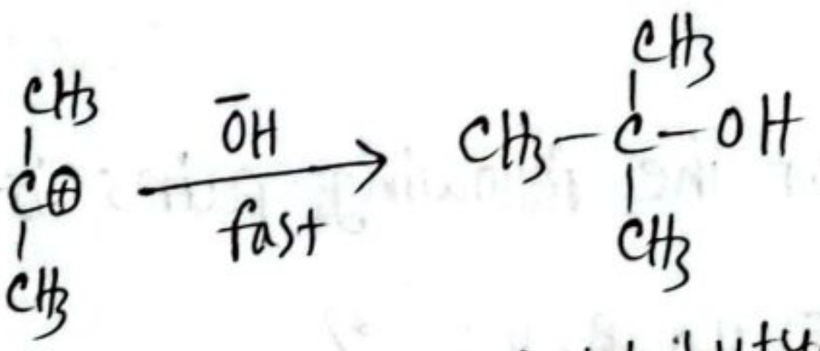
(11) $H_2\ddot{S}$: is more ~~electropl~~ nucleophile than $H_2\ddot{O}$:. Both O & S has two lone pair but O is more electronegative than S. So O can't easily donate its lone pair than S. So H_2S more nucleophile.

(X) 2017 Give the mechanism of the hydrolysis of t-butyl bromide in presence of base?

$\Rightarrow$ 

t-butyl bromide

Mechanism: step-1: 

step-2: 

tert-butyl alcohol

It follow S_N1 mechanism & occur in 2 step.

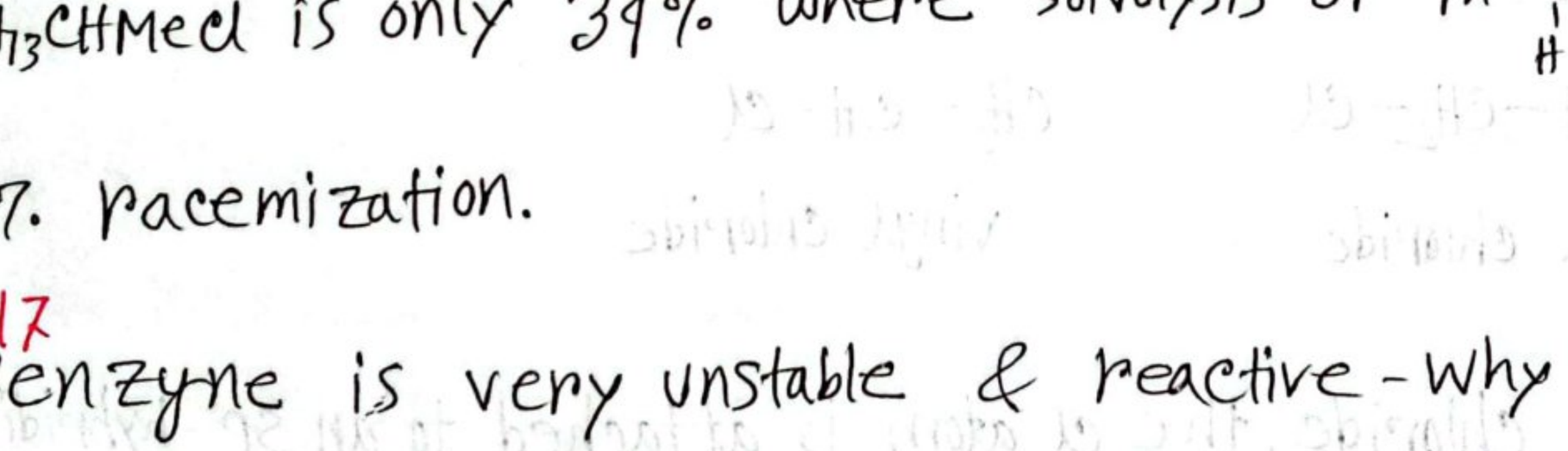
(X) 2017 Solvolysis of (+)PhCHMeCl leads to 98% racemization in aqueous acetone while (+)-C_6H_{13}CHMeCl leads to only 34% racemization-why?

$\Rightarrow$ we know, when the carbocation is more stable, then the racemization is more.

(±)-phCHMeCl $\xrightarrow[\text{slow}]{-Cl^-}$ Ph–C(Me)(H)–$\overset{\oplus}{}$ + Cl^-

(±)-C_6H_{13}CHMeCl $\xrightarrow[\text{slow}]{-Cl}$ C_6H_{13}–C(Me)(H)–$\overset{\oplus}{}$ + Cl^-

among these carbocation Ph–C(Me)(H)–$\overset{\oplus}{}$ is more stable due to presence of benzene ring. For benzene ring there delocalization of e^- or resonance is possible for this reason it is stable.



but, C_6H_{13}–C(Me)(H)–$\overset{\oplus}{}$ is less stable than Ph–C(Me)(H)–$\overset{\oplus}{}$ so solvolysis of (±)C_6H_{13}CHMeCl is only 34% where solvolysis of Ph–C(Me)(H)–$\overset{\oplus}{}$ leads to 98% racemization.

2017

Benzyne is very unstable & reactive - Why?

⇒ due to its highly strained triple bond in a cyclic compound. Benzyne is typically formed by removing two H or substituting groups from adjacent C-atom in benzene, creating a triple bond. This triple bond introduces significant strain & disrupts the aromatic stability of benzene means it loses aromaticy.

The triple bond in benzyne is not a typical linear triple bond. It is a bent bond, meaning the two π orbitals involved in the bond are not perfectly aligned. This misalignment creates significant angle strain, making the molecule unstable. This strain results in a high energy state for benzyne, making it highly reactive. The π-e^- in the triple bond are highly localized making benzynes electrophilic & reactive toward nucleophiles. Nucleophiles readily attack benzynes at the strained triple bond.

Benzyne

2017

(*) compared to ~~alkyl~~ allyl chloride, vinyl chloride is very unreactive towards nucleophilic substitution? Why?

⇒ $CH_2=CH-CH_2-Cl$ allyl chloride

$CH_2=CH-Cl$ vinyl chloride

① In vinyl chloride, the Cl atom is attached to an sp^2-hybridized carbon atom in a double bond system. This bond has more s-character making it shorter, stronger & less polar, which reduces the tendency of Cl to leave as a leaving group.

In allyl chloride, the Cl atom is attached to an sp^3 hybridized

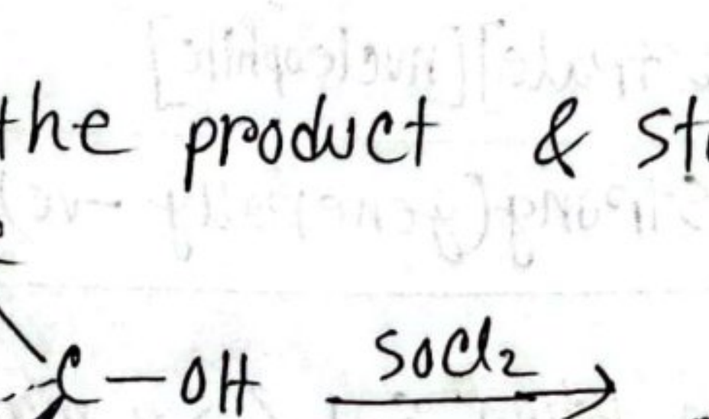
carbon, making the bond weaker & more prone to cleavage.

(II) In vinyl chloride, resonance delocalization occurs btwn the Cl lone pairs & the π-systems of the double bond. This gives the C-Cl bond partial double bond character, further stabilizing the bond & making it less reactive. But in Allyl chloride no delocalization occur.

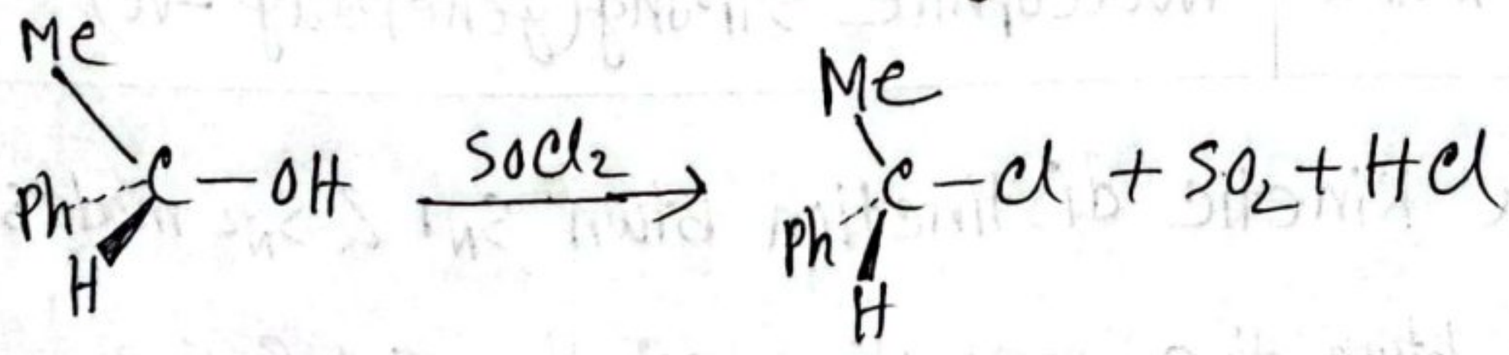
(III) The allyl carbocation is resonance-stabilized, making the reaction more favorable. Vinyl less reactive due to lack of resonance stabilization.

(IV) $CH_2=CH-CH_2-Cl \xrightarrow[\text{slow}]{-Cl^-} CH_2=\overset{\curvearrowright}{CH}-\overset{\oplus}{C}H_2 \longleftrightarrow \overset{\oplus}{C}H_2-CH=CH_2$

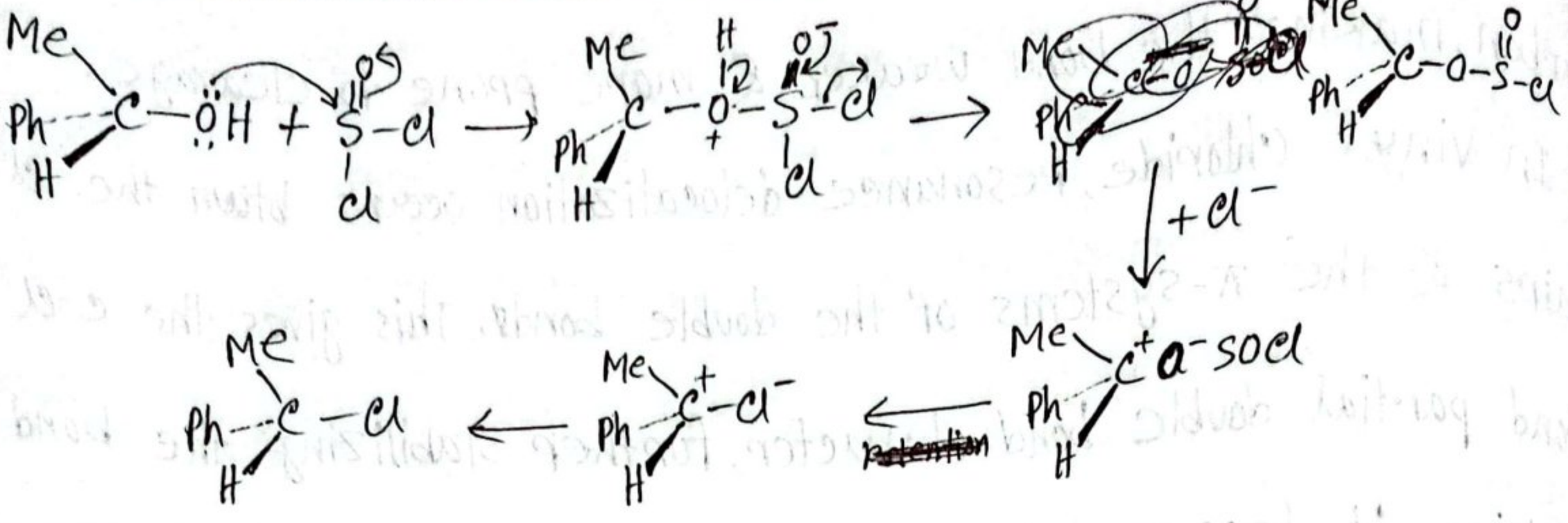
(V) 2017 what is S_Ni ? complete reaction & show its mechanism:



⇒ S_Ni means Nucleophilic Substitution intra-molecular. & in this reaction retention of configuration occurs that means configuration of the product & starting material is same.



Mechanism:



2016, 2015

(A) What is the salient difference btwn S_N1 and S_N2 pathway?

S_N1	S_N2
two step reaction	one step reaction
carbocation formed	no carbocation formed
intermediate formed (carbocation)	~~intermediate~~ formed transition state
polar protic solvent is used	polar aprotic solvent is used
It is unimolecular reaction	bimolecular
follow first order kinetics	follow second order kinetics
partially racemized product is formed + inverted (mix)	only inverted product is formed.
Rx reactivity order $3^\circ > 2^\circ > 1^\circ > CH_3X$	$CH_3X > 1^\circ > 2^\circ > 3^\circ$ ← reactivity order
Rate = K [substrate]	Rate = K [substrate] [nucleophile]
Nucleophile weak (generally neutral	Nucleophile strong (generally -ve)

2016, 2015

(A) How doo you make a kinetic distinction btwn S_N1 & S_N2 modes?

⇒ A kinetic distinction btwn the operation of the S_N1 & S_N2 modes can be made by adding a competting nucleophile, such as azide anion (N_3^-).

In S_N1 reaction the rate of the reaction depends on only substrate so after adding nucleophile the rate of reaction is unchanged.

But in S_N2, the rate of reaction depends on both substrate & nucleophile. so by adding nucleophile the rate of reaction also increases. with increasing conc. of nucleophile increasing the rate of reaction in S_N2 modes.

2016, 2024(t), 2015

⊛ The rate of solvolysis of Me_3CBr is found 3×10^4 times faster in 50% aq. ethanol than ethanol (normal) - explain?

⇒ Due to the effect of solvent. H_2O (aqueous) is more polar than ethanol & dielectric constant of H_2O is higher than ethanol so ~~H2O~~ the carbocation will be stable. For this reason rate of reaction is higher than ethanol alone.

$$(CH_3)_3CBr \xrightarrow[-Br^-]{\text{slow}} CH_3-\overset{CH_3}{\underset{CH_3}{\overset{|}{\underset{|}{C}}}}\oplus + Br^-$$

(carbocation)

water molecule can solvate carbocation stabilized by H-bonding. This stabilization lowers the energy barrier for carbocation formation, making it easier for the reaction to proceed. Increased polarity of the solvent mixture enhances the ionization of the halide

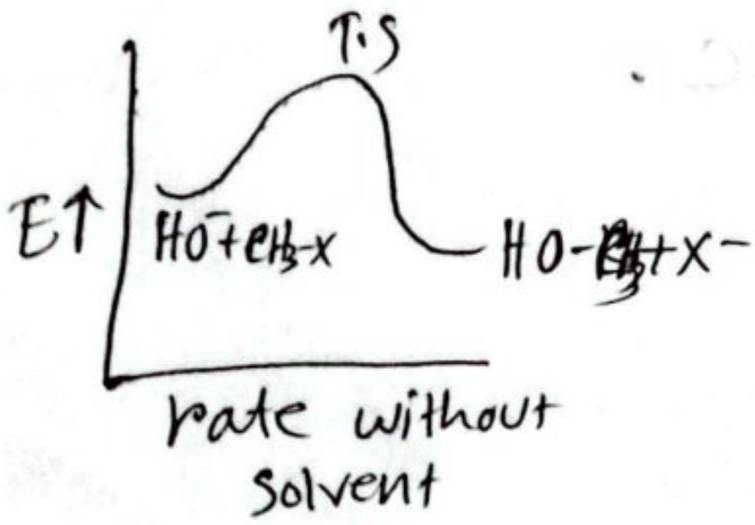
ion, facilitating the formation of the carbocation.

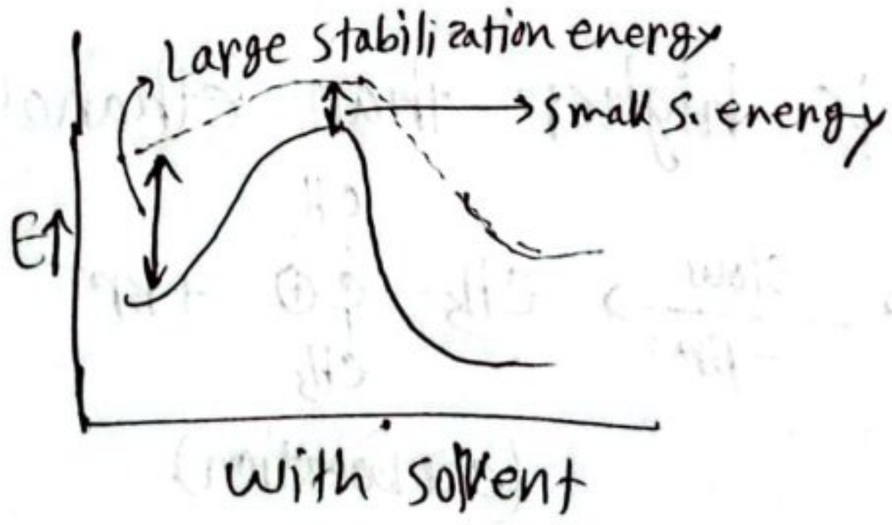
⊗ 2016, 2015
For S_N2, increasing solvent polarity is found to result in slight decrease in r. rate - why?

⇒ Because in S_N2, new charge is not developed in the transition state & the charge is dispersed in the Transition state comparing the starting material

$$CH_3-X + \bar{O}H \rightarrow \left[HO^{\delta-}\cdots CH_3\cdots X^{\delta-}\right]^{\neq} \rightarrow HO-CH_3 + X^-$$

Polar solvent solvated starting material stabilized because concentrated charge present but in transition state the charge is dispersed. when solvent polarity increases the more stronger H-bond with form starting material that means stronger solvation is occur in starting material.





⊗ 2016, 2015
why does polar non hydroxylic DMF have marked effect on (increased 4.5×10^9) S_N2 reaction of MeI with N_3^- at 0°C compared the reaction with MeOH?

⇒ The S_N2 reaction of MeI with N_3^- :

$$MeI + N_3^- \xrightarrow[0°C]{S_N2} MeN_3 + I^-$$

For S_N2, the greater the polarity of solvents, the slower the reaction. Again the changing form of protic to aprotic solvents also affect the r. rate. In MeOH, the nucleophile N_3^- is highly solvated through H-bonding; whereas it is very much poorly solvated in DMF. The largely unsolvated N_3^- ion in DMF is a very much stronger nucleophile than the solvated ones which slowers the r. rate. Hence N_3^- in DMF can easily attack MeI & make the reaction 4×10^9 times faster comparing the reaction in MeOH.

2016, 2015, 2021, TU

✱ Why does S_N2 r. rate decreases as the following series is traversed : CH_3-Br, $CH_2(Me)Br$, Me_2CHBr, Me_3C-Br ?

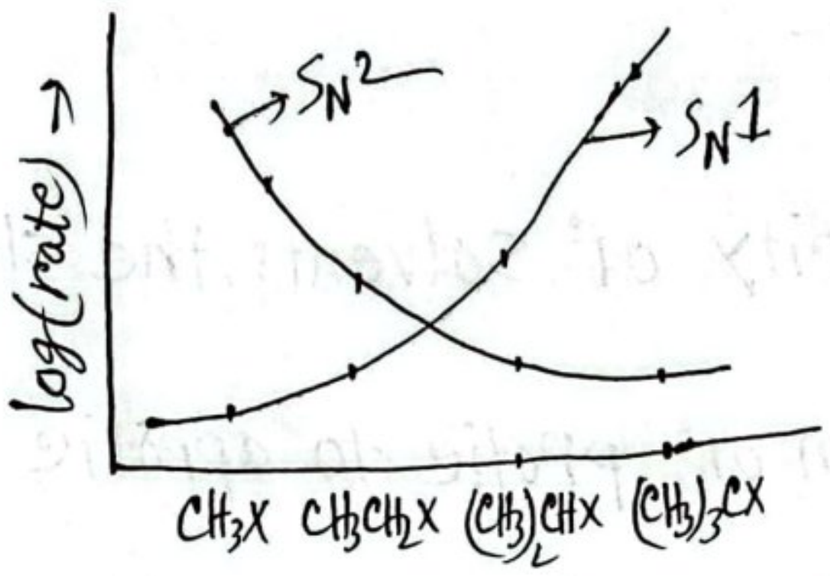
⇒ rate of reaction order :

$CH_3-Br > MeCH_2Br > Me_2CHBr > Me_3C-Br$

① Nucleophile attack on the C^{st} carbon & transition state is formed. But more methy group decreas partial positive charge on this carbon, so nucleophile can not easily attack on this carbon.

② more methyl group make more crowding or more steric

~~hin~~ hinderence so the r. rate decreased.



2016

⊗ The π-orbital system of benzene can act as a neibouring group in nucleophilic substitution?

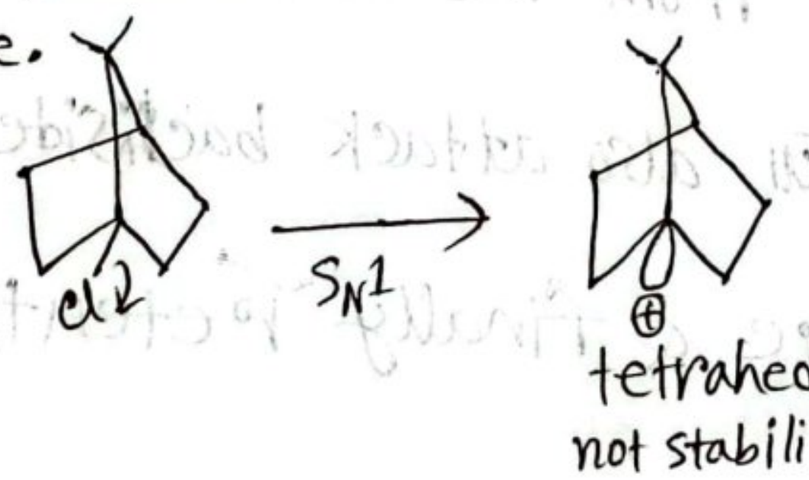
⇒ π-orbital system of benzene can act as a neibouring group by stabilizing intermediates or transition states through deloclization. This occurs when benzene ring is adjacent to the reactive center. If the carbocation is adjacent to a benzene ring, the π-e^- system of the ring can delocalize the +ve charge through resonance. This stabilization makes the intermediate more favourable, increasing the r. rate. The benzene ring's π-e^- can also influence the orientation & approach of the nucleophile due to electronic & steric effects. Ex. solvolysis of benzyl chloride in water. The π-e^- of the benzene ring can participate in the reaction, stabilizing the carbocation intermediate & leading to a faster r. rate.

2015, 10

(*) The following bicyclic halogenoalkane is almost unreactive to S_N1 & S_N2 – why?

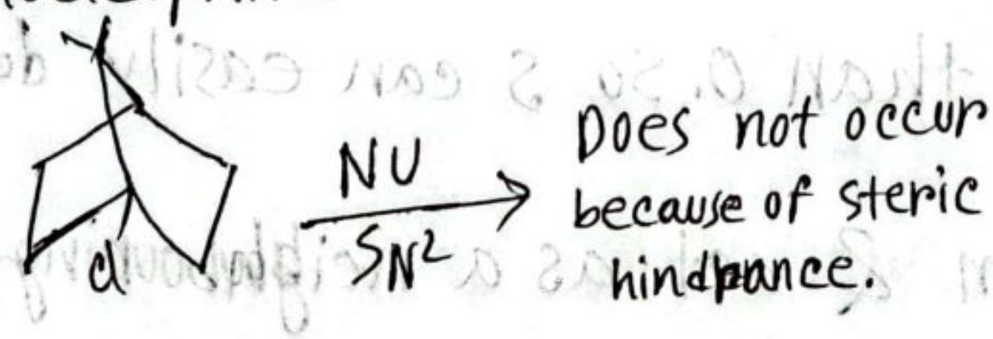
⇒ due to their unique structural constraints, which hinder both the formation of intermediates & the approach of nucleophiles.

Bicyclic structures, particularly those with bridgehead carbons, do not allow for the formation of a stable carbocation due to angle strain (Bredt's rule). normally carbocation planner in geometry but ~~bicy~~ bicyclo cannot for planner structure. so it is unstable & S_N1 mode is not possible. carbocation normary sp^2 but bridgehead 'C' cannot sp^2 due to strain increase.



$$\text{Strain} \propto \frac{1}{\text{stability}}$$

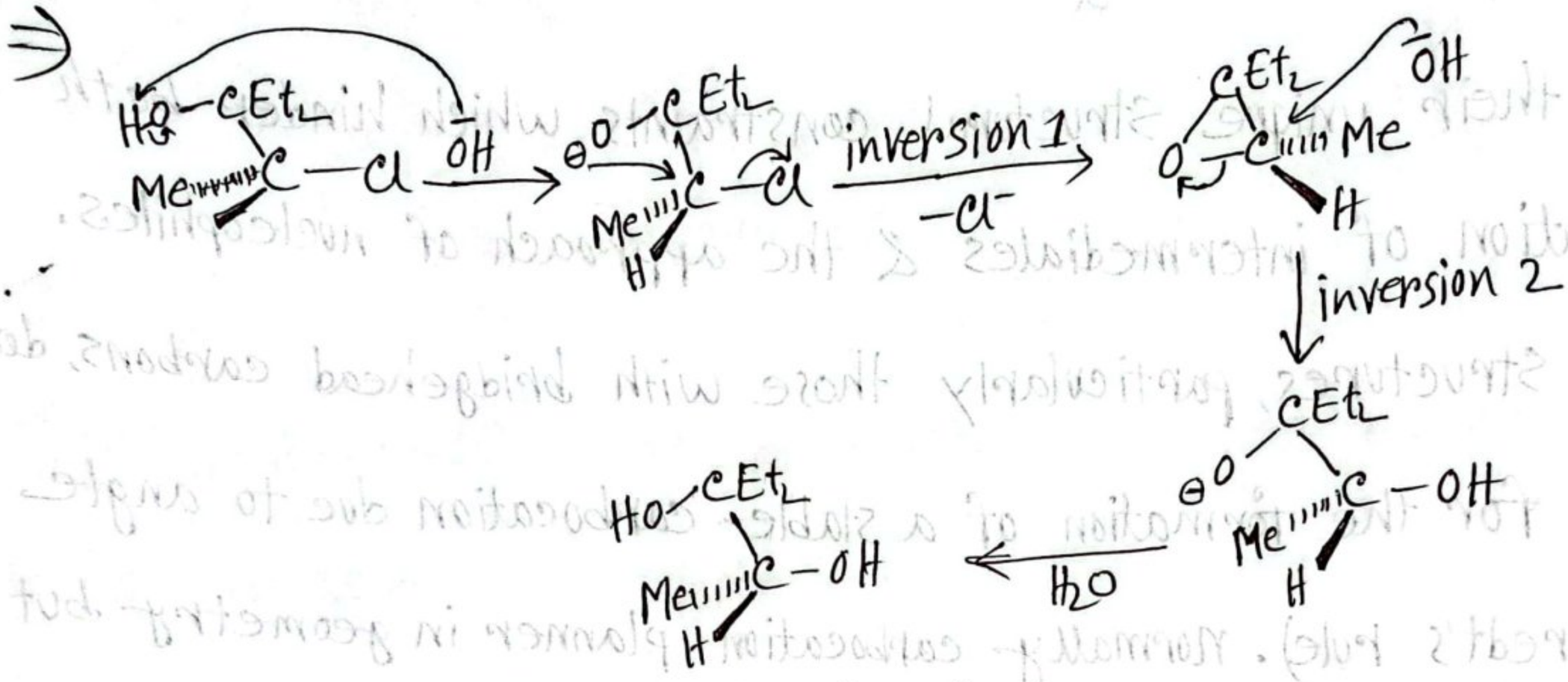
The crowded & rigid bicyclic structure blocks the backside attack of the nucleophile. so S_N2 is very difficult or impossible:



(*) 2015
In S_N2 order of reactivity $CH_3X > 1° > 2° > 3°$ – why?

⇒ because of steric hindrance.

⊗ 2015 The alkaline hydrolysis of 1,2-chlorohydrin ($MeCHClCEt_2OH$) is found to yield 1,2-diol ($MeCH(OH)CEt_2OH$) with retention - explain?

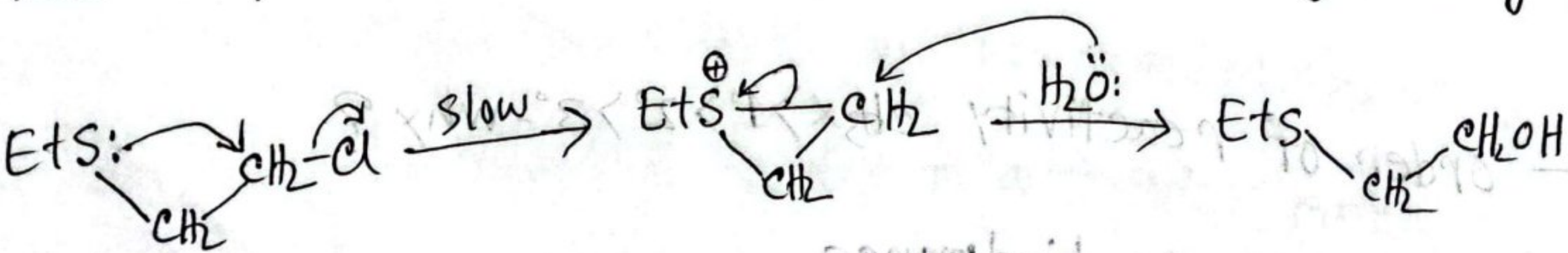


Retention of configuration

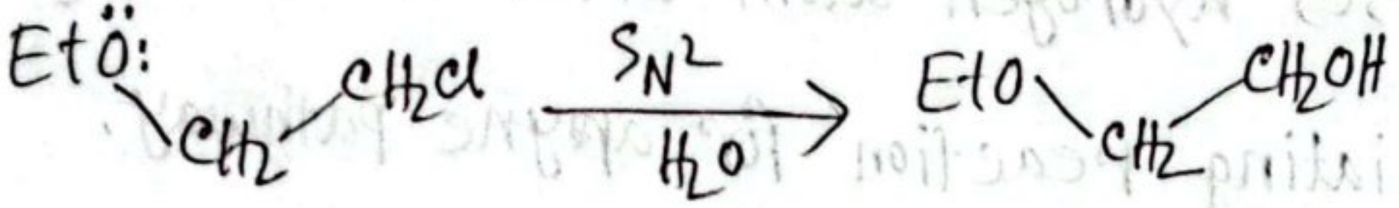
At first $\bar{O}H$ base take H^+ from $HO-CEt_2-$ & O^- is formed. which is neibouring ion & attack from the backside & first inversion take place. The nucleophile $\bar{O}H$ also attack backside of C-O bond & 2nd inversion also take place & finally retention of conf. found.

⊗ 2015, 2017 (+) $EtSCH_2CH_2Cl$ is found to undergo hydrolysis 10^4 times faster than $EtOCH_2CH_2Cl$ under compareable condition – explain-why?

⇒ S is less electronegative than O. so S can easily donate the e⁻ pair to carbon atom & act as a neighbouring group.

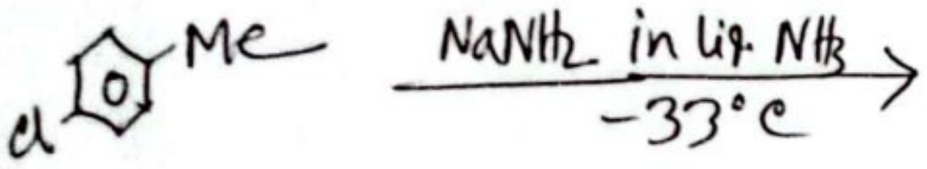


But in $EtOCH_2CH_2Cl$, O (3.5) is sufficiently electronegative & do not want to donate e^- pair & hydrolysis of $EtOCH_2CH_2Cl$ follow S_N2.

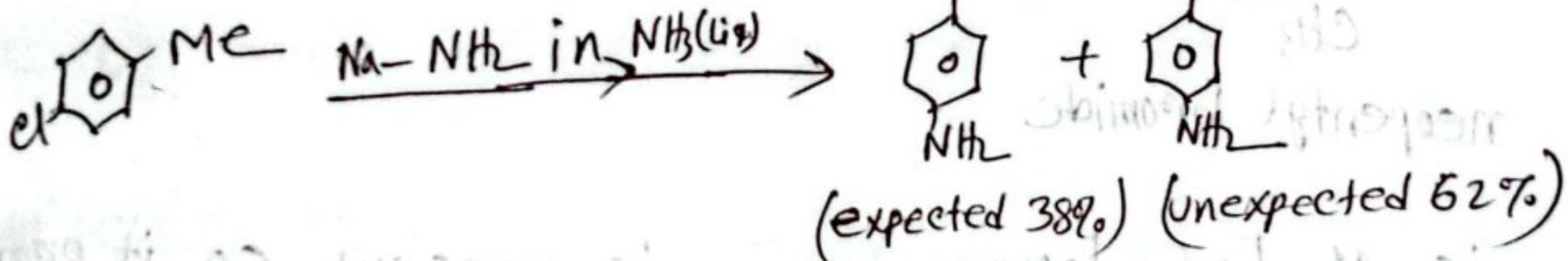


for this difference in electronegativity hydrolysis of $EtSCH_2CH_2Cl$ is 10^4 times faster than $EtOCH_2CH_2Cl$.

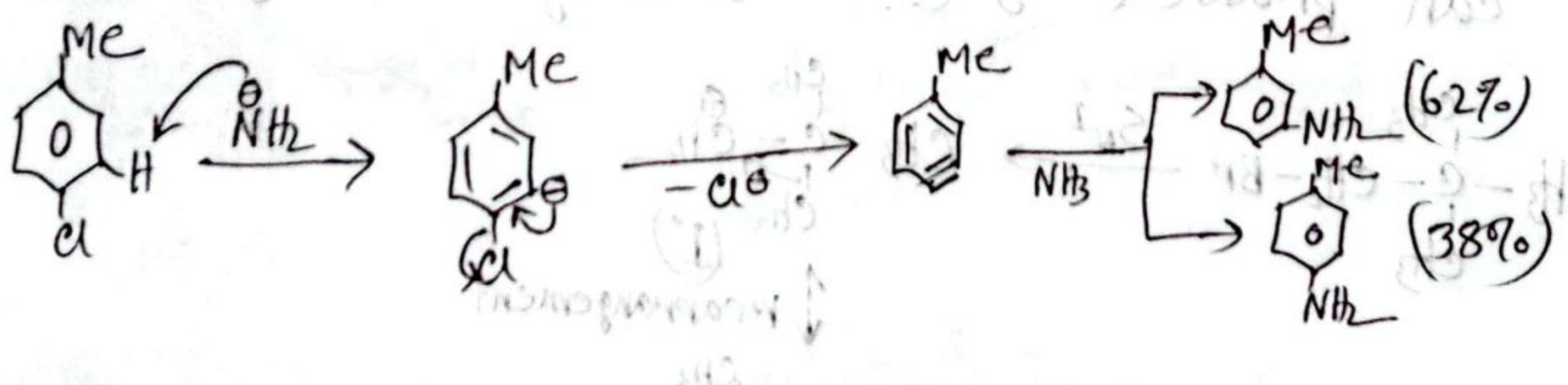
(4) 2018 complete the reaction & mechanism with evidence:



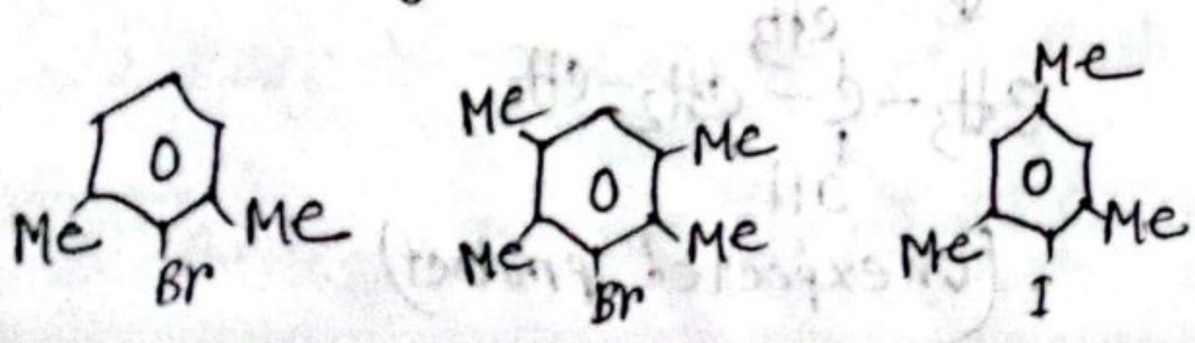
⇒



Mechanism:



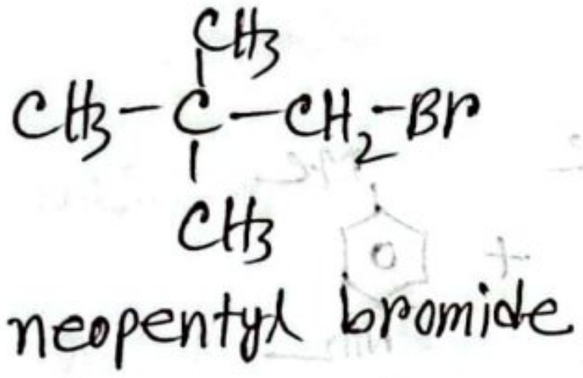
Evidence: Direct evidence in support of the aryne pathway is provided by the fact that the following halides:-



react with $\overset{\ominus}{N}H_2$ only under conditions much more vigorous than can be explained by any steric effect exerted by their o-Me group. Because none possesses hydrogen atom ortho to the halogen which is essential for initiating reaction for aryne pathway.

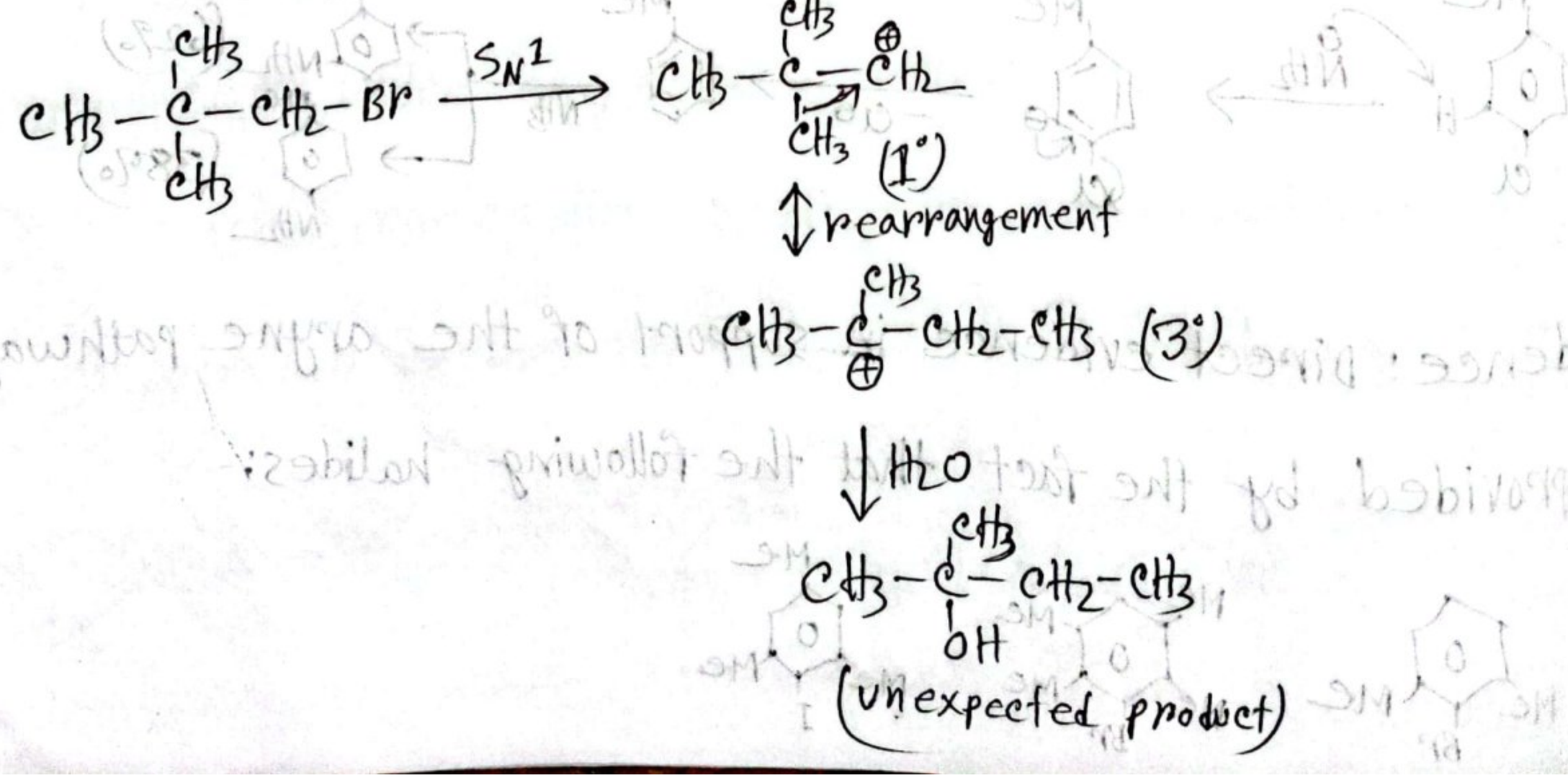
2018
⊛ The S_N2 hydrolysis of neopentyl bromide is slow, but under S_N1 mode the rate is increased & found unexpected product, 2-methyl-2-butanol – explain?

⇒



neopentyl bromide

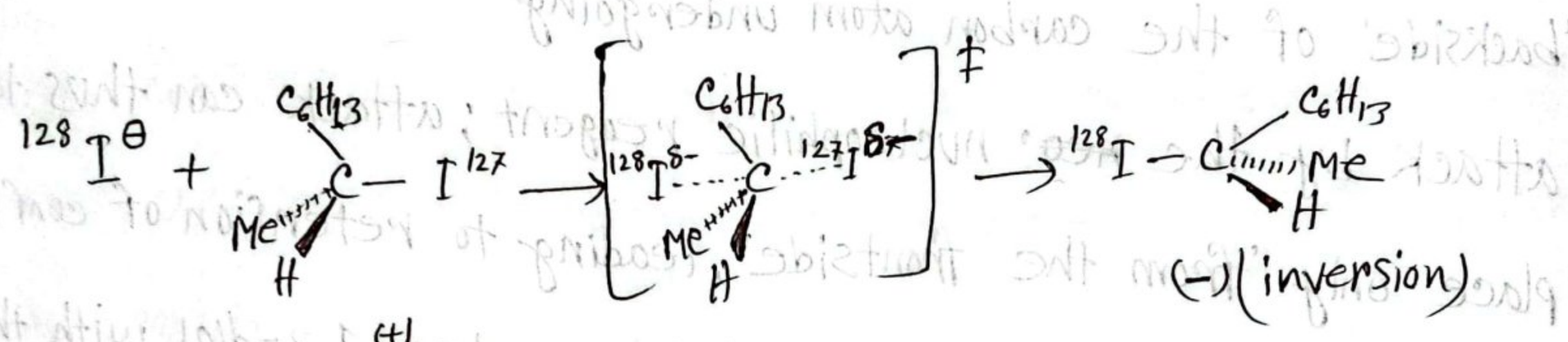
This halide is 1°, but larger group is present so it produce larger steric effect. ~~because~~ so S_N2 becomes slow. But it can produce 3° carbocation by rearrangement,



3° more stable than 1° so rearrangement occur & unexpected 2-methyl-2-butanol is formed.

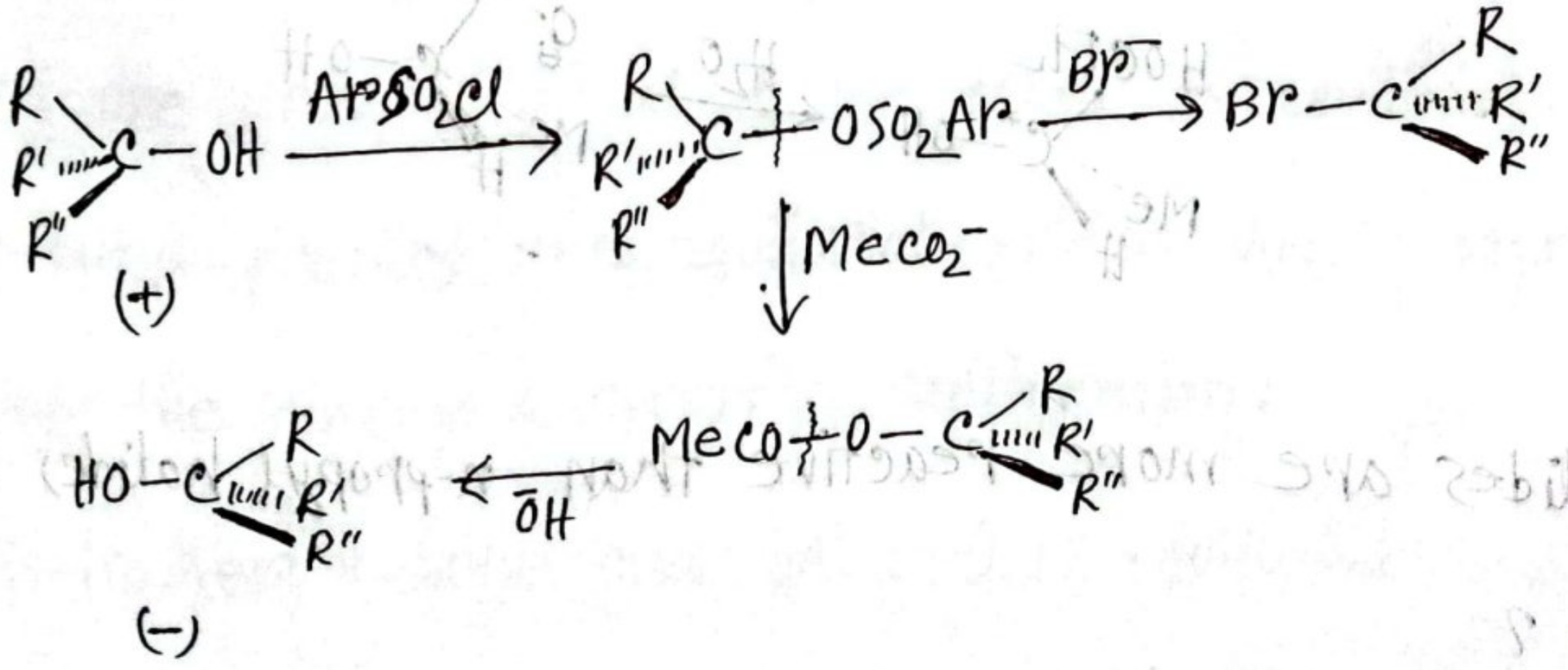
2018
(*) How can you prove S_N2 involve inversion configuration?

⇒ evidence-1:



Here optically active alkyl halide reacts with labelled $^{128}I^-$ & after reaction opical activity is zero means racemization occur. So S_N2 involve inversion.

evidence-2:



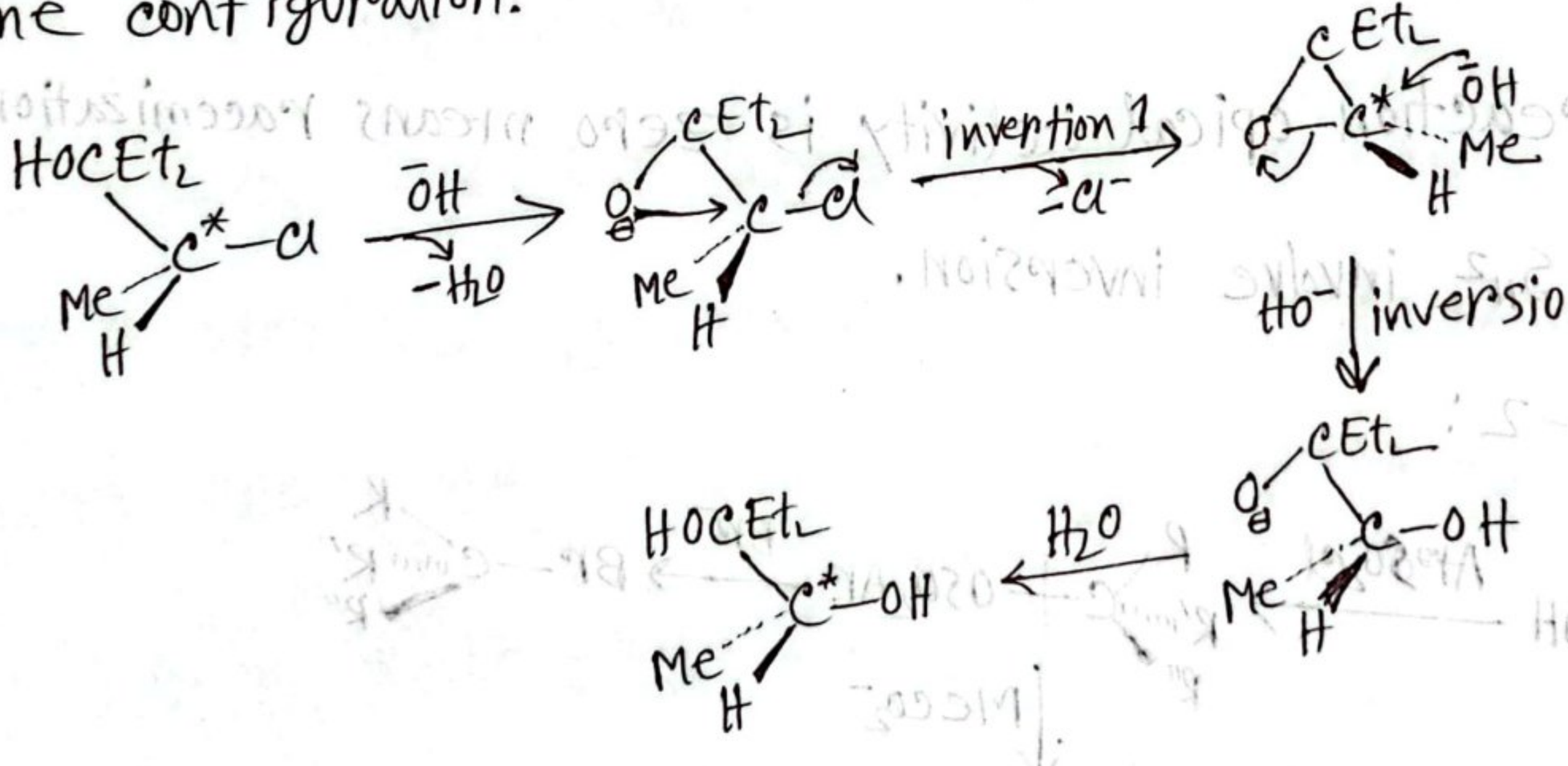
In this reaction (+) Alc react with tosyl chloride ($ArSO_2Cl$) & C–O bond breaking is not involved (C–O bond of Alc is not broken). Their

configuration also same. Then $MeCl^-$ nucleophile attack from th backside in which C-O bond is breaking of nucleophile & inversi occurs & $ArSO_3^-$ expelled.

2018
(*) In Substitution, NGP leads to retension of configuration - explain

⇒ Neibouring group can use its e- pair to interact with the 'backside' of the carbon atom undergoing substitution, thus preventing attack by the ~~neo~~ nucleophilic reagent; attack can thus take place only 'from the frontside', leading to retension of configurat

Ex. base hydrolysis of 1,2-chlorohydrin produce 1,2-diol with the same configuration.



2018
(*) Allyl halides are more reactive than n-propyl halides in S_N1 ? - why ?

⇒ Due to stability of carbocation. Allylic carbocation is more stabilized by resonance. But n-propyl halide for primary halide

& it has no resonance stability.

$CH_2=CH-CH_2X \xrightarrow{\text{slow}} CH_2=CH-\overset{\oplus}{C}H_2 + X^-$

allyl halide

$CH_2=CH-\overset{\oplus}{C}H_2 \leftrightarrow \overset{\oplus}{C}H_2-CH=CH_2$

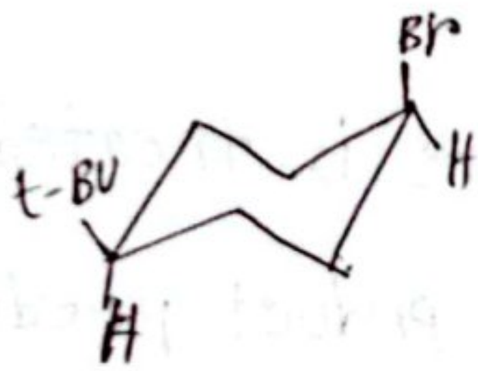
RAJIB
che-50

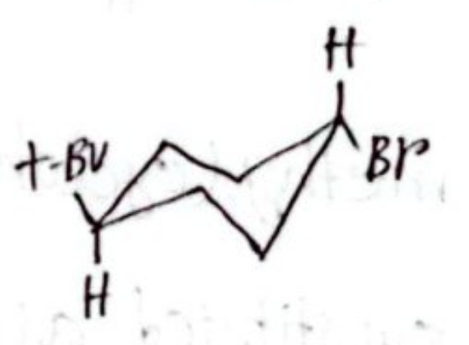
$CH_3-CH_2-CH_2-Cl \xrightarrow{\text{slow}} CH_3-CH_2-\overset{\oplus}{C}H_2 + Cl^-$

(no resonance)

2015, TU, 2020

⊛ which diasteriomer of 1-bromo-4-t-butylcyclohexane, the cis or trans, undergoes elimination more rapidly when treated with sodium epoxide?



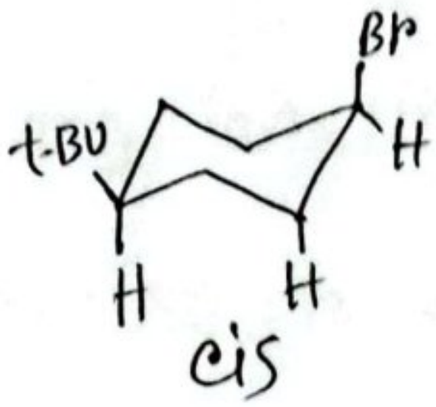


?

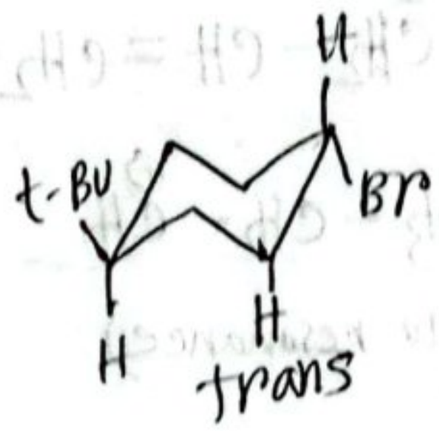
⇒ cis isomer undergo ~~no~~ elimination more rapidly, E_2 requires the β-H to be anti-periplanar to the leaving group (180° apart in the newman projection). Antiperiplanar is easier to achieve when one group is axial & the other is equitorial. The bulky tert-butyl prefers the equitorial position due to steric hindrance, locking the ring in a specific conformation.

In cis-isomer, t-butyl group (at C-4) is equitorial & Br (at C-1) must be axial. this arrangement places at least on β-Hydrogen on the adjacent carbon antiperiplanar to the axial Br, allowing elimination

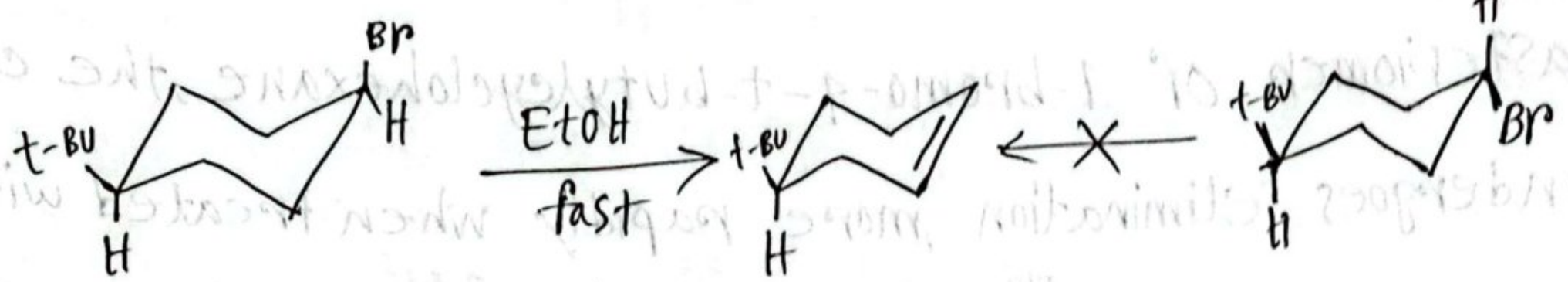
rapidly. In trans, t-butyl (at c-1) is equitorial, & Br (at c-1) is also equitorial. This arrangement prevents the β-Hydrogen from being anti-periplanar to the Br, making elimination more difficult.



cis



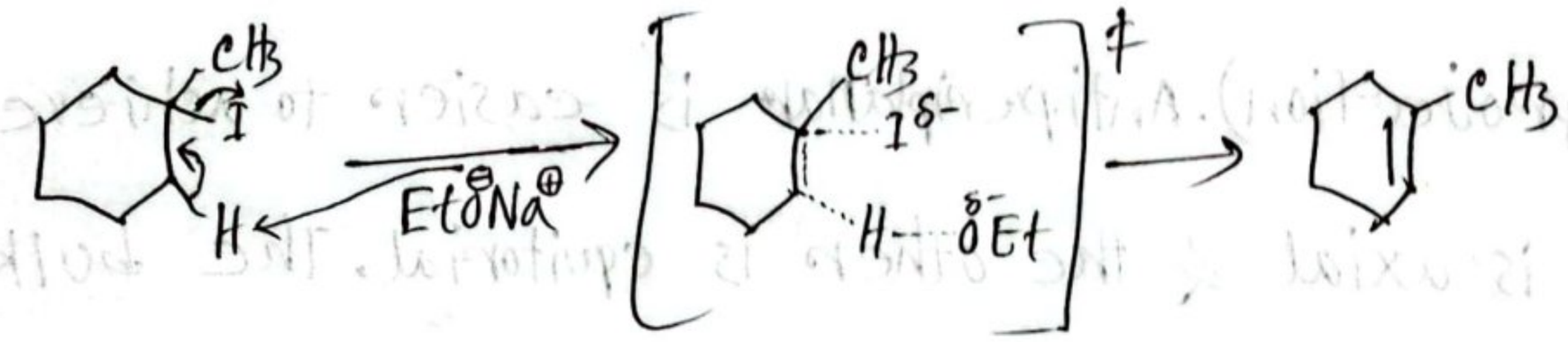
trans



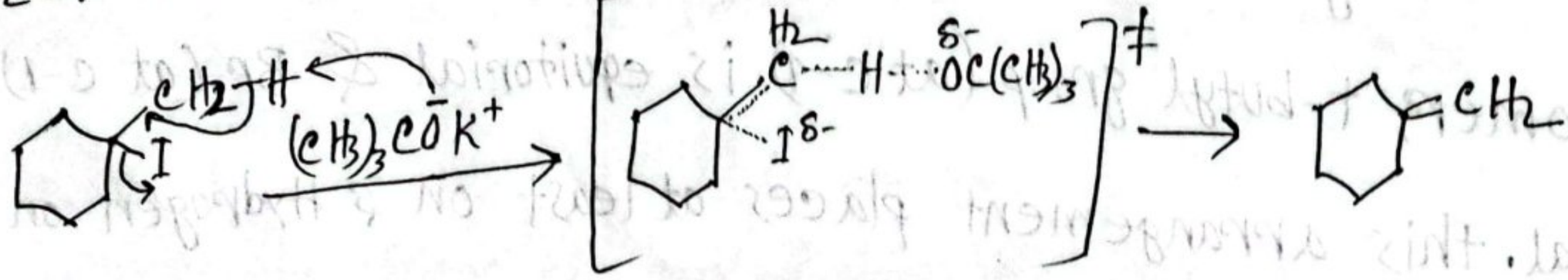
2015, TU, 2020

(✗) When 1-iodo-1-methylcyclohexane is treated with sodium ethoxid the more highly substituted alkene product predominates whereas th same reaction with potassium tert-butoxide gives less substituted alkene predominantly - Give structure of two product & explain?

⇒



highly substituted product formed because this reaction follow Saytzev elimination.



Less substituted alkene is formed becase this reaction follow

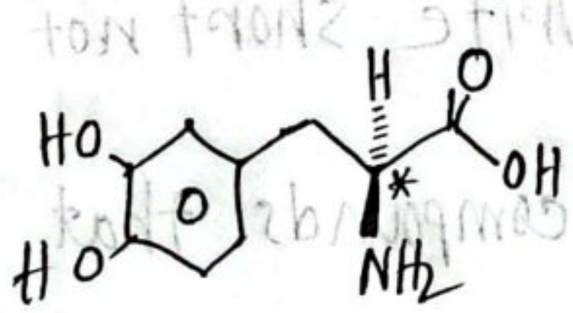
Hoffmann rule. Because this base is larger & present steric factor so it absorbs proton from $-CH_3$ group & less substituted product (alkene) is formed.

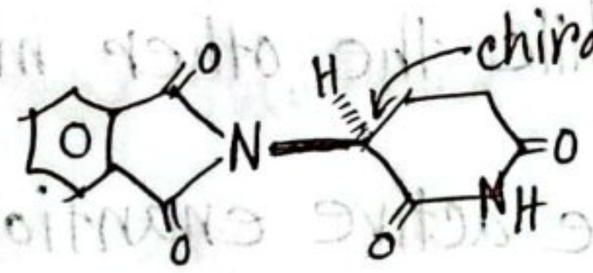
2021
(*) "Importance of ~~chemical~~ chiral drugs" write short not about it ?

⇒ chiral drugs are pharmaceutical compounds that exist in two mirror-image forms, known as enantiomers. The importance of ~~two~~ chiral drugs lies in their ability to enhance therapeutic outcomes & reduce side effects. one enantiomer may provide the desired therapeutic effect, while the other might be less effective. By isolating & using the active enantiomer, drug efficiency can be improved & patient safety ensured.

Drugs having chiral centers are called chiral drugs. ⓘ chirality is a crucial factor which can impact drug's safety, efficacy, pharmacological activity & metabolism. ⓘⓘ chirality affects a drug's interactions with biological targets, such as proteins, nucleic acids. ⓘⓘⓘ chiral drugs have applications in treating a variety of conditions including cardiovascular disease, CNS disorders, local anesthesia, anti-~~inf~~ inflammatories, antimicrobials & anticancer drugs. ⓘⓥ chiral drug Simplifies the interpretation of the Basic pharmacology ⓥ our body-

contains large number of chiral molecule so chiral drug can interact with them properly. (VI) By using only the active enantiomer, lower dosage may achieve the desired effect, reducing side effects & improving patient outcomes.

EX. L-Dopa:  which is used to treat parkinson's disease (L-enantiomer), causes decreasing dopamine production. Brain converts L-Dopa to dopamine, associated with movement & emotions. while D-enantiomer is inactive.

Thalidomide :  R-enantiomer acts as a sedative, while the S-enantiomer caused severe birth defects.

2018
(★) chemical shift value of Aldehydic proton is found 9 to 10- why?

⇒ (I) Aldehydic H+ directly attached to a carbon that is part of carbonyl group. 'O' is highly electronegative, pulling e- density away from the aldehydic proton. This results in significant deshielding of the proton, shifting its chemical shift downfield (higher PPM)

(II) π-e cloud of the carbonyl group creates an anisotropic magnetic field. This field deshields Ald. proton, contributing to its downfield shift. This increase the deshielding effects,

making signal in 9 to 10 ppm ranges.

2017
(**) Differentiate btwn transition state & intermediate with example?

→

Transition State	intermediate
It has partial bonds	It has complete bonds.
It has higher energy than intermediate	It has lower energy than T.S.
It has no definite geometry	It has definite geometry.
can not be isolated	can be isolated
Extremely unstable	Relatively stable
Represented by a double dagger symbol (‡).	Represented as a regular molecular structure

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330.-Solve (addition)

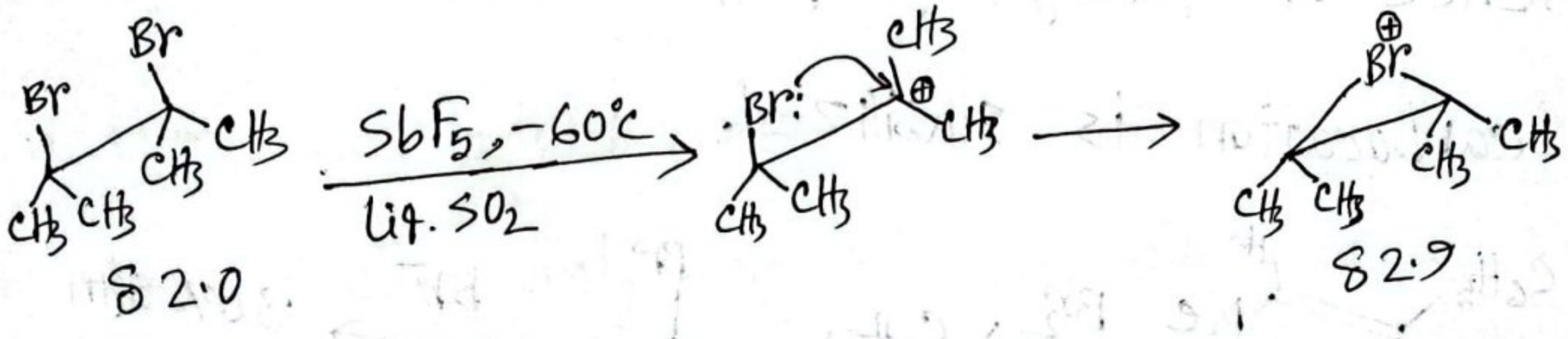
Rajib
che-50

2018, 2017, 2016, 2020

(*) Give two evidence of formation of cyclic bromonium ion during Br_2 addition in alkene?

Evidence-1:

when 2,3-dibromo-2,3-dimethyl butane react with SbF_5 in liq: SO_2 at -60°C led to the formation of an ion pair, but this exhibited not the two signal (one from each of two different groups of six equivalent protons) N.M.R. Spectrum expected of two signal, Instead one signal only was observed (δ 2.9) indicating that all 12 protons were equivalent. i.e. what is being observed is almost certainly the bromonium ion.



δ 2.0 δ 2.9

Evidence-2:

when Br_2 treated with unusual alkene, produce cyclic bromonium ion which can isolated and no further attack

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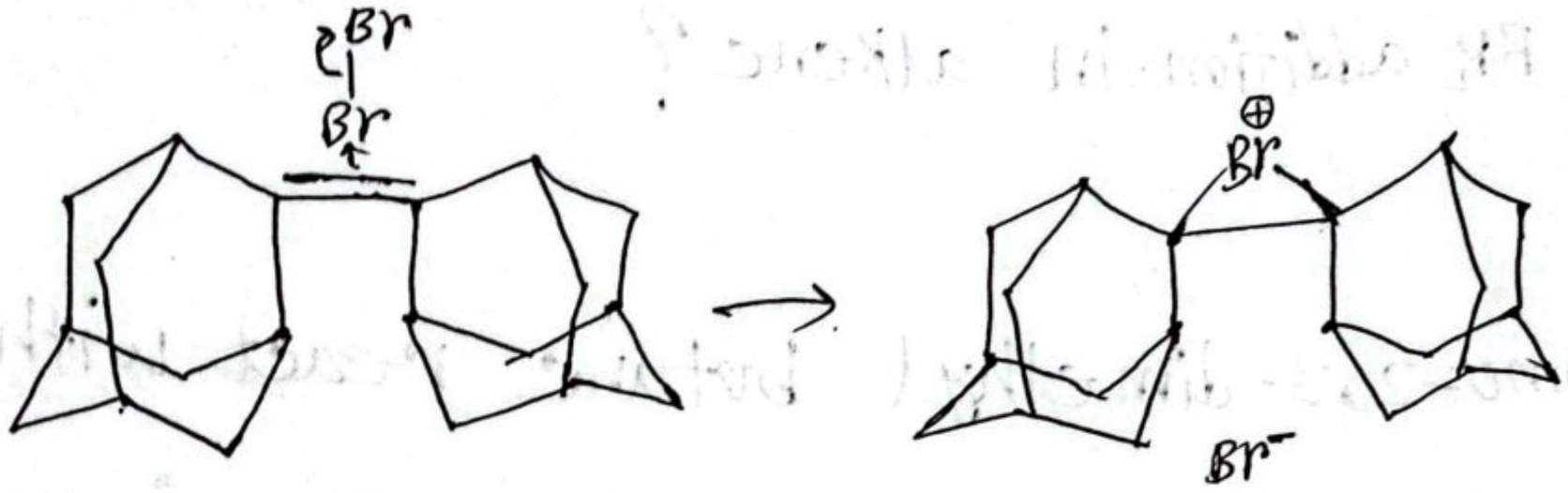
By Br^- because of its extreamely bulky & case like structure ~~and~~ at each end of the original double bond. Also the overall addition of Br_2 cannot occur.

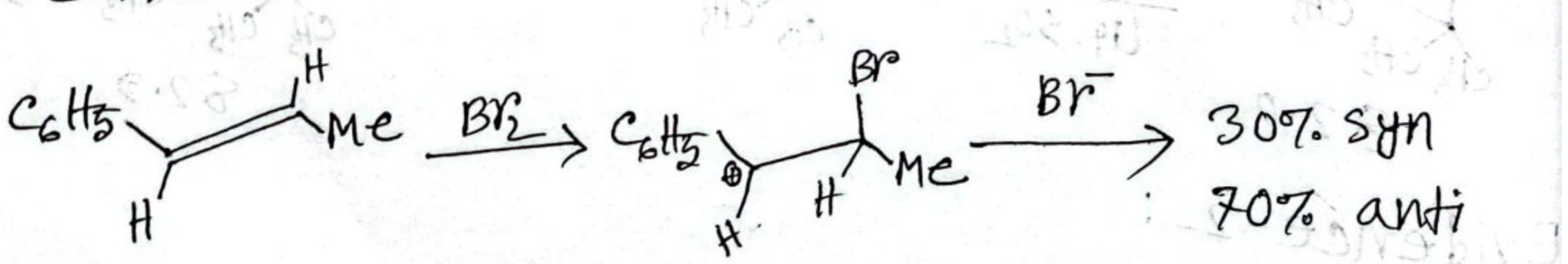


Thus the formation of cyclic bromonium ion is proved

(2017, 2016)

Why does trans 1-phenyl propene afford only 70% ANTI addition product when treated with Br_2?

⇒ trans-1-phenyl propene reacts with Br_2 & gives 70% anti-addition. Anti addition is decreases because of stability of carbocation is increased ~~be~~due to the presence of phenyl group. Here, delocalization occurs and carbocation is stabilized.



But when trans-2-butene reacts with Br_2, the 100% anti product is found because carbocation is not

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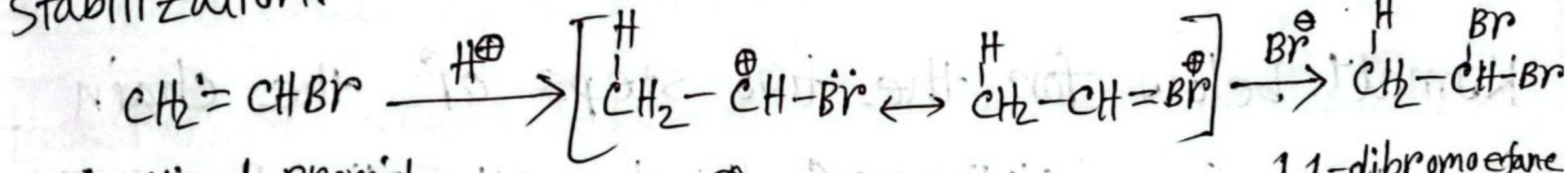
stabilized, cyclic bromonium ion is formed.

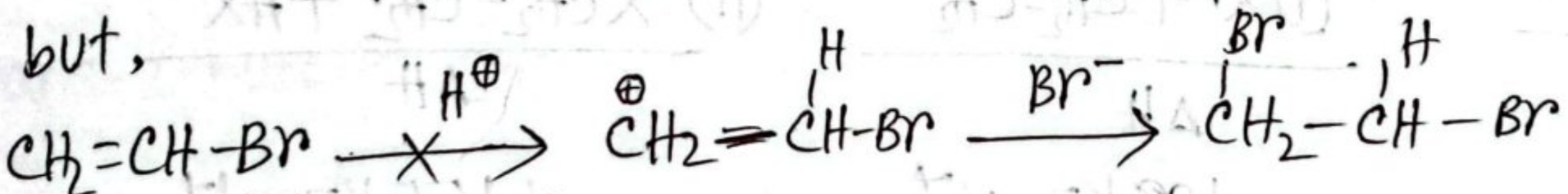
2016, 2018, 2018(t)

⊛ 1,1-dibromoethane is the only product obtained from HBr addition to vinyl bromide - explain?

⇛ The carbocation ① formed from addition of HBr to $CH_2=CHBr$ is stabilized by the involvement of the e⁻ pairs on Br but in carbocation ⑪ there is no stabilization.



but,



Hence 1,1-dibromethane is the only product.

2018, 2018(t)

⊛ what is anti markonikov rule? Explain the fact that HBr is the only one of the four Hx which will add readily to alkene via radical pathway?

⇛ when an unsymmetrical reagent add to an unsymmetrical alkene in the presence of peroxides

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the positive part of the reagent goes to the double bonded carbon atom that already has the lower number of hydrogen atom. EX.

$$CH_3-CH=CH_2 + HBr \xrightarrow{\text{peroxide}} CH_3-CH_2-\overset{Br}{\overset{|}{C}}H_2$$

⇒ HBr is the only one of the four HX that will add readily to alkene via a radical pathway. The reason for this is reflected in the ΔH value in KJmol⁻¹ below for the two steps of the chain reaction for addition of HX to $CH_2=CH_2$. For exmpl

H-X	(i) $X\cdot + CH_2=CH_2$ ΔH	(ii) $XCH_2-CH_2\cdot + HX$ ΔH
HF	−188 KJmol⁻¹	+155 KJmol⁻¹
H-Cl	−109 KJ	+21 KJ
HBr	−21 KJ	−96 KJ
H-I	+29 KJ	−113 KJ

only for HBr, both chain steps are exothermic. For HF the first step is exothermic but 2nd step is highly ~~endo ther~~ endothermic reflecting

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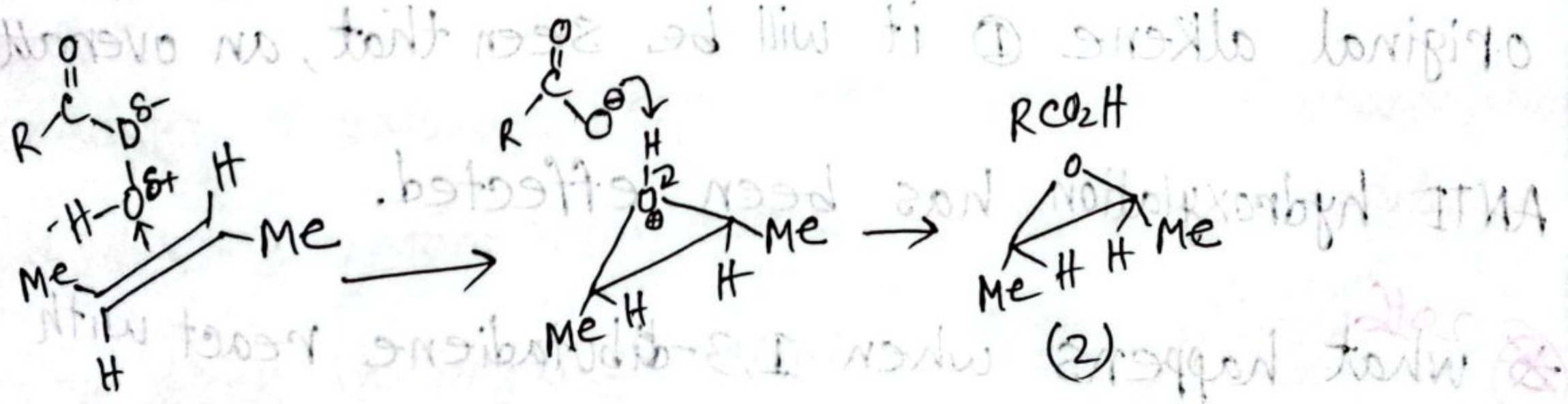
the strength of the HF bond & the difficulty of breaking it. For HCl it is again the 2nd step that is endothermic though not so great extent. while for, HI, it is the first step that is endothermic reflecting the fact that the energy gained in forming. so HBr is only one.

⊛ 2018, 2017 (H)
An overall anti hydroxylation take place during the reaction of trans-2-butene with a peroxyacids - explain?

⇒ peroxyacids, RCO·OOH will also oxydise alkenes. e.g. trans-2-butene by adding an oxygen atom across the double bond to form an epoxide (2)



Epoxide (1) undergo nucleophilic attack under either acid or base catalysed conditions to yield 1,2-diol. In

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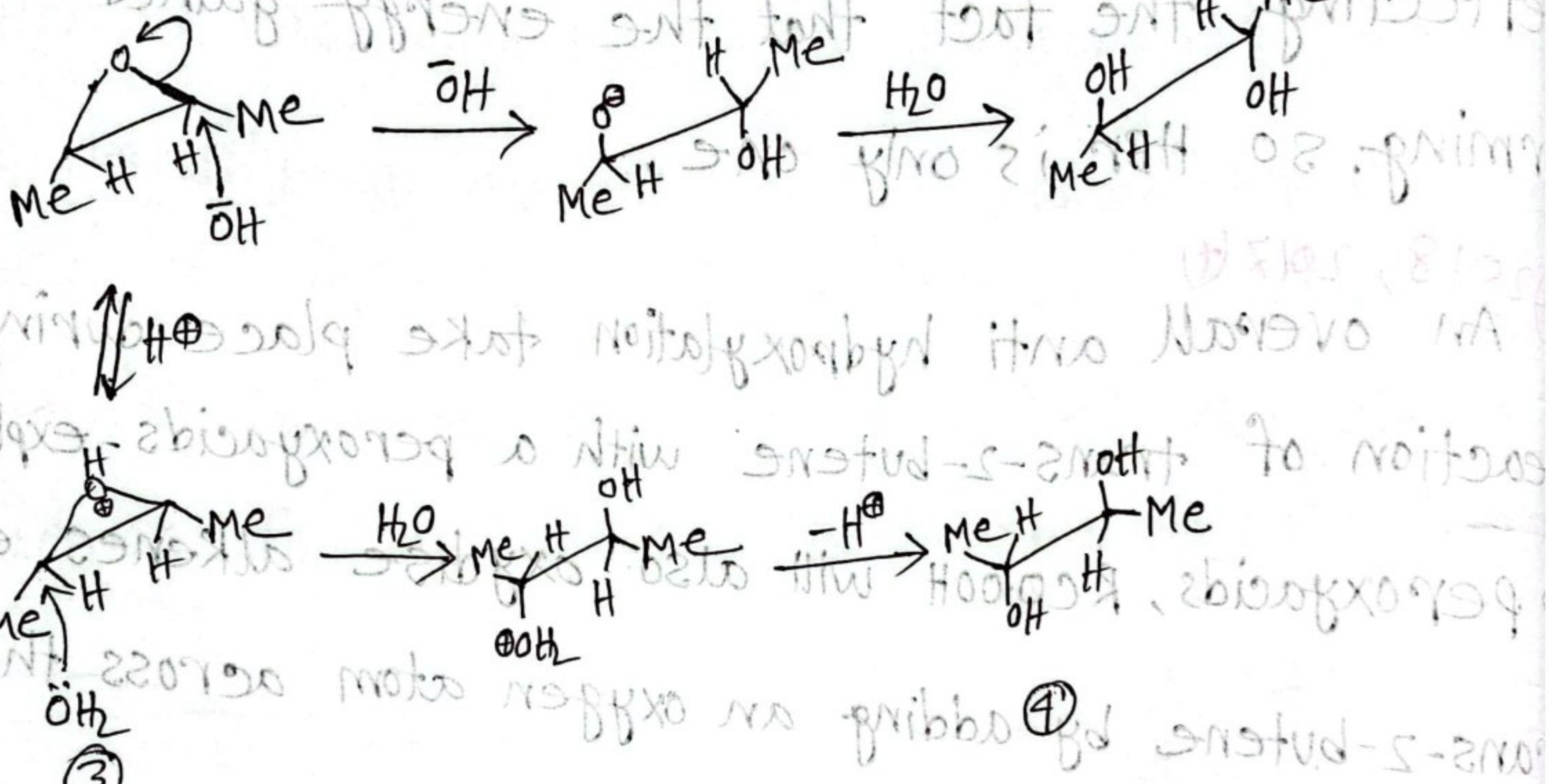
either case, attack by the nucleophite on a carb
atom will be on the side opposite to the oxygen
bridge, such attack on the epoxide will involve
inversion of configuration.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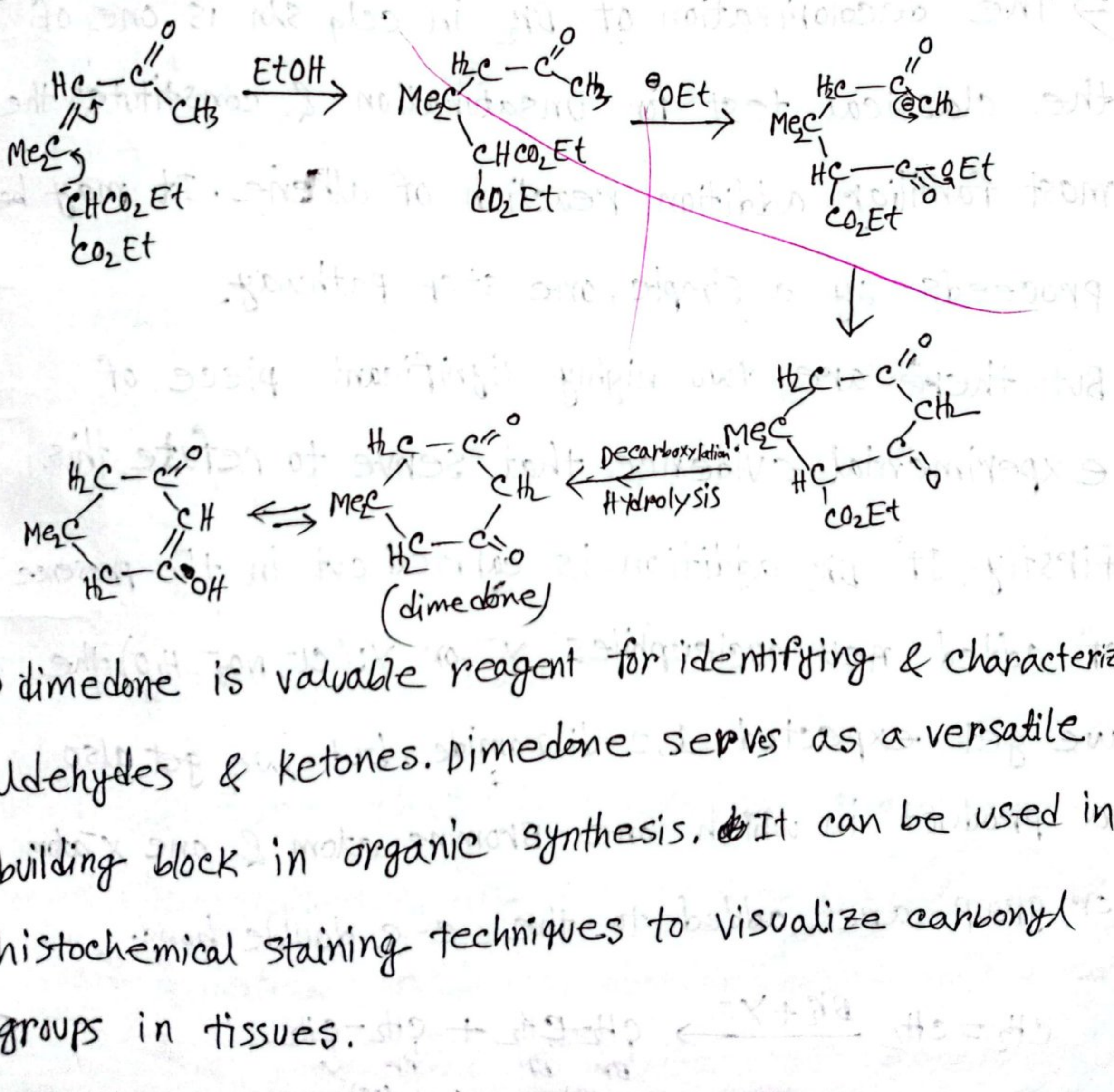
⊛ 2016, 2018 (t)
Give a synthesis of dimedone from mesityl oxide? & write down its importance?

⇒ The synthesis of 1,1-dimethylcyclohexane-3,5-dione (dimedone) starting from 2-methylpent-2-ene-4-one (mesityl oxide) & the carbanion $^{\ominus}CH(CO_2Et)_2$:



⇒ dimedone is valuable reagent for identifying & characterizing aldehydes & ketones. Dimedone serves as a versatile building block in organic synthesis. It can be used in histochemical staining techniques to visualize carbonyl groups in tissues.

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(★) 2017 Discuss experimental evidence that refuses the assumption that addition of Br_2 to c-c double bond may take place in one step?

or, → 2017 (H) what are the experimental evidence to support the argument that the addition involves two steps?

⇒ The decolorization of Br_2 in ccl_4 sln is one of the classical test for unsaturation & constitutes the most familiar addition reaction of alkene. It may be proceeds by a simple, one step pathway.

But there are two highly significant piece of experimental evidence that serve to refute this.

Firstly, If Br_2 addition is carried out in the presence of added ~~neu~~ nucleuphiles Y^- or $Y{:}(Cl^-, NO_3^-, H_2O)$ the we get expected 1,2-dibromide but we get also a product in which one Bromine atom & one Y^- atom or group are added to the c-c double bond.

$$CH_2=CH_2 \xrightarrow{Br_2+Y^-} \underset{Br}{\overset{}{CH_2}}-\underset{Br}{\overset{}{CH_2}} + \underset{Br}{\overset{}{CH_2}}-\underset{Y}{\overset{}{CH_2}}$$

(I) (II)

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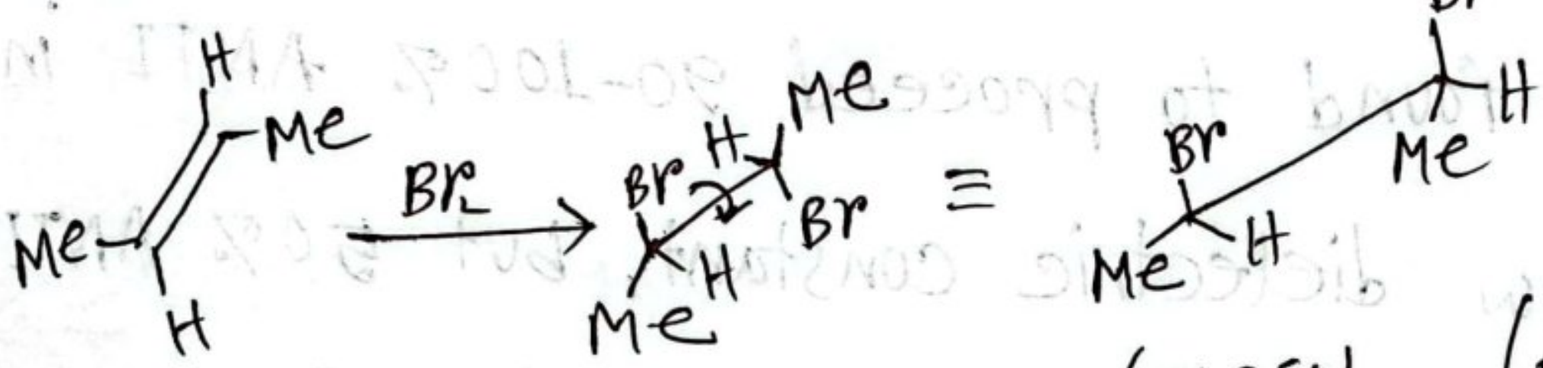
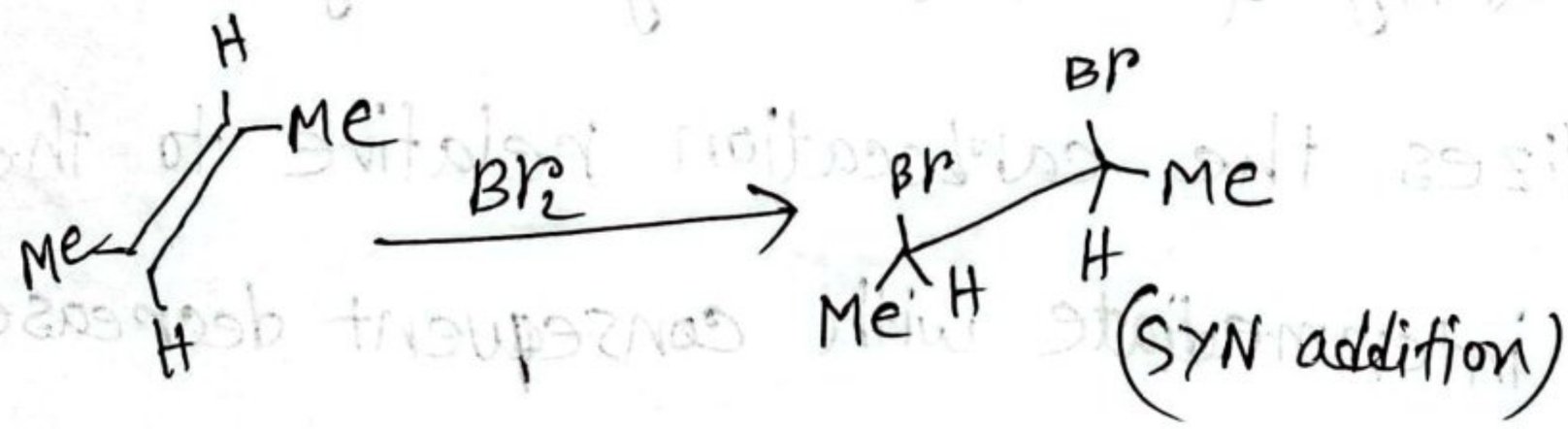
This is clearly incompatible with one step pathway (একধাপ) in which there would be no opportunity for attack by ye

secondly, It is found that, In trans-2-butene the two bromine atoms add on from opposite sides, i.e. ANTI addition.



The product is the symmetrical meso dibromide whereas if syn. addition would take place then we might get unsymmetrical (±) dibromide.



But it is found that the addition is almost completely steriioselective i.e. ≈100% ANTI addition, which indicates that addition of halogen to alkene is a two step process like as follows:

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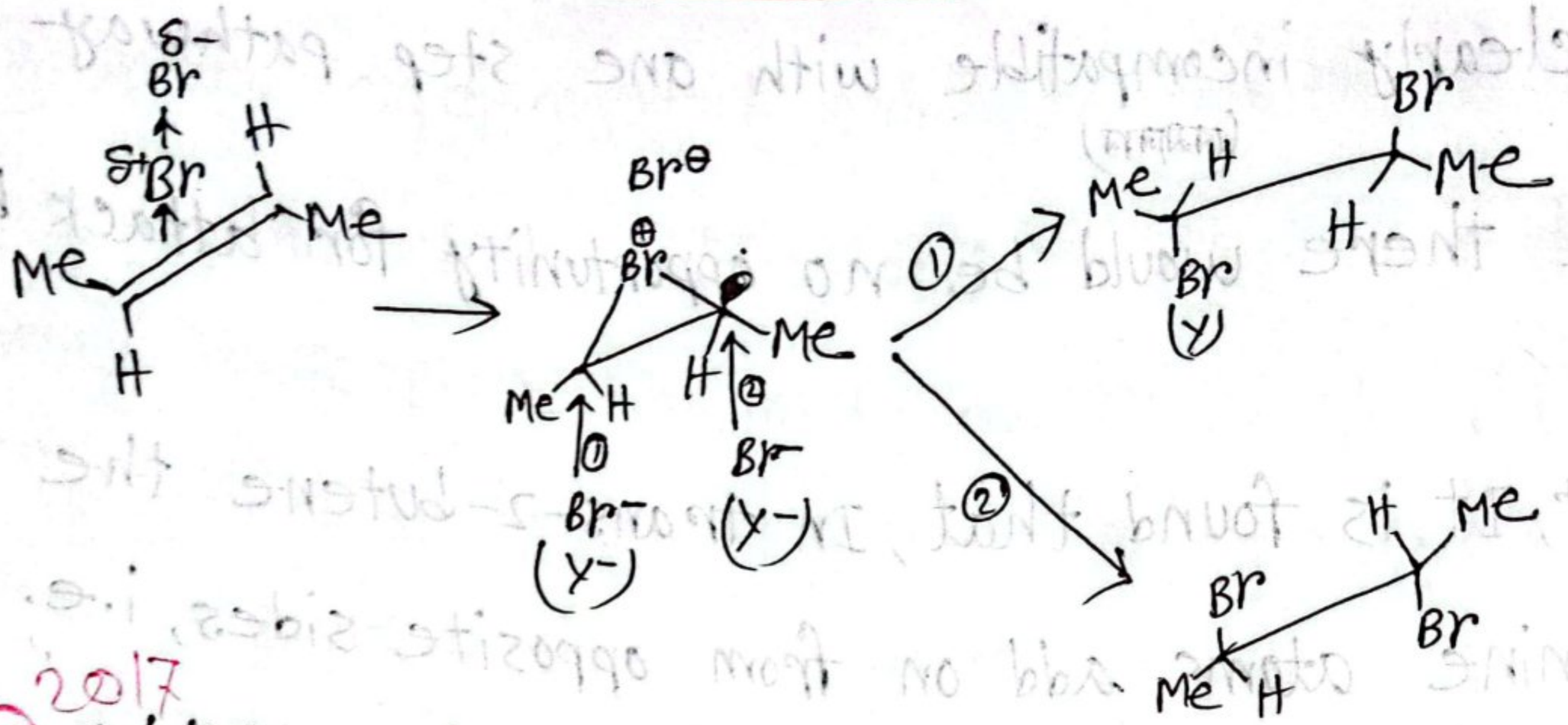

2017

⊛ Addition of bromine to 1,2,-diphenylethene (Stilbene) was found to proceed 90-100% ANTI in Solvents of low dielectric constant, but 50% ANTI only in a solvent with ($\epsilon = 35$) - explain ?

⇒ Because higher dielectric constant, higher polarity & greater the ability of stabilization.

Increasing polarity & ion solvating ability of the Solvent stabilizes the carbocation relative to the bromonium ion intermediate with consequent decrease in anti sterioselectivity.

~~2017, 2018~~

~~⊛ Provide a support for the suggestion that addition of Halide does normally proceed via carbocation intermediates under polar condition ?~~

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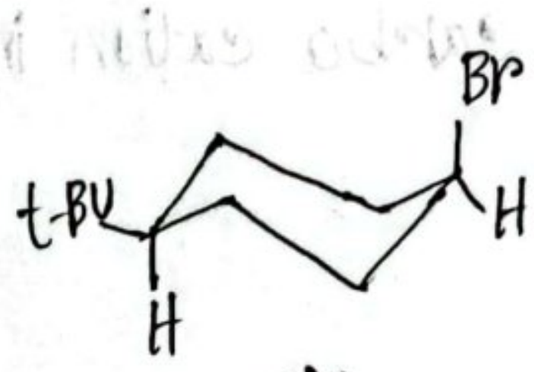
TU
Which diasteriomer of 1-bromo-4-t-butylcyclohexane, the cis or the trans, undergoes elimination more rapidly when treated with sodium ethoxide? explain.

⇒ trans undergo rapidly. In cis, the Br at carbon 1 (C1) and the t-butyl group at carbon 4 (C4) are on the same side of the cyclohexane ring. the t-butyl group is very bulky and due to steric reasons reaction is slow. But in trans isomer Bromine at C1 and t-butyl group at C4 are on opposite sides of the ring and form antiperiplanar conformation favourable for elimination. So elimination occur rapidly

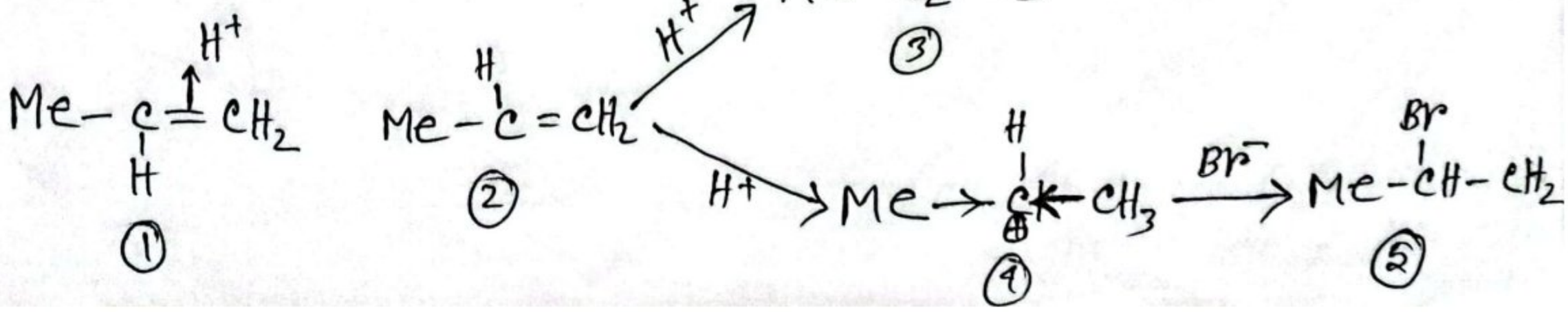


cis

trans

2017, 2018
Give one evidence in support of the statement that addition of HHal does normally proceed via carbocation intermediates?

⇒ Markonikov rule:



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In non-polar solvents the proton is no doubt provided by H-Hal but in polar solvents & especially in hydroxylic solvents, more likely by its conjugate acid e.g. by H_3O^+ in H_2O: so H^+ with no e^- pair available, can form in some cases a π-complex ① as the intermediate. It is the relative stabili of possible alternative carbocation ③ & ④ that determines the overall orientation of addition e.g. in the addition of HBr to propene ② under polar conditions, ④ will be formed in preference to ③. only addition product obtained is 2-bromopropane ⑤. So we can see that the addition of HHal does normally proceed via carbo cation intermediates under polar condition.

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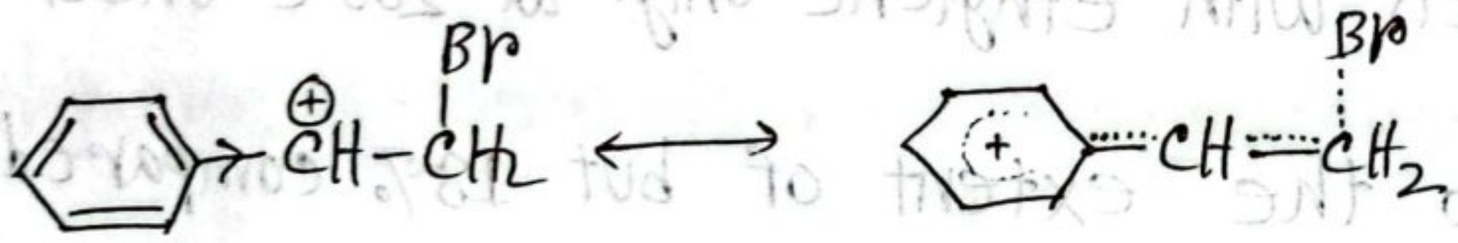
2017 (t)

How would you explain an increase on the rate of addition (4×10^3 times) of bromine to $phCH=CH_2$ with respect to its addition to ethylene?

⇒ phenyl group increase the rate of electrophilic addition considerably 4×10^3 times, due to the stabilization that can induce in the intermediate or in the transition state. The possible delocalization of benzyl type carbocation leading to specific stabilization of carbocations.



on the other hand, ethylene can not for such type Stabilization that can induce in the intermediate,

$$CH_2=CH_2 \longrightarrow \overset{\oplus}{C}H_2-\underset{Br}{C}H_2 \xrightarrow{Br^-} \underset{Br}{C}H_2-\underset{Br}{C}H_2$$

2017 (t)

Complete the reaction & give mechanism:

$$Me-\underset{Me}{\overset{Me}{C}}-CH=CH_2 \xrightarrow{HI} ?$$

$$\Rightarrow Me-\underset{Me}{\overset{Me}{C}}-CH=CH_2 \xrightarrow{HI} Me-\underset{Me}{\overset{Me}{C}}-\underset{I}{CH}-CH_3 \text{ (minor)} + Me-\underset{I}{\overset{Me}{C}}-\underset{Me}{CH}-CH_3$$

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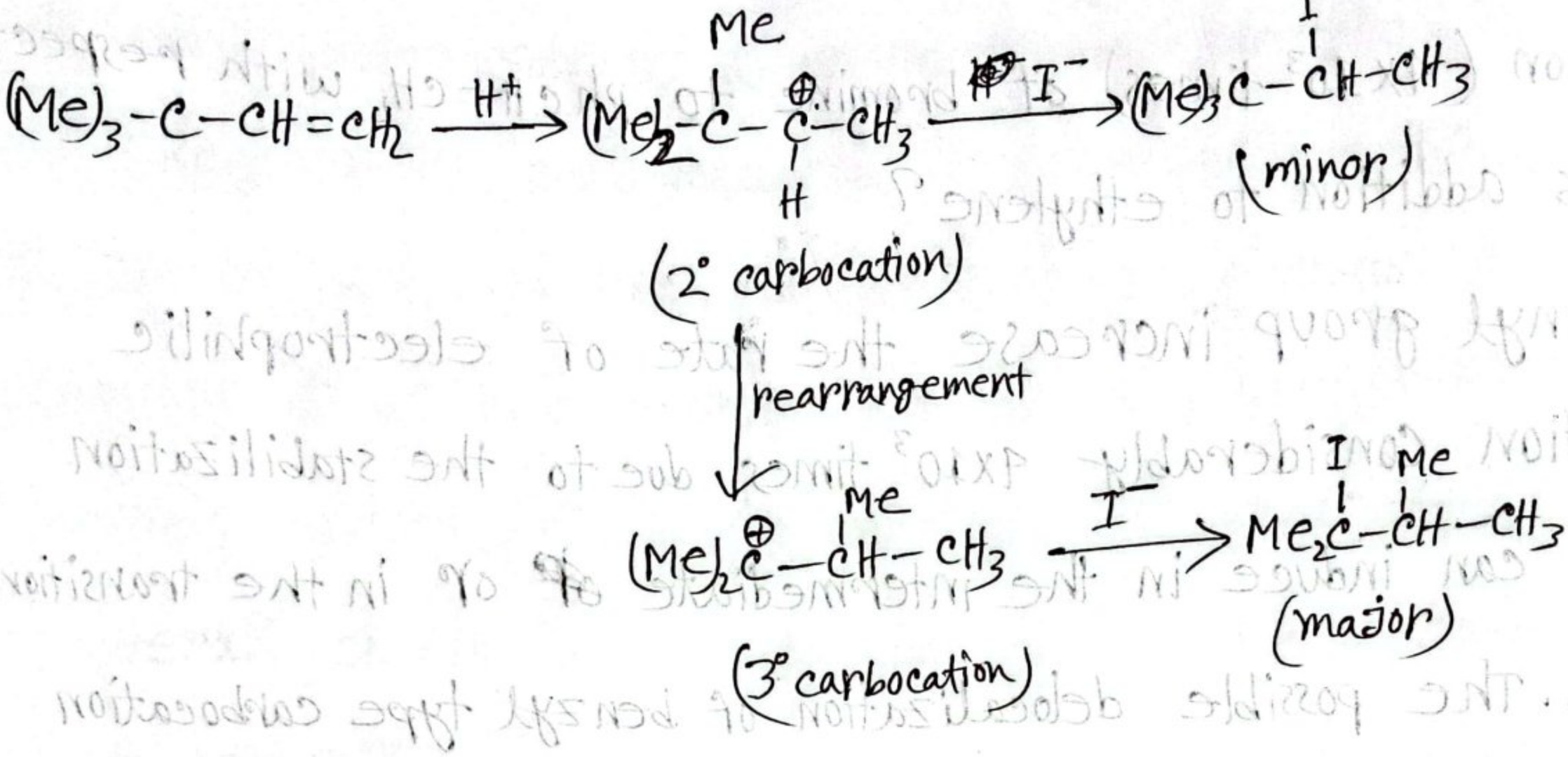
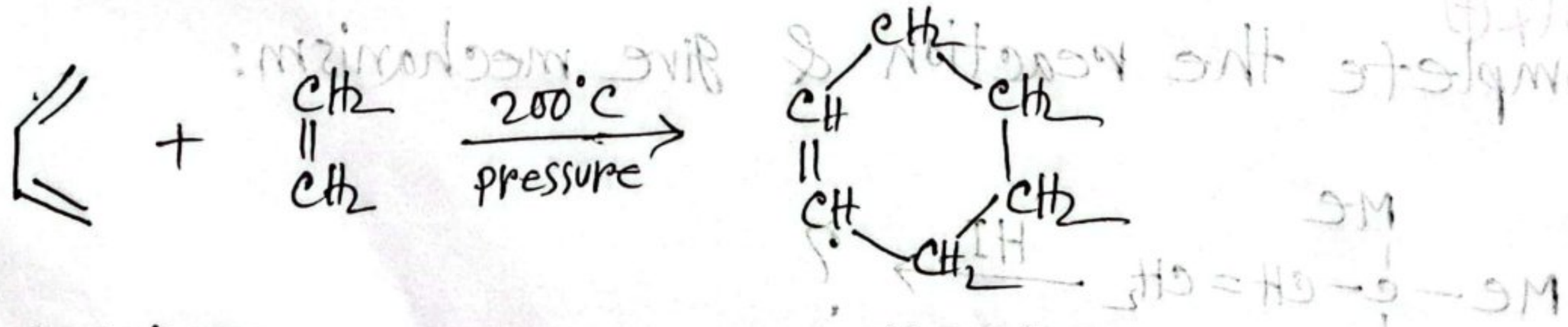
Mechanism:



2018, 2017(t)

Ⓧ Butadiene reacts with ethylene only at 200°C under pressu & even then to the extent of but 18% compared with almost 100% yield with maleic anhydride in benzene a 15°C - explain?

⇒ Butadiene react with ethylene at 200°C under pressur where 18% product form :-

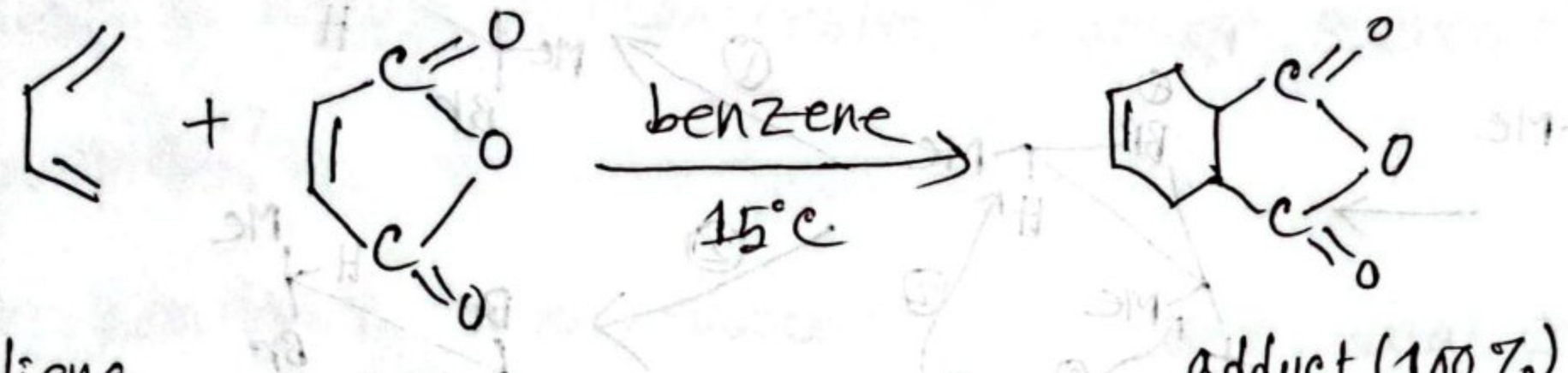


1,3-butadiene

Adduct (18%)
(cyclohexene)

Butadiene react with maleic ~~acid~~ anhydride in benzene

at @ 15°C.



diene

maleic anhydride
(dienophile)

adduct (100%)

These reaction are found to be promoted by e⁻ donating substitution in the diene & by e⁻ withdrawing substituents in the dienophile.

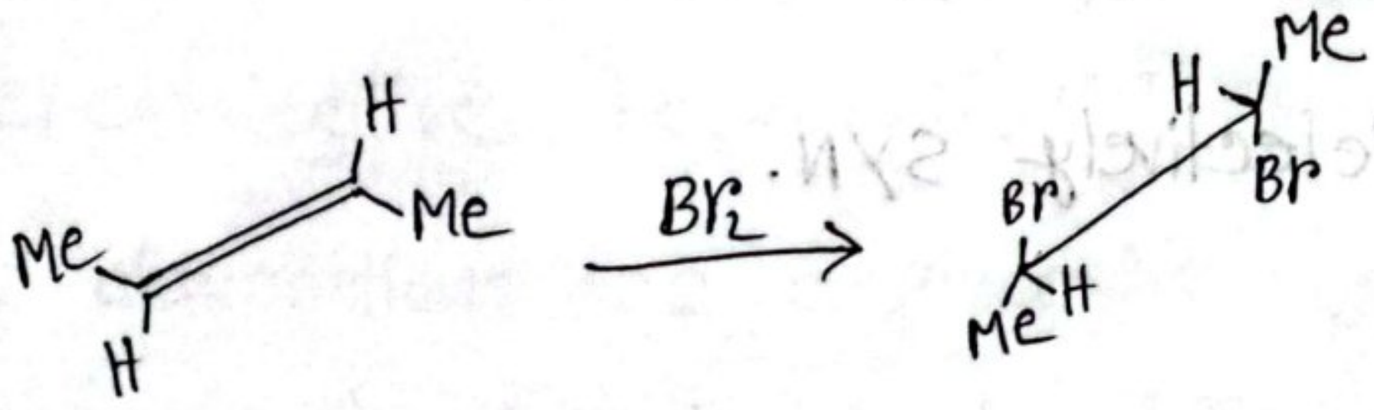
In ethylene, there is no e⁻ withdrawing group whereas maleic anhydride has e⁻ withdrawing group.

Therefore reaction of butadiene with maleic anhydride in benzene gives 100% adduct whereas ethylene gives only 18% ~~cyb cy~~ adduct.

2018(t)

⊛ Give the reaction & mechanism of electrophilic addition of Br_2 to trans-2-butene?

⇒



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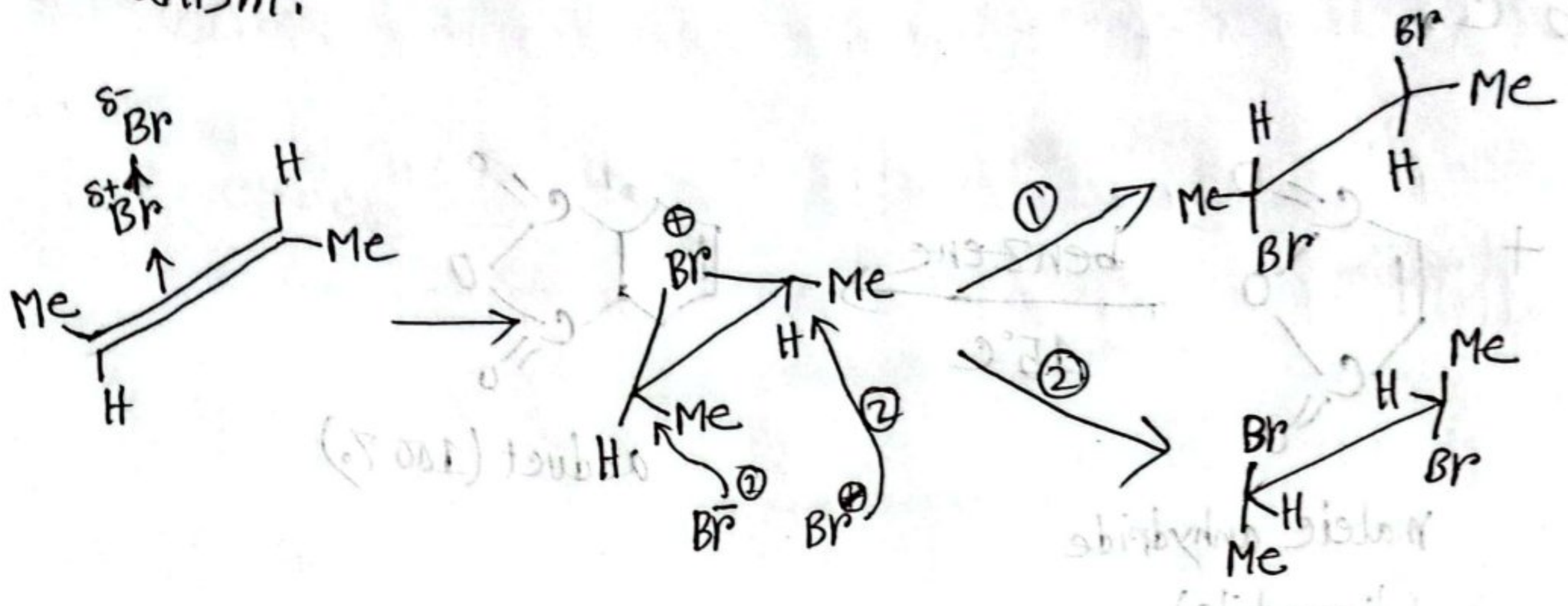
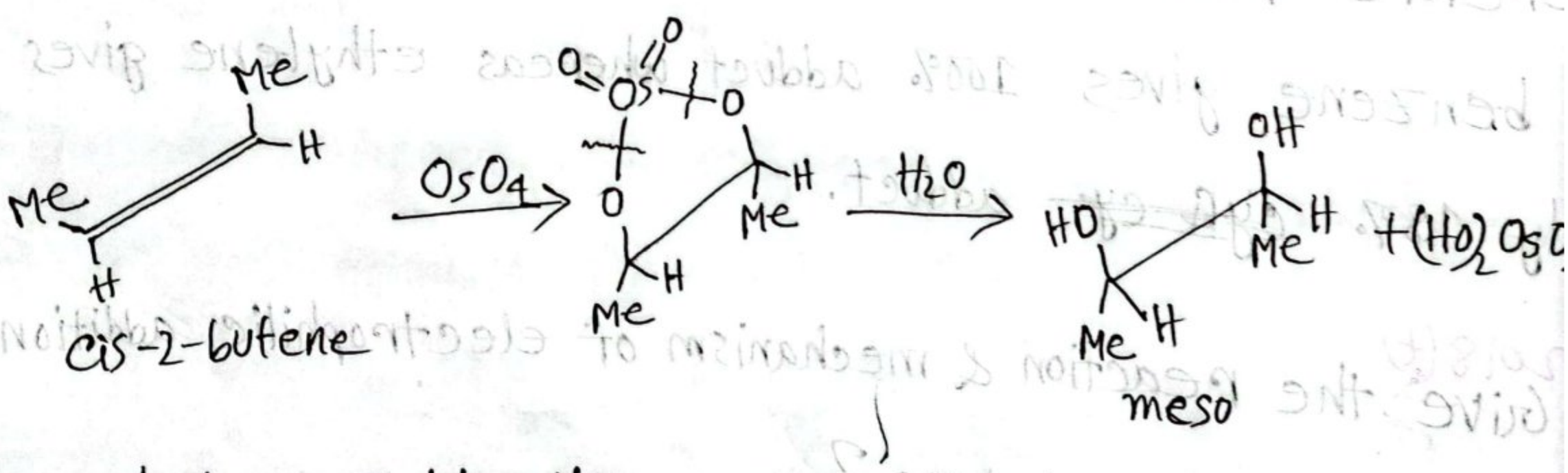
I Mechanism:



2018(H)

⊗ Hydroxylation of cis-2-butene yields meso-1,2-diol- Justify the statement?

⇒ For cis-hydroxilation of alkene OsO_4 or $KMnO_4$ are usually used. The OsO_4 adds to yield cyclic osmic ester which can be made to undergo readily hydrolytic cleavage of the Os–O bonds to yield 1,2-dial.



cis-2-butene yields the meso 1,2-diol i.e. the overall hydroxylation is stereoselectively SYN.

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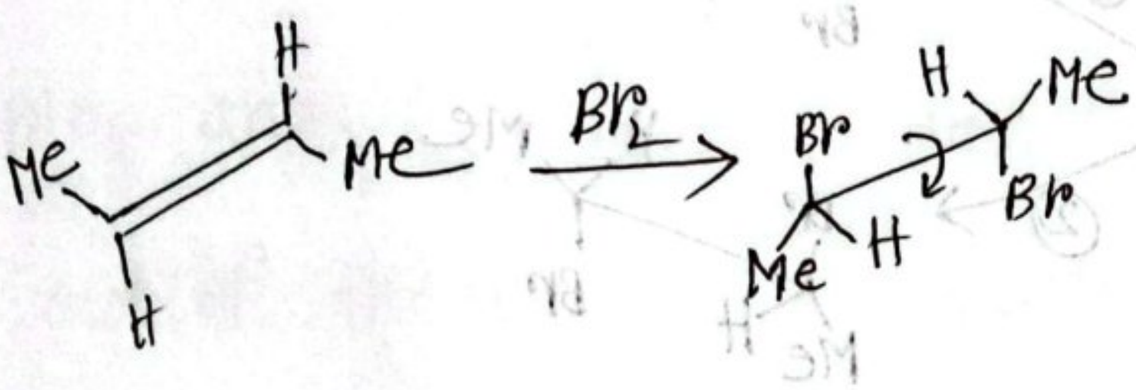
2018, 2022

(x) Trans-2-butene reacts with bromine to give dibromo product in highly stereoselective? Explain stereoselectivity with example?

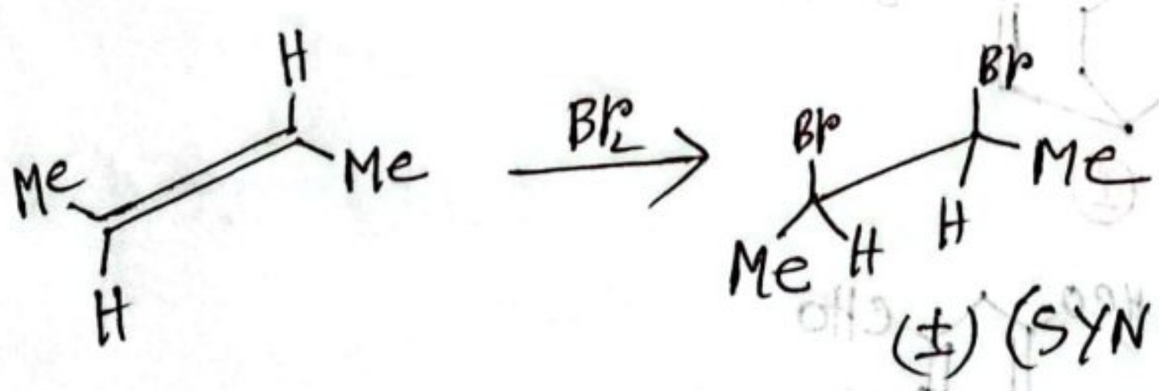
or Br_2 addition to trans-2-butene 100% anti. what conclution you can draw from statement? Explain with reasoning?

⇒ Two bromine atoms add on from opposite sides of the planar trans-2-butene, it is called anti addition.



(meso) (anti addition)

The product is the symmetrical meso dibromide. If addition had been SYN it would have been unsymmetrical (±) dibromide.



(±) (SYN addition)

It is found that in practice with trans-2-butene & with other simple acyclic alkenes, that the addition is almost completely Stereoselective, i.e. ≈100% ANTI addition.

This observations are explainable by a pathway in which one end of a bromine molecule becomes positively polarised

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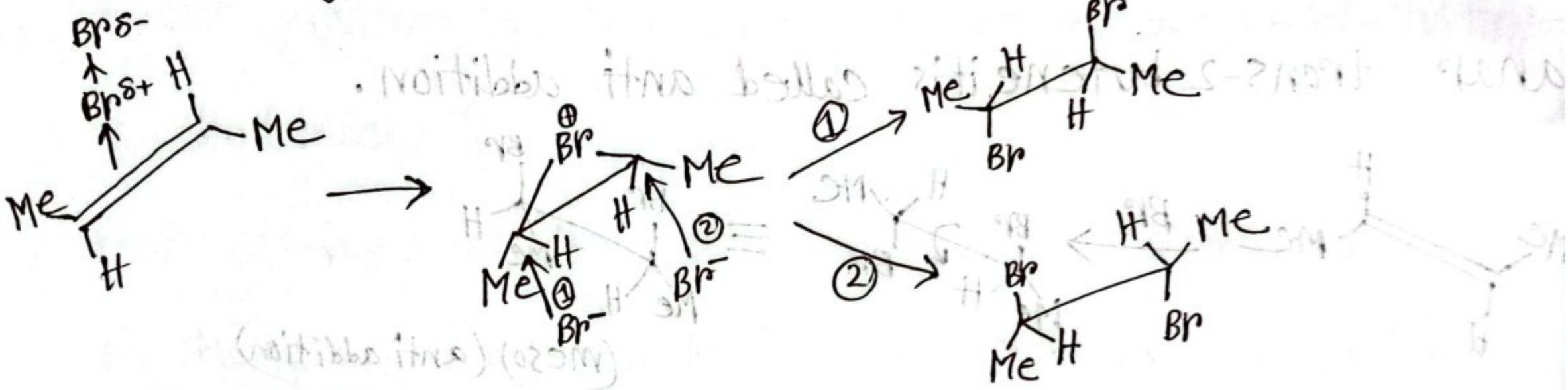
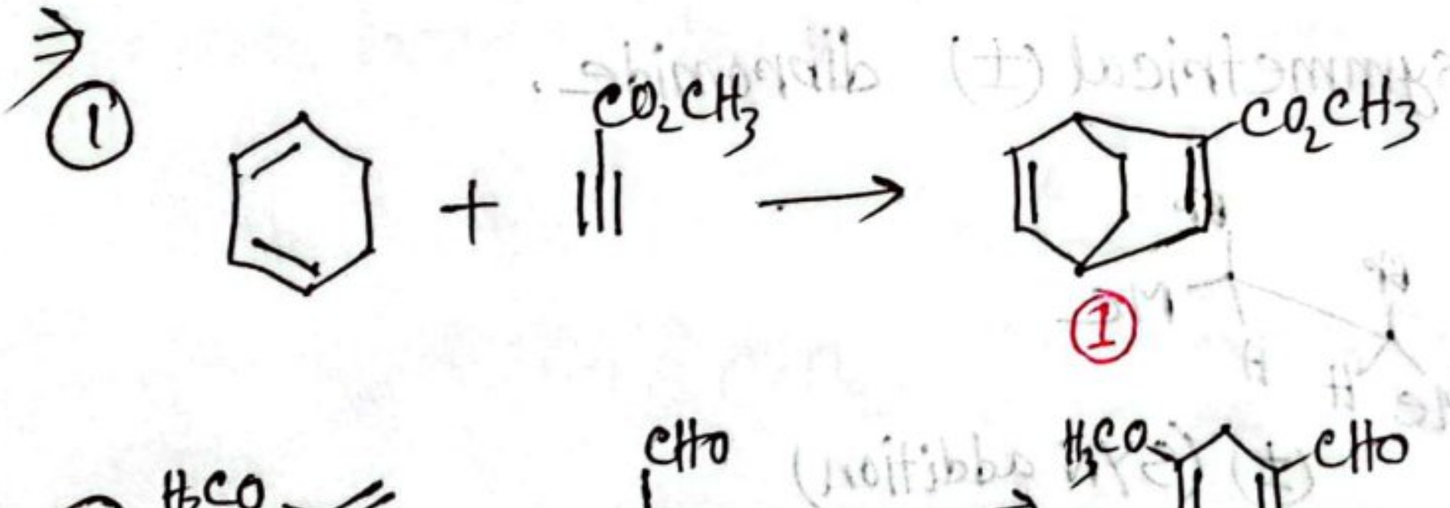
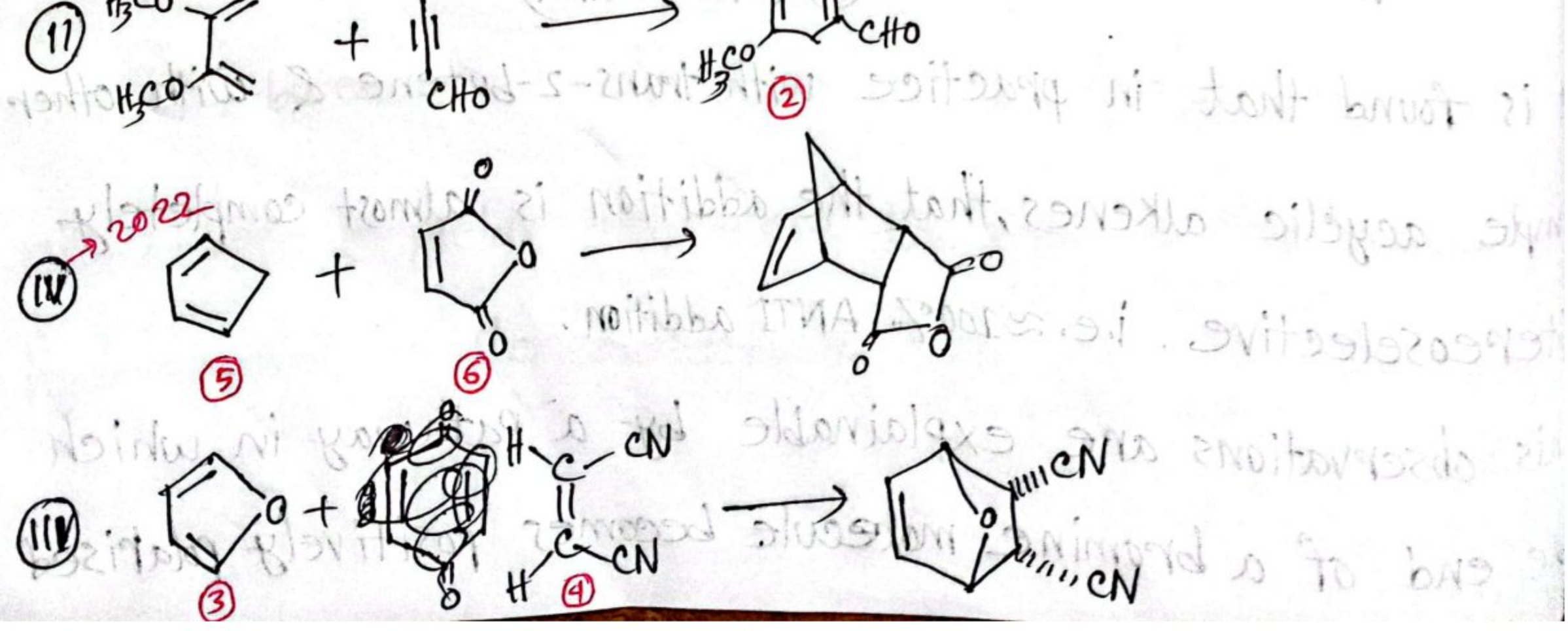
through e^- repulsion by the π e^- of the alkene, thereby forming a π complex with it. this then breaks down to form a cyclic bromonium ion: Addition is completed through nucleophilic attack by the residual Br^- on either of the original double bond carbon atoms, from the side opposite to the large bromonium ion Br^+ to yield the meso dibromide



2016, 2020

★ complete the reaction:





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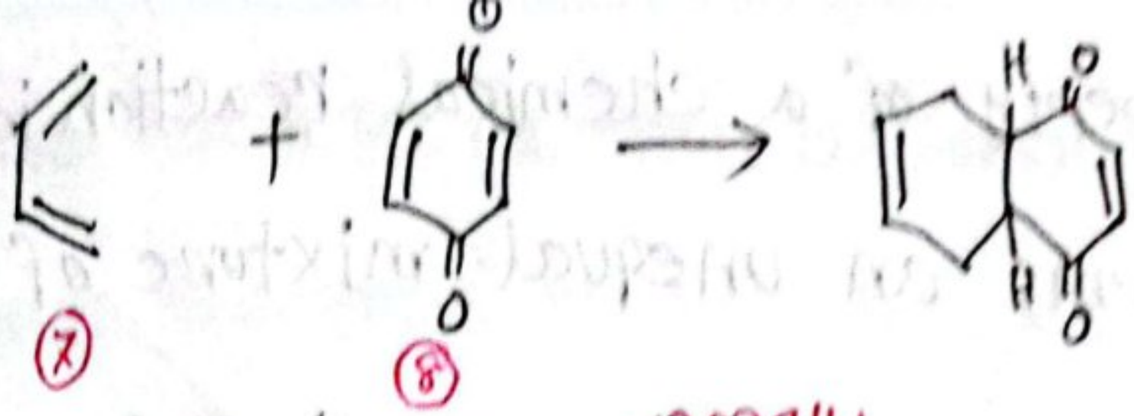
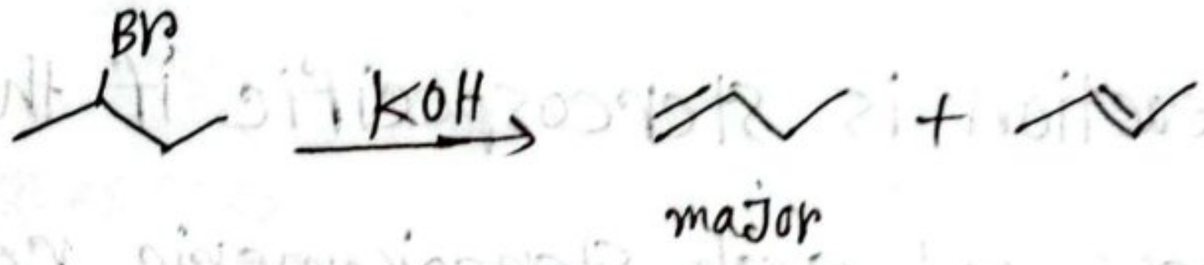
(v) 

(7) (8)

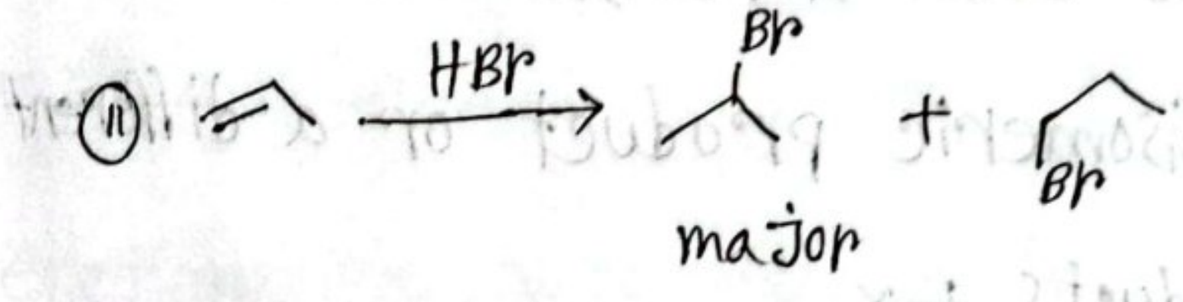
(x) Explain Regioselectivity, chemoselectivity, sterioselectivity, stereospecificity? (2029(4))

⇒ Regioselectivity is the preference of one direction of chemical bond making or bond breaking over all the other possible directions.

During any c. reaction when two or more products are obtained & out of these one is major & all products are constitutional isomers then such reactions is called. regioselective reactions.

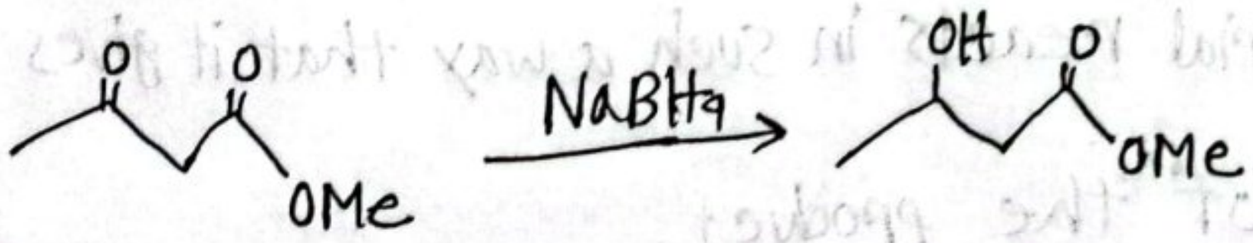
Ex. (i) 

major

(ii) 

major

chemoselectivity: is the preferential reaction of a c. reagent with one of two or more different functional group. Ex. $NaBH_4$ is highly chemoselective which reduces only ketonic group.



(Selectively যাকোন 1টি ... এরে chemoselectivity বলে)

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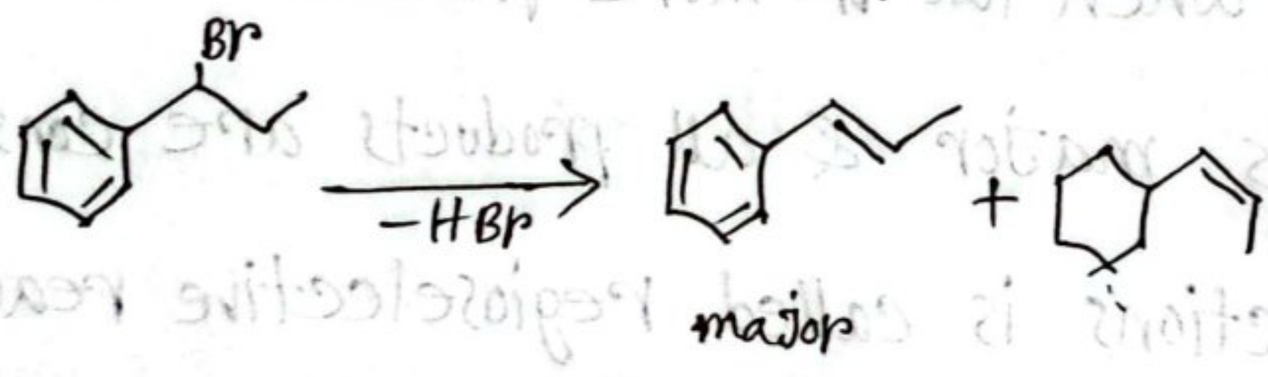
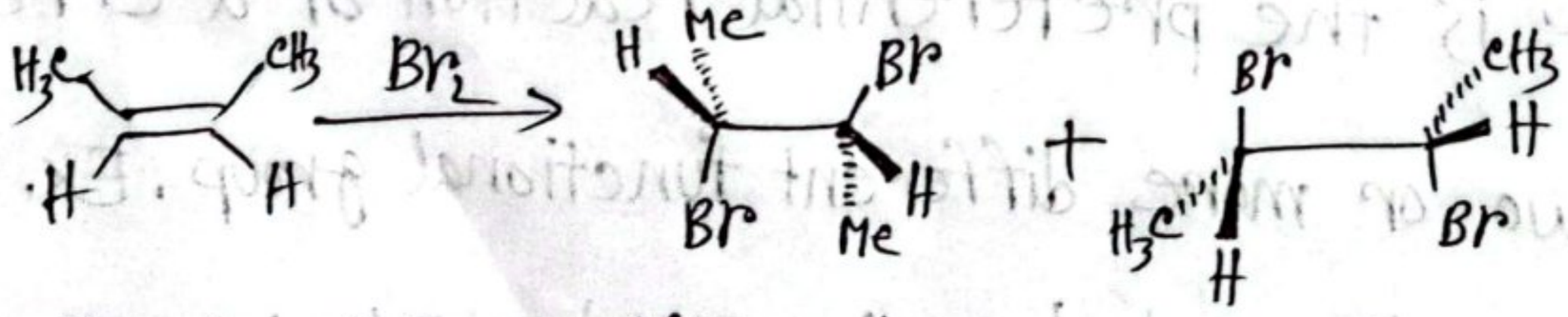
Stereoselectivity: is the property of a chemical reaction in which a single reactant forms an unequal mixture of stereoisomers during a non-stereospecific creation of a new stereocenter or during a non-stereospecific transformation of a pre-existing one.

Simply, It is the preference of a chemical reaction to produce one stereoisomer over another when two or more stereoisomers are possible.



Stereospecificity: A reaction is stereospecific if the reac is exist as stereoisomers and each stereoisomeric reactant leads to a different stereoisomeric product or a differen set of stereoisomeric products. Ex.



A reaction is stereospecific when a particular stereoisomeric form of the starting material reacts in such a way that it gives a specific stereoisomeric form of the product.

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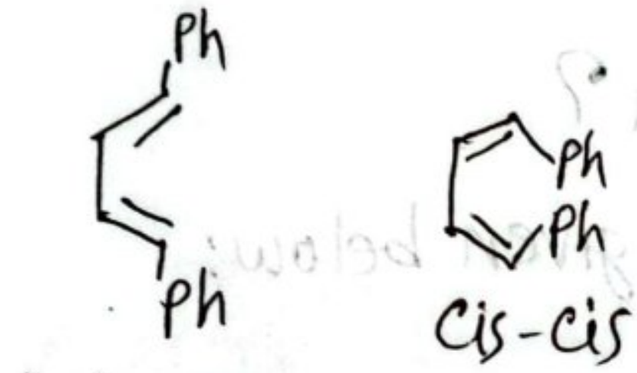
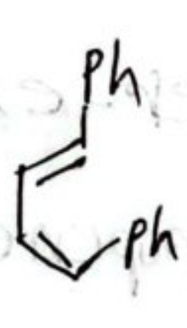
(2018) Diels alder reaction is ~~stere~~ sensitive to steric effects. Give example?

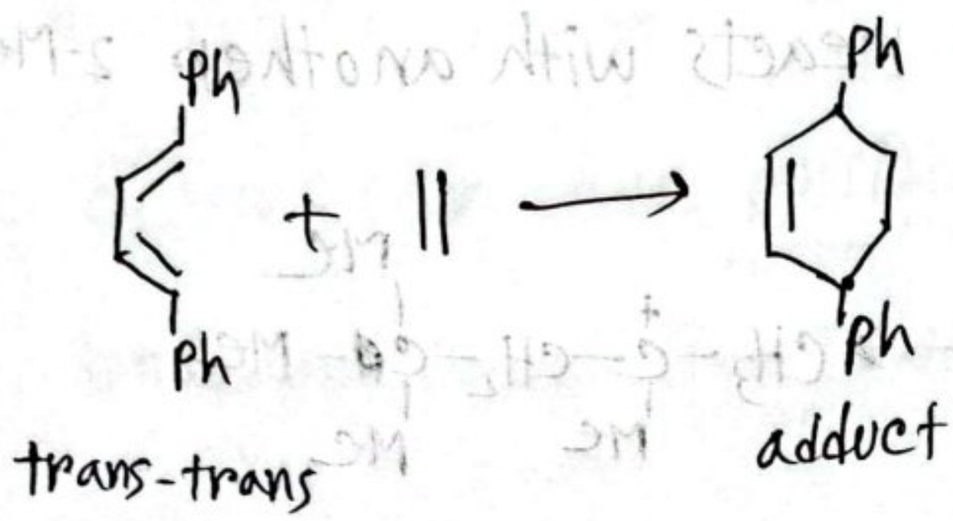
⇒ Diels alder reaction is highly sensitive to steric effects, means the the size & orientation of substituents on the diene & dienophile can significantly impact the r. rates & selectivity.

The reaction in which conjugate diene reacts with dienophile & gives a cyclo compound this r. is called diels alder reaction. There are three possible isomers of 1,4-diphenyl butadiene:-





In these three isomer only trans-trans show diels alder reaction. So, Diels alder r. is sensitive to steric effects.



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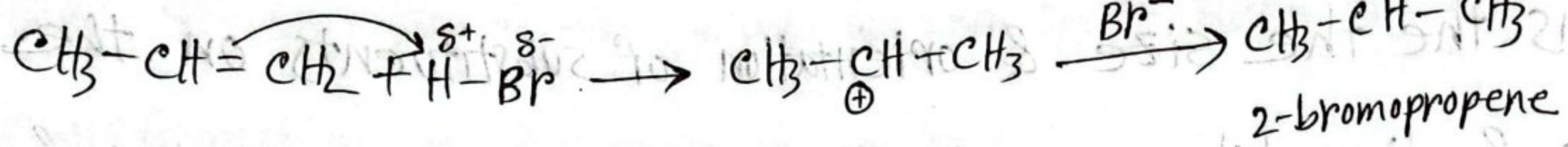
2018
In the addition of HBr to propene under polar conditions the only product is 2-bromopropane - explain?

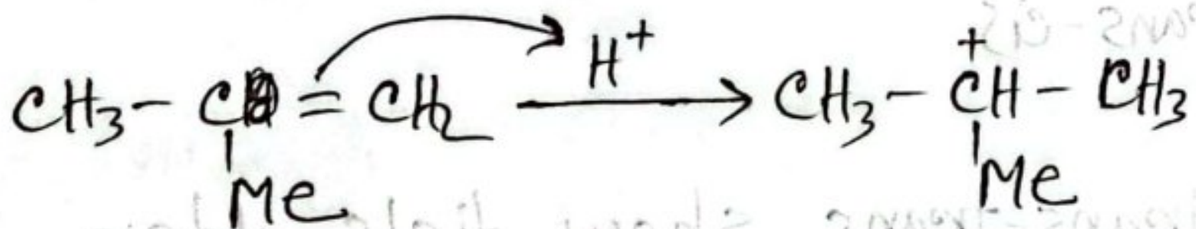
⇒ due to the stability of the carbocation.



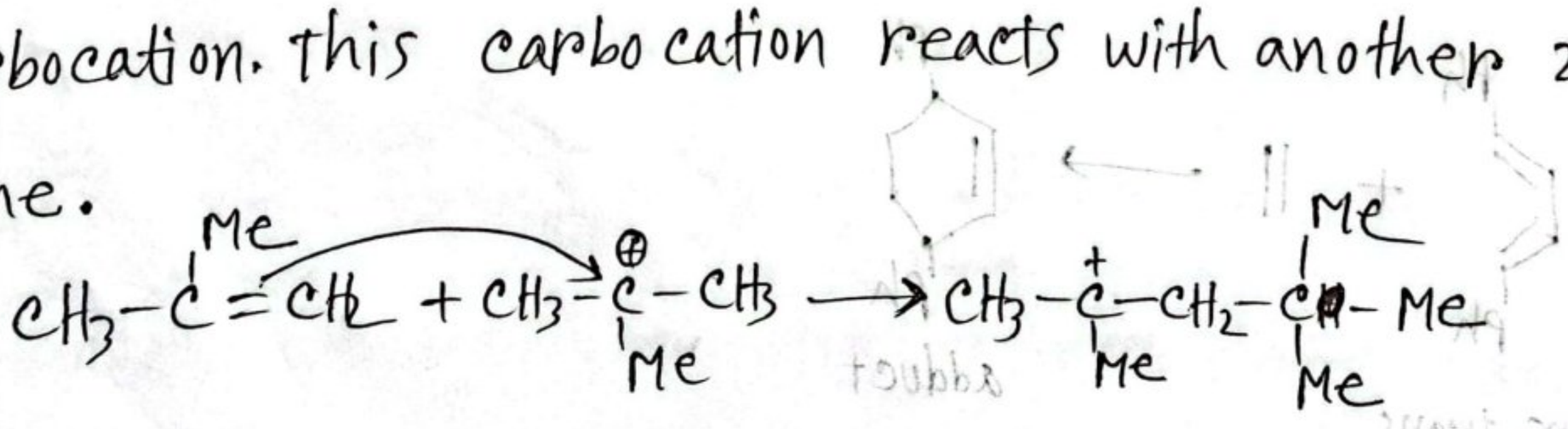
there 2° & 1° carbocation possible. As 2° carbocation is more Stable than 1° carbocation so the only product 2-bromoprop is produced.

2018
protonation of 2-Methylpropene can eventually lead to yield a high cationic polymer - explain?

⇒ protonation of 2-Methylpropene is given below:



Here, 3° carbocation is formed because 3° is more stable tha 1° carbocation. This carbocation reacts with another 2-Methyl propene.

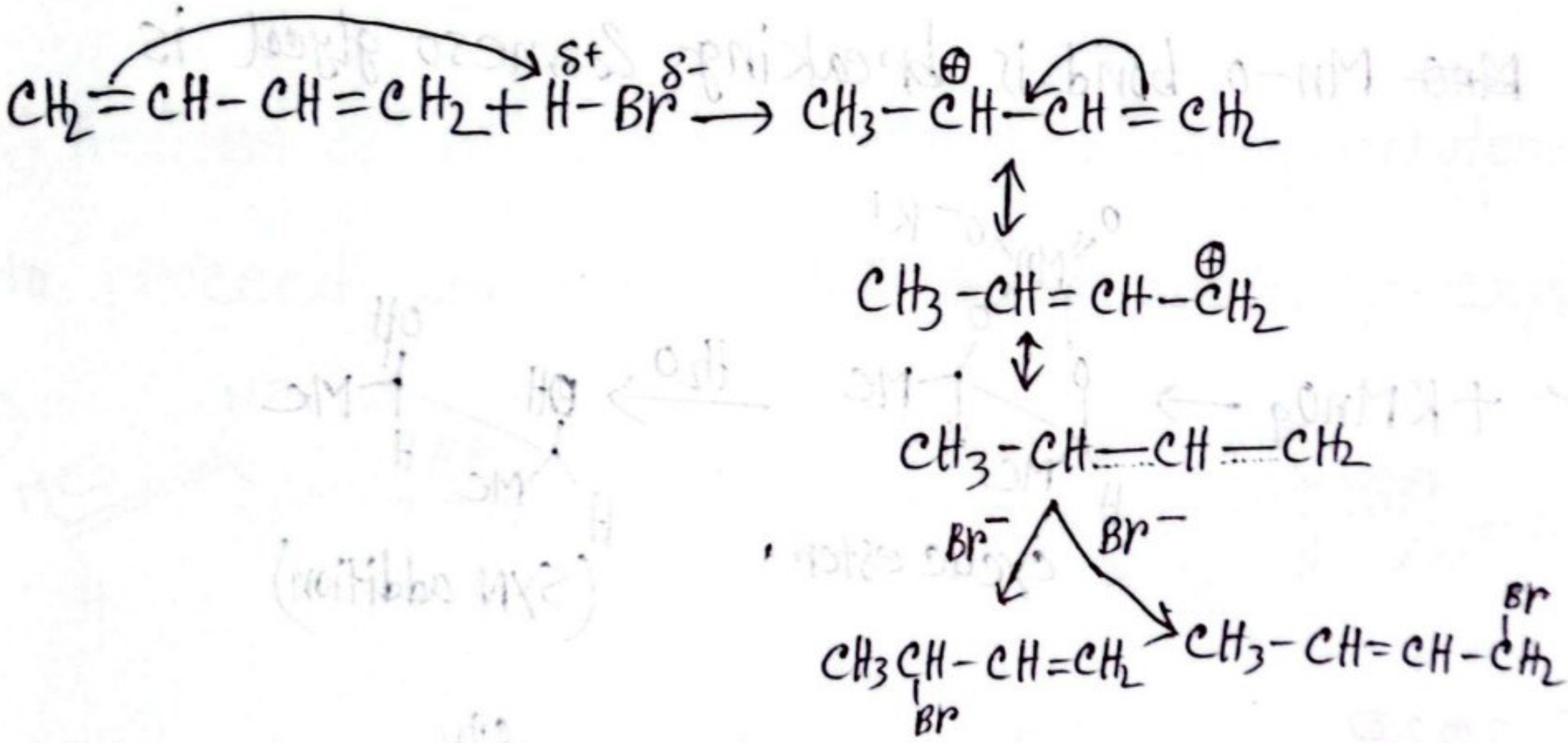


when this carbocation react with another 2-Methyl

propene & give high cationic polymer.

2018
(*) conjugate dienes are found to undergo addition reactions somewhat more rapidly than non-conjugated dienes-explain?

⇒ when conjugated diene reacts HHal gives addition reaction this reaction occur more rapidly than other diene.



In this reaction carbocation is formed, which is more stable because of resonance. we know, when the carbocation is more stable then the reaction is so faster.

CH2=CH-CH2-CH=CH2 + HBr → CH3-CH-CH2-CH=CH2

there is no resonance possible so it is not stable so non-conjugated diene undergo slowly than conjugated diene.

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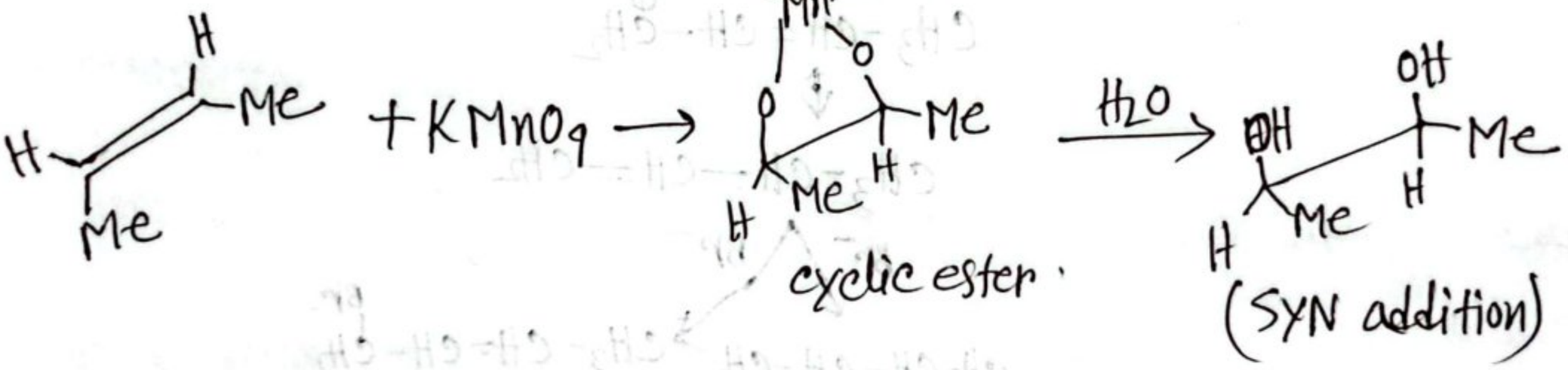
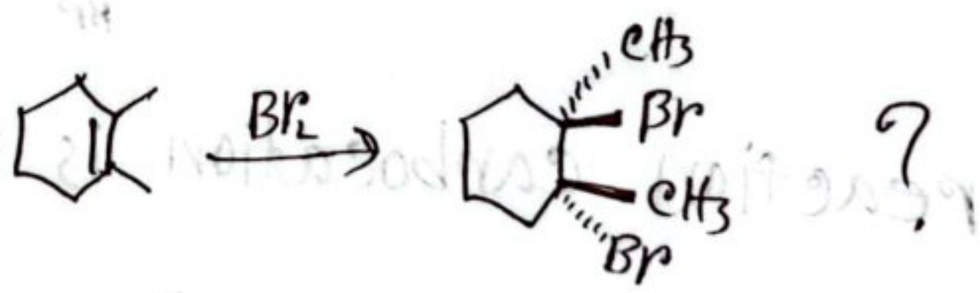
2016, 2015, 2020

The reaction of cis-2-butene with cold alkaline $KMnO_4$ gives meso glycol. propose a mechanism showing all intermediates to justify the statement?

⇒ cis-2-butene reacts with cold $KMnO_4$ & this reaction is SYN addition. In this reaction cyclic permaganate ester is formed. This ~~Mn-O~~ Mn-O bond is breaking & meso glycol is formed.

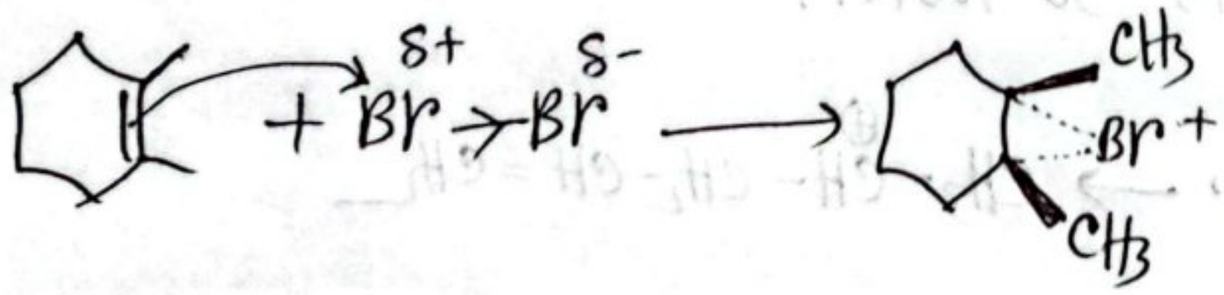


2015, 2016, 2020

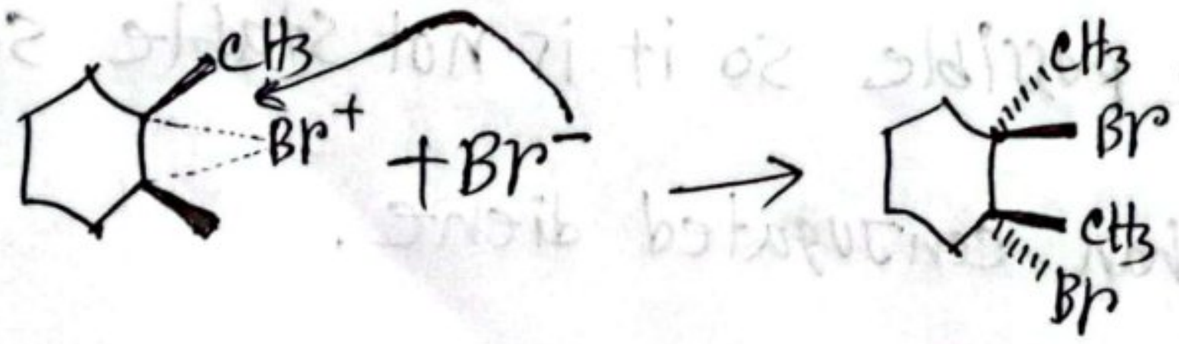
Suggest a mechanism for: 

?

⇒ Mechanism: Step-1



Step-2:



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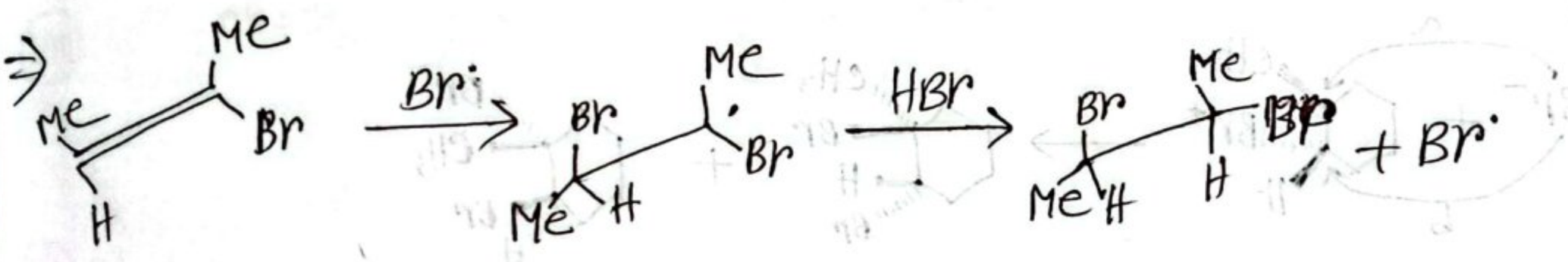
(*) How you prepare primary alc from alkenes?

⇒ primary Alc. can be prepared from alkenes by anti-markonikov addition of B_2H_6 (hydroboration) followed by oxidation of the resultant trialkyl boron with alkaline H_2O_2.

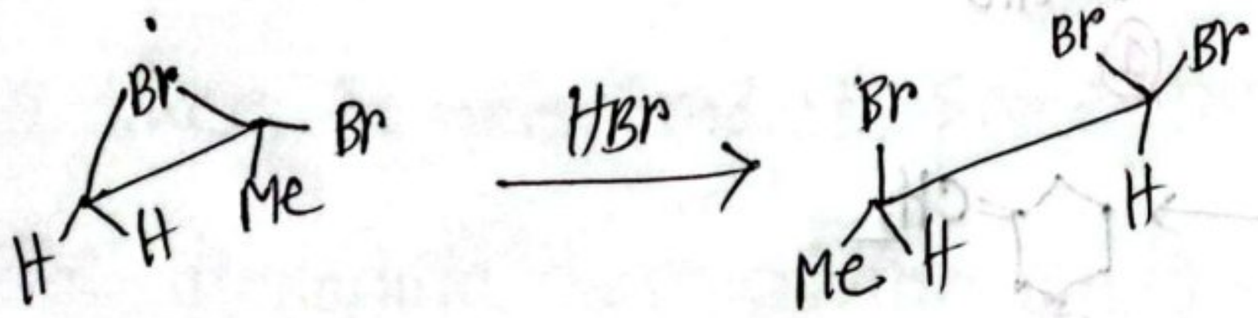
$$Me-CH=CH_2 \xrightarrow{B_2H_6} (MeCH_2CH_2)_3B \xrightarrow{H_2O_2} MeCH_2CH_2OH + B(OH)_3$$

(*) Addition of liq. HBr at −80°C to cis-2-bromobutene was found to proceed with high TRANS stereoselectivity-explain?

⇒



To account high Trans stereoselectivity, it has been suggested that addition proceed via a cyclic bromonium ~~cation involved in~~ the radical analogous to the cyclic bromonium cations involved in the polar addition of Br to alkenes



the overall addition would then completed through attract by HBr from the less hindered side to yield the product of overall trans addition.

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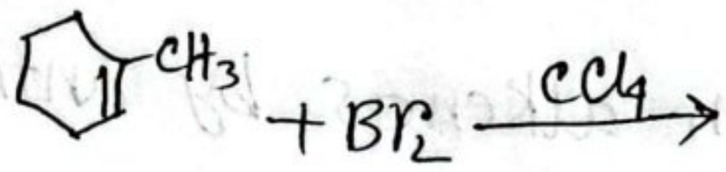
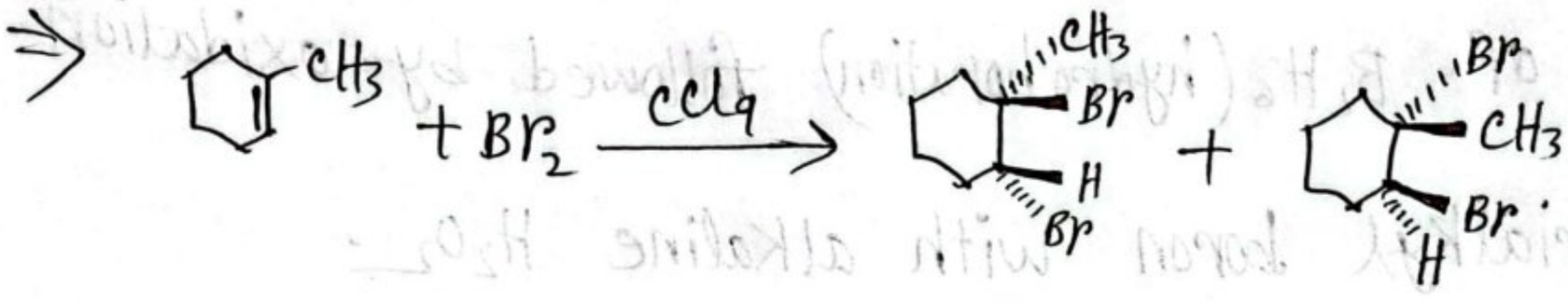
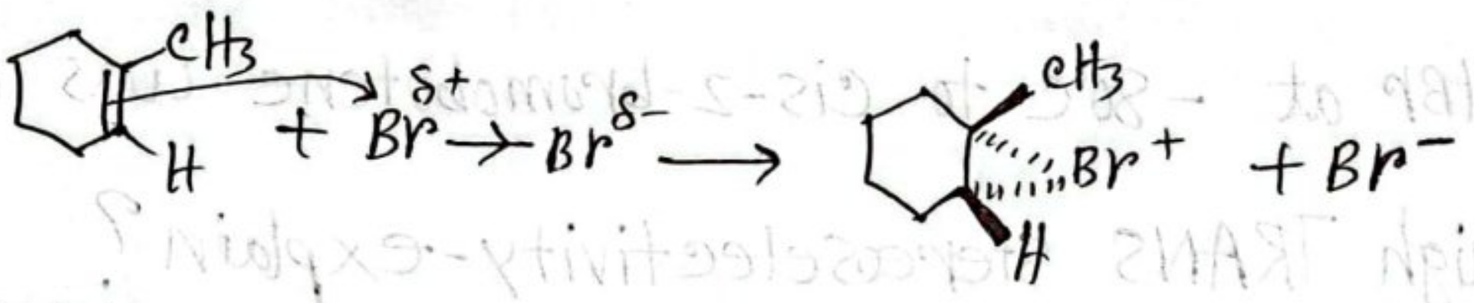
⊛ 2013
Complete the reaction with mechanism:



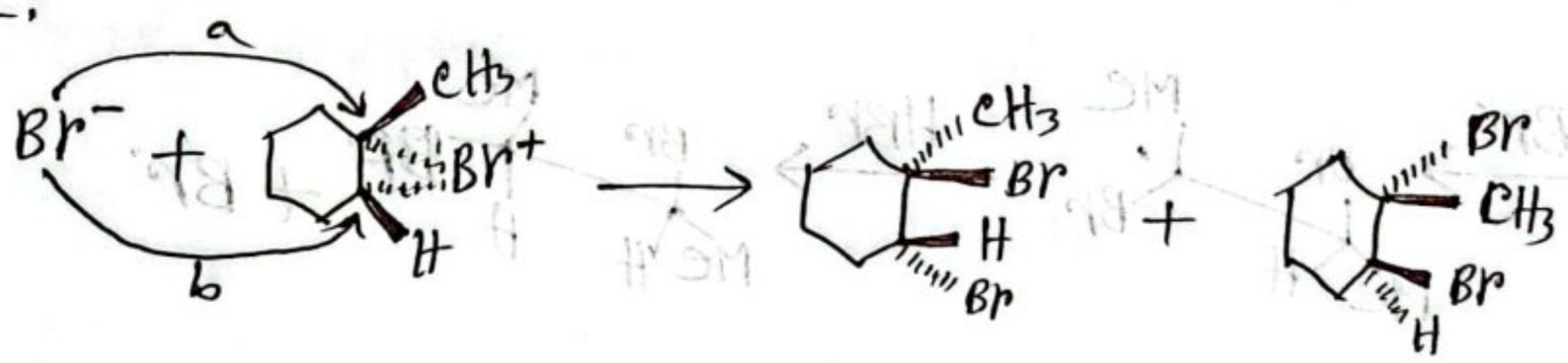


(±) Recimic mixture

Mechanism: Step-1:

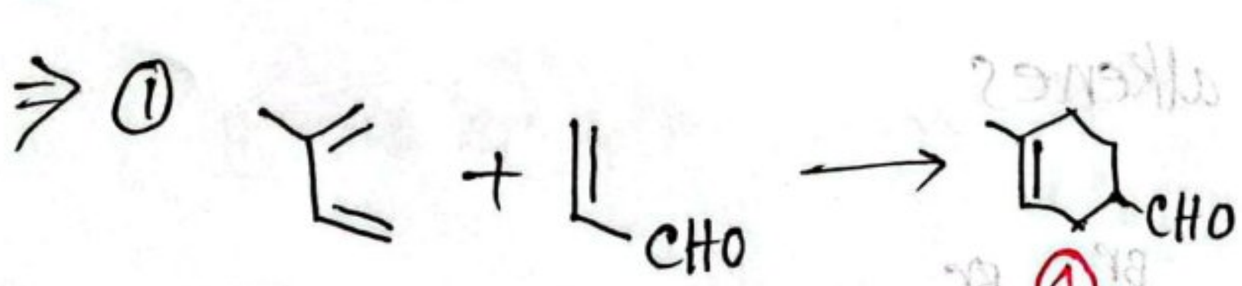


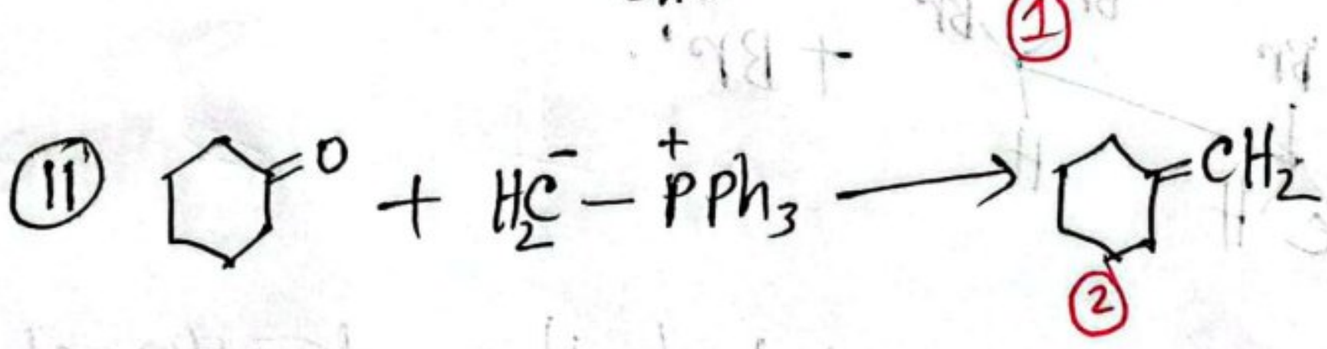
Step-2:

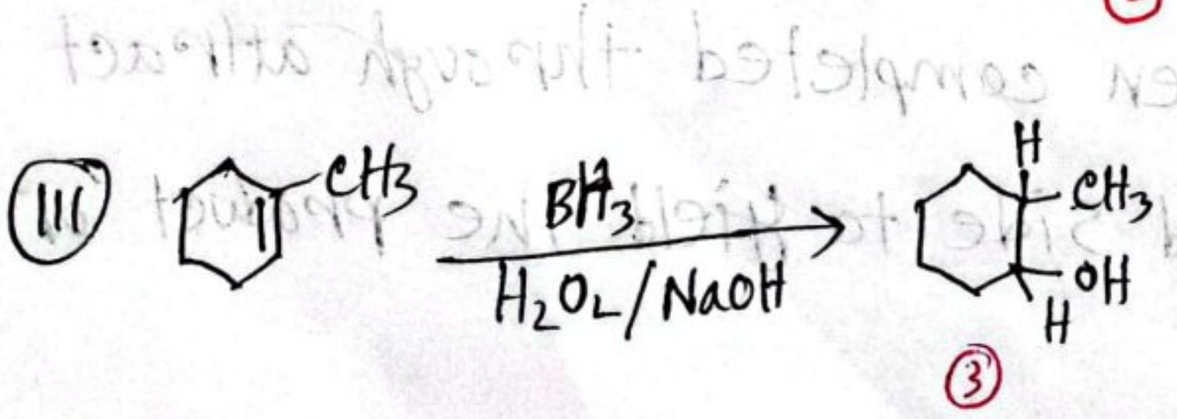


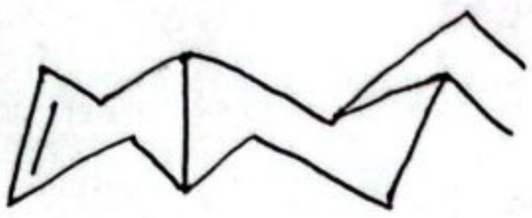
This reaction is followed by ANTI addition & the product is (±) dibromide.

⊛ 2013, 2012
Complete the reaction:

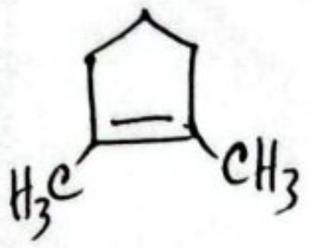
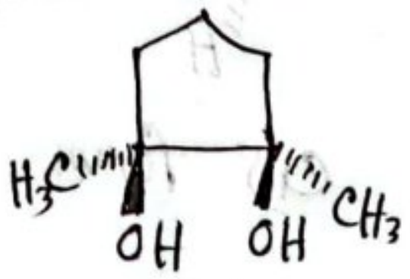




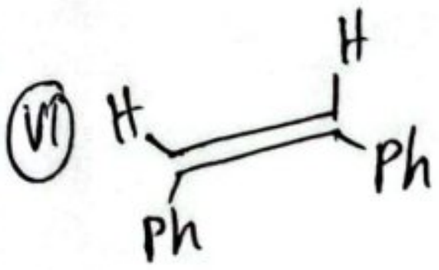
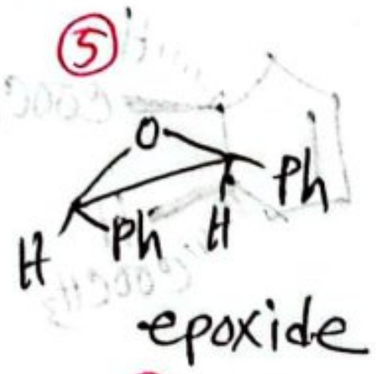


(IV)  $\xrightarrow{Br_2(CCl_4)}$

(9)

(V)  $\xrightarrow[\text{ether, pyridine}]{OsO_4}$ 

(5)

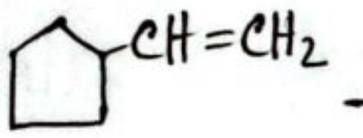
(VI)  $+RCOOH \longrightarrow$ 

epoxide

(6)

(VII) $C_2H_5-CH=CH_2 \xrightarrow[H_2O]{\text{conc. } H_2SO_4} C_2H_5-\underset{OH}{CH}-CH_3$

(7)

(VIII)  $\xrightarrow[NaBH_4]{Hg(OAc)_2\,H_2O}$

(8)

2016, 2020

(X) Diels alder reaction is stereospecific w.r.to both diene & dienophile - explain with example?

⇒ because it preserves the stereochemistry of both the diene & the dienophile in the product. If the diene is in the cis conformation, the substituents on the diene with remain on the same face in the product. same for trans-diene.

If the dienophile has substituents in a cis relationship, the product retains the cis relationship in resulting six membered ring. same for trans. For example:

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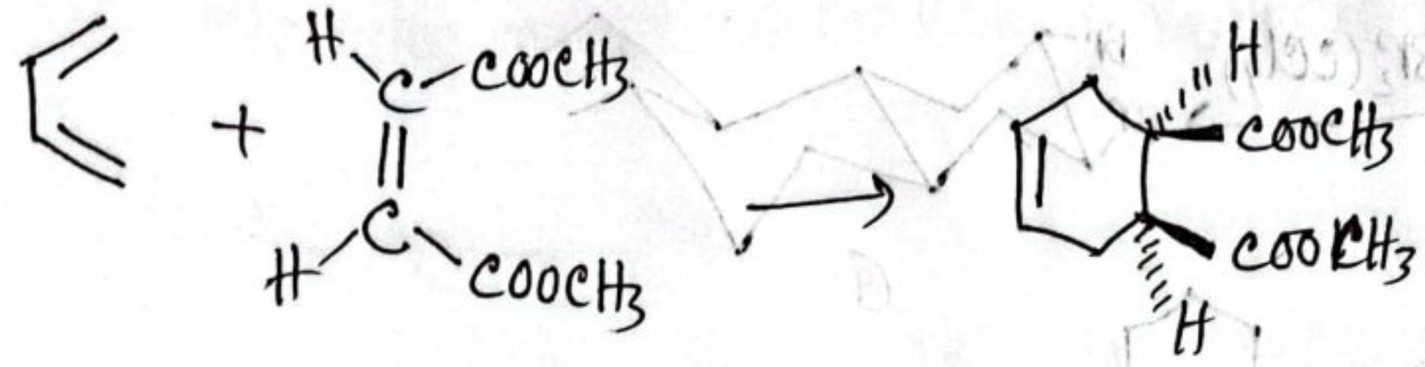
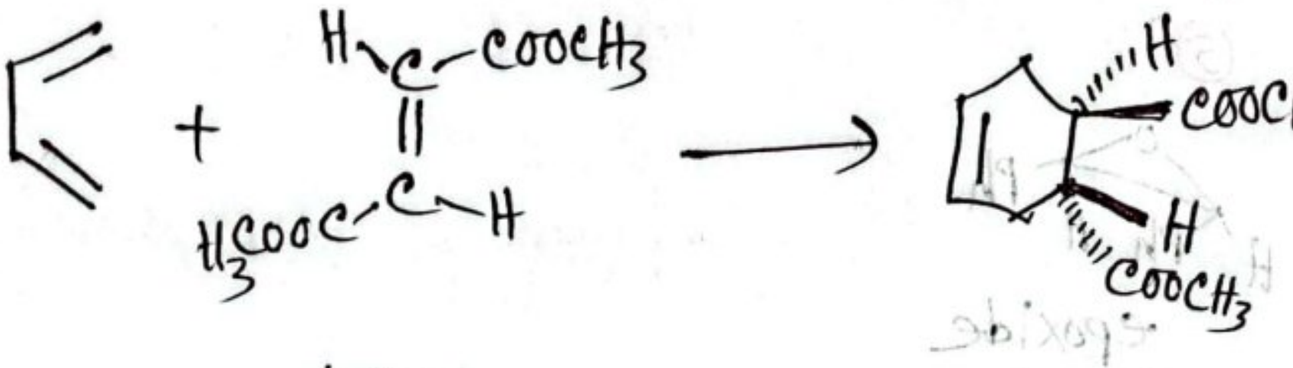
① 

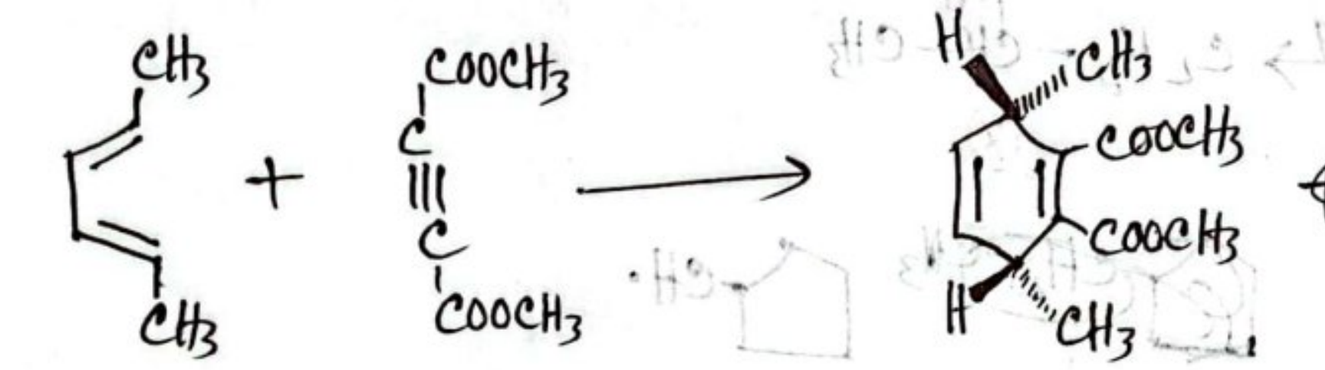
cis form

cis form

② 

trans

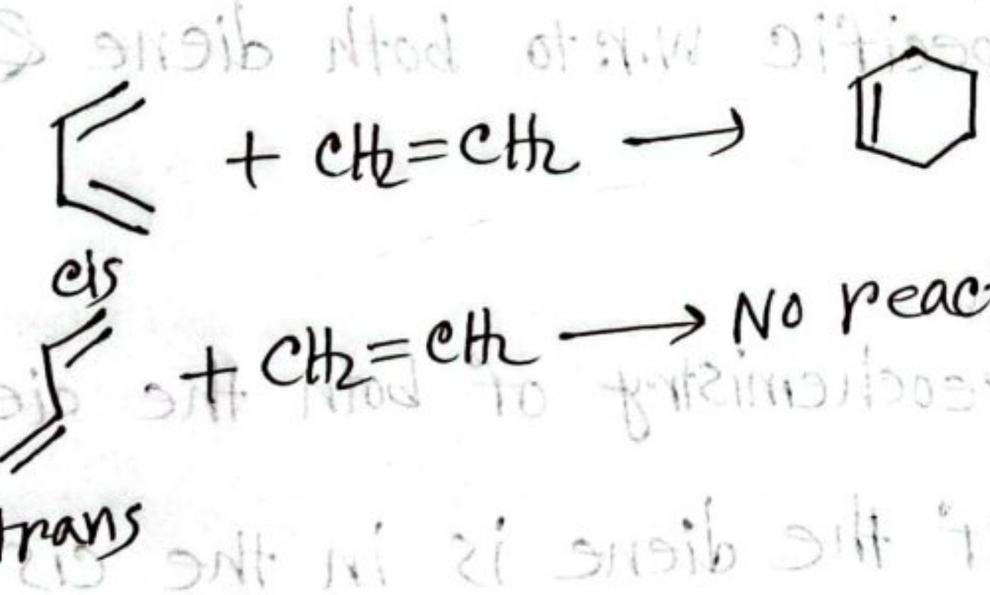
trans

③ 

cis

cis

~~(endo) (major)~~

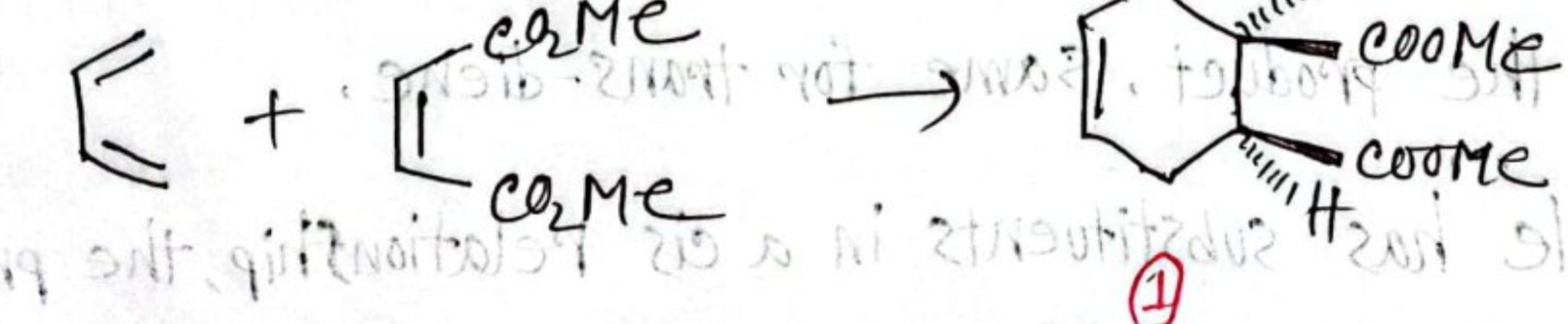
④ 

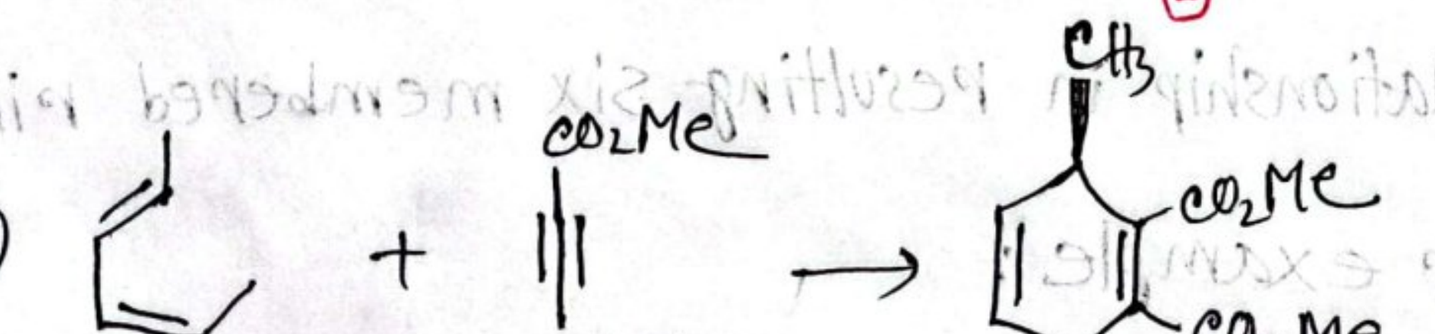
cis

trans

2022

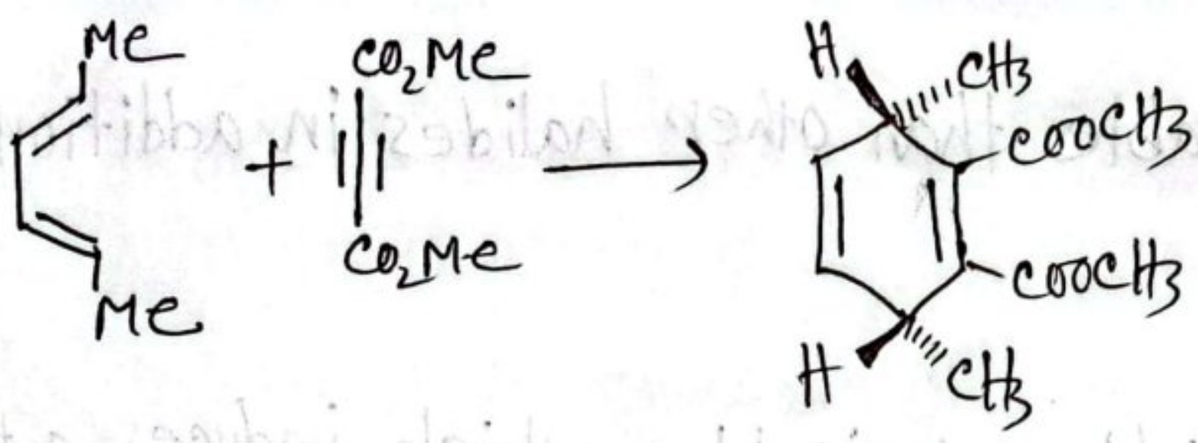
⑤ completed the reaction:

⇒ ① 

② 

(anti)

(III)



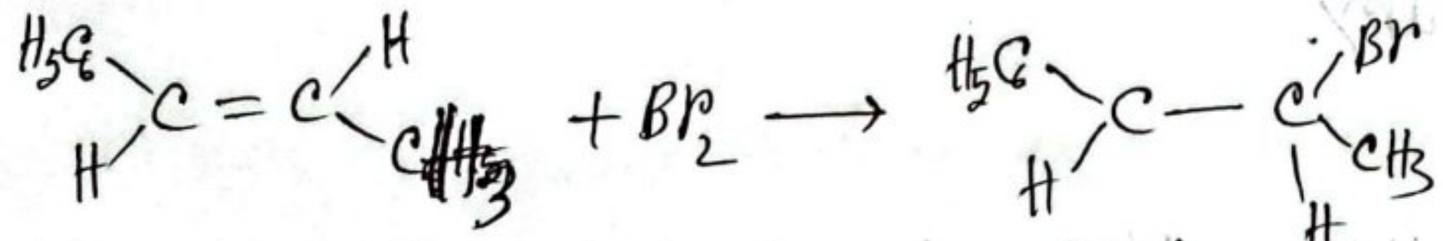
(3)

(IV)

(4)

(8) when trans-1-Phenylpropene is treated with Br_2, Some SYN addition is observed. Why it results in a lower stereoselectivity?

⇒ This is due to phenyl group. Normally Br^- ~~attacks in a~~ leads to anti addition but in the presence of phenyl group can stabilize a carbocation which carbocation is planar & Br^- can attack both above & under the plane, so SYN addition occur.



The presence of phenyl group disrupts the Br^- due to its ability to delocalize +ve charge & so carbocation is formed & both SYN & ANTI can take place, which means it results lower Stereoselectivity.

CamScanner

(*) Why Br_2 is more suitable than other halides in addition of alkene?

⇒ (I) Br_2 molecules are highly polarizable which induces a temporary dipole moment in the Br_2 molecule, making it more reactive towards the alkene.

(II) Br_2 forms cyclic bromonium ion intermediate which is relatively stable due to delocalization of +ve charge over three membered ring

(III) Br_2 is liquid at room temperature, making it easier to handle.

(IV) Br_2 is less reactive than F_2 & Cl_2 which can lead to side reacti but more reactive than iodine(I_2) which reacts too slowly.

(V) Br atoms are good leaving groups, which facilitates further reactions if necessary.

(*) Why R. rate of HBr addition to vinyl bromide is 30 times slower than the addition of ethene?

⇒ addition of HBr to ethene or vinyl bromide proceed via the formation of carbocation.

$$CH_2=CH_2 \xrightarrow{H^+} CH_3-\overset{\oplus}{C}H_2 \xrightarrow{Br^-} CH_3-CH_2Br$$

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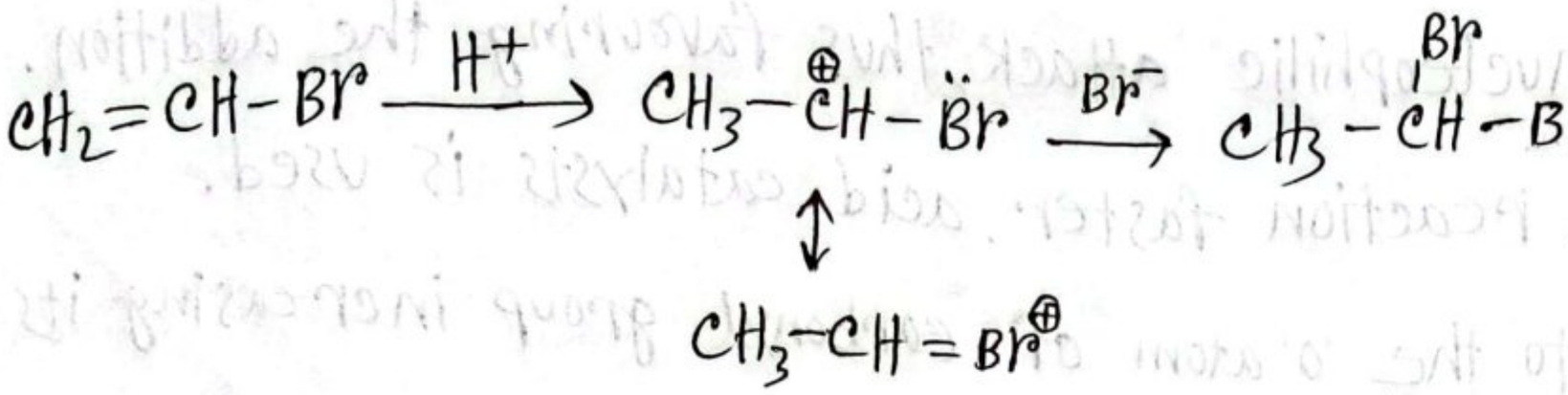
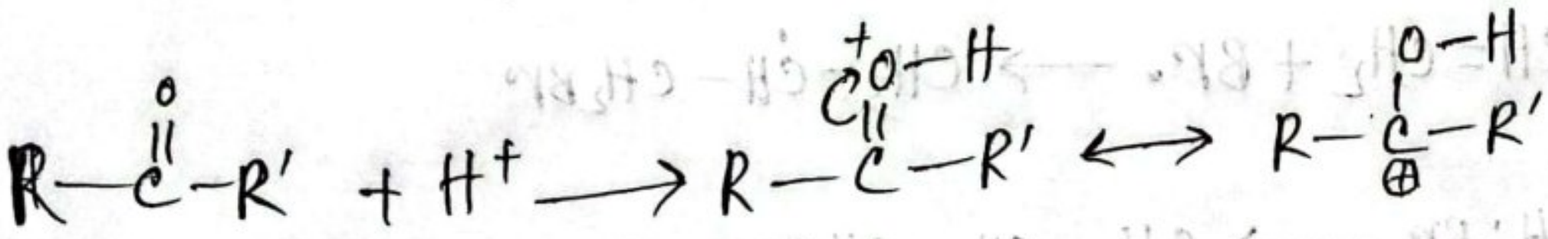


Here, the first step of the reaction is the rate determining step, where H^+ acts as electrophile & proceed to attack. In vinyl bromide, the e^- withdrawing nature of Br atom reduces e^- density via negative inductive effect & makes it less susceptible to attack by electrophiles. But in ethene, no such things happen & the reaction occur at normal pace. Thus inductive effect of Br makes HBr addition in vinyl bromide is 30 times slower than ethene.

(x) Why acid used in the addition reaction of ald or ketone?

⇒ Generally weak acid is used as catalysts in the addition reaction of Ald & Ketones. The carbonyl group in ald. & ketones contain a lone pair which reacts with weak acid.



Product formed in the reaction show resonance which increase the +ve charge on the carbonyl carbon which makes the

c prone to a nucleophilic attack, thus favouring the addition.

(ii) to make the reaction faster, acid catalysts is used.

(i) Acid donate H+ to the 'O' atom of carbonyl group increasing its positive charge. (iii) increase electrophilicity of carbonyl carbon enhances its susceptibility to attack by nucleophile, which is the key of addition reaction. (iv) Acid helps stabilize intermediate species that may form during the reaction.

(x) Discuss reaction of HBr with propene in presence of peroxide & in absence of peroxide?

⇒ Absence of peroxides follow markonikov rule:

Step-1: $CH_3-CH=CH_2 \xrightarrow{H-\ddot{B}r} CH_3-\overset{\oplus}{C}H-CH_3$

step-2: $CH_3-\overset{\oplus}{C}H-CH_3 \xrightarrow{Br^-} CH_3-\underset{Br}{\overset{|}{C}}H-CH_3$ (2-bromo propane)

presence of peroxides follow anti merkonikov rule:

Initiation: $R-O-O-R \xrightarrow{\Delta} 2\,R-O\cdot$

$R-O\cdot + H-Br \longrightarrow R-OH + Br\cdot$

propagation: $CH_3-CH=CH_2 + Br\cdot \longrightarrow CH_3-\dot{C}H-CH_2Br$

$CH_3-\dot{C}H-CH_2Br + H:Br \longrightarrow CH_3-CH_2-CH_2Br + Br\cdot$

Termination: $Br\cdot + Br\cdot \longrightarrow Br_2$

CamScanner

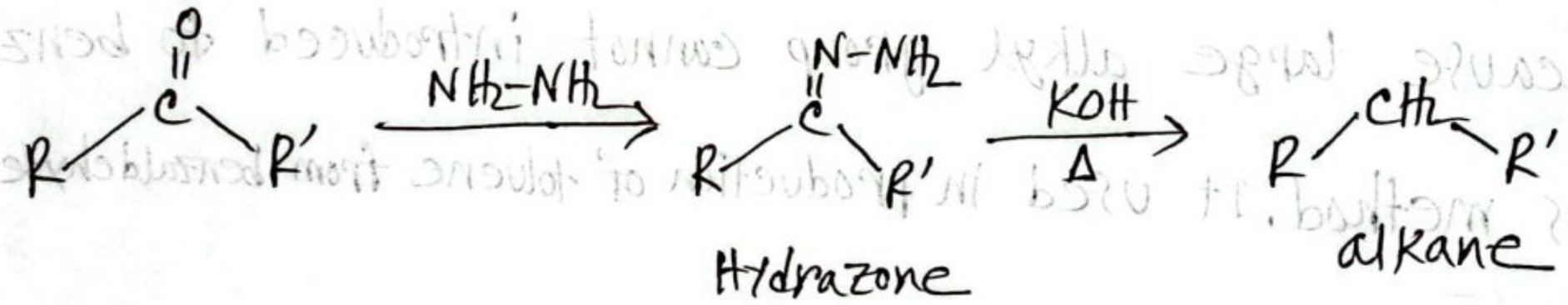
Name reaction

RAJIB
Che-50

(*) 2021 Write short not on 'Wolf-Kishner reduction'?

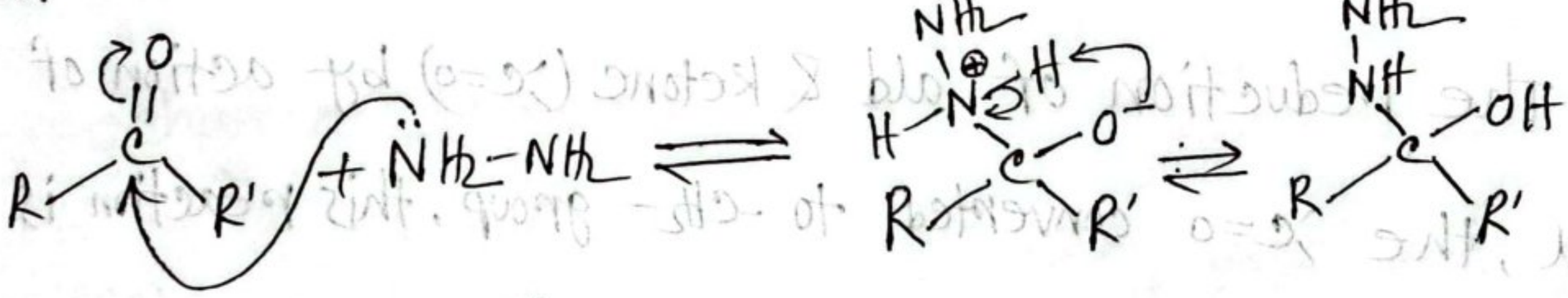
⇒ The reduction of aldehyde & ketone ($>C=O$) by action of hydrazine (NH_2-NH_2) in the presence of base (KOH/NaOH) then $>C=O$ is converted to $-CH_2-$ group, This reaction is called wolf kishner reduction reaction.

Reaction:

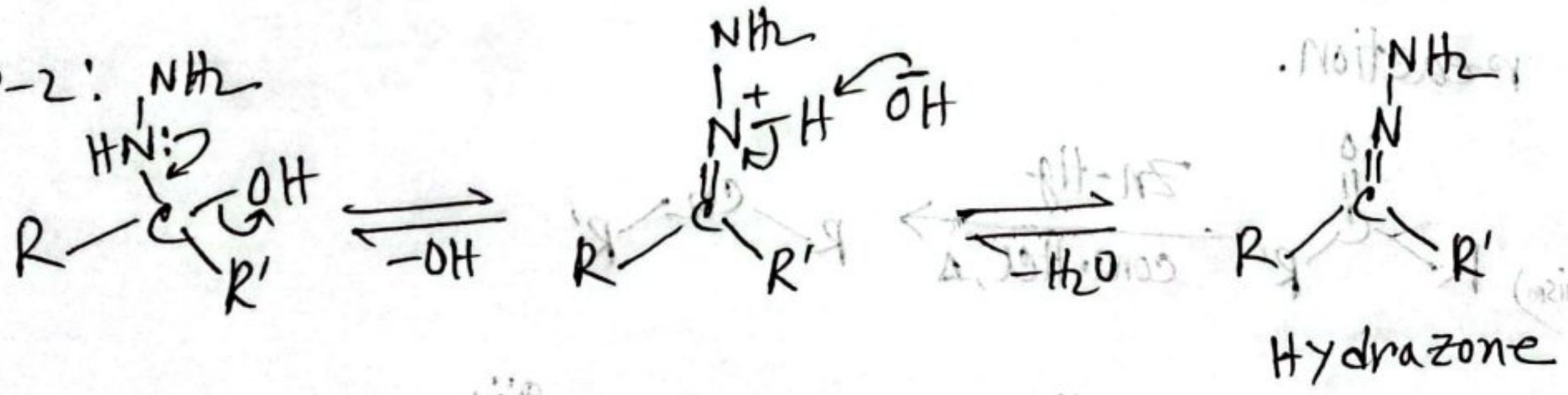


Mechanism:

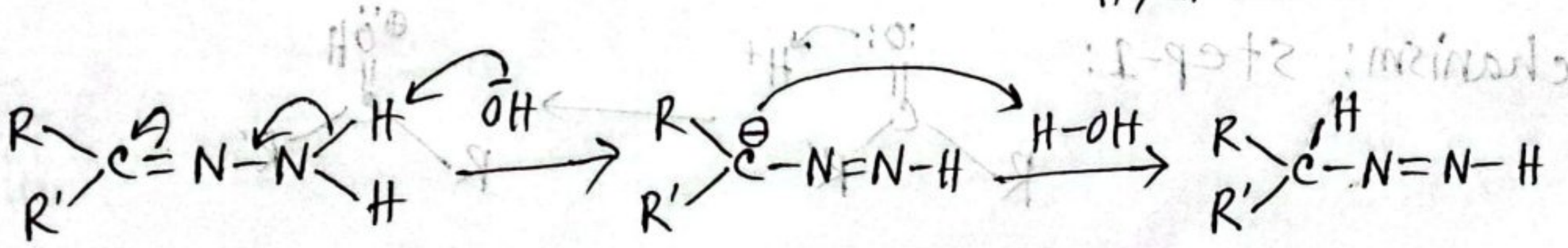
Step-1:



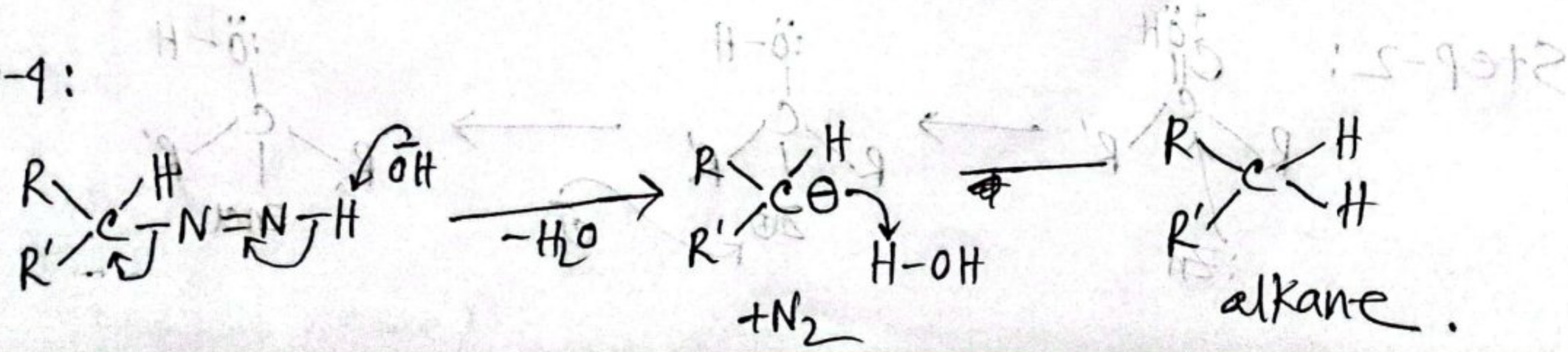
Step-2:



Step-3:



Step-4:



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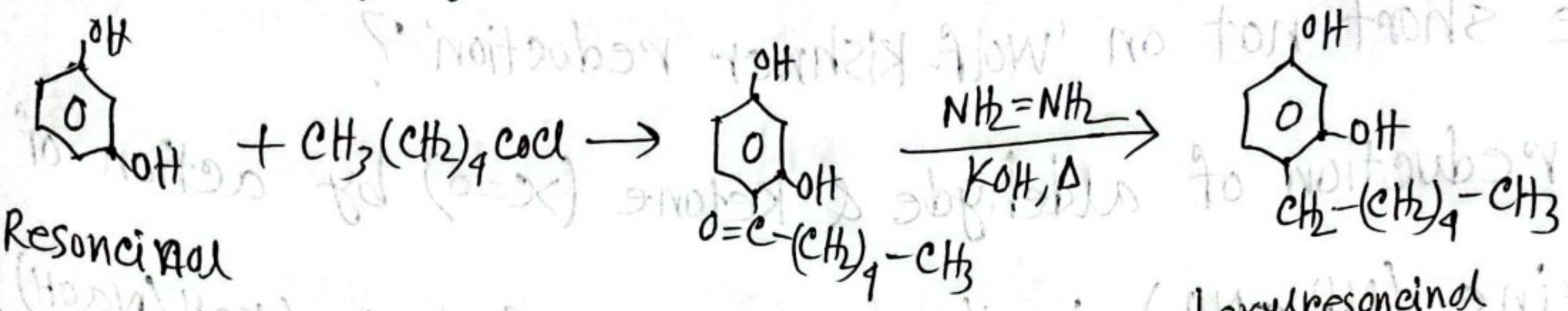
Application: This reduction is particularly important when applied to the alkyl/aryl ketones obtained from friedel-craft acylation.



Resoncinol

hexylresoncinol

Application:

This hexylresoncinol used as an anticeptic in our daily life. By friedel-craft alkylation we cannot prepare this hexylresoncinol because large alkyl group cannot introduced to benzene ring by this method. It used in production of toluene from benzaldehyde. But Hydrazine is tox

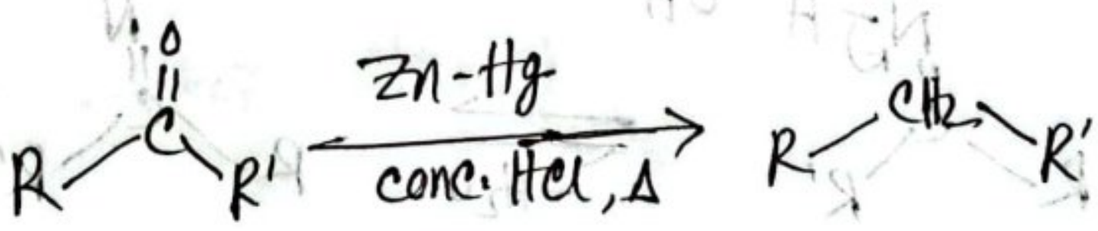
2015

write short note on 'clemmension reduction' ?

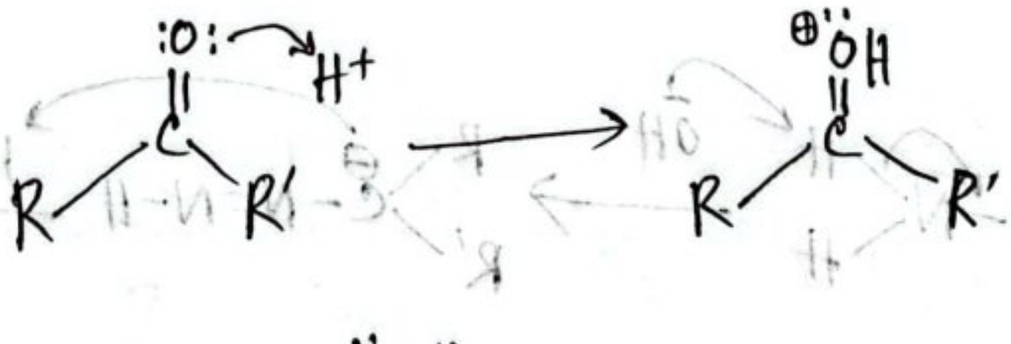
⇒ The reduction of ald & ketone (>c=o) by action of Zn-Hg & conc. Hcl, the >c=o converted to -CH2- group, this reaction is called clemmensin reduction reaction.

Reaction:

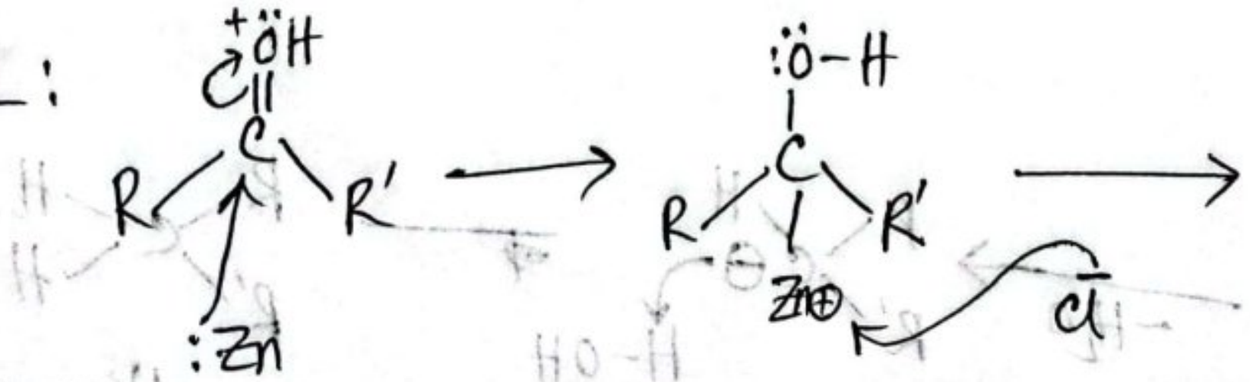
(carbanionic mechanism)



Mechanism: step-1:



Step-2:



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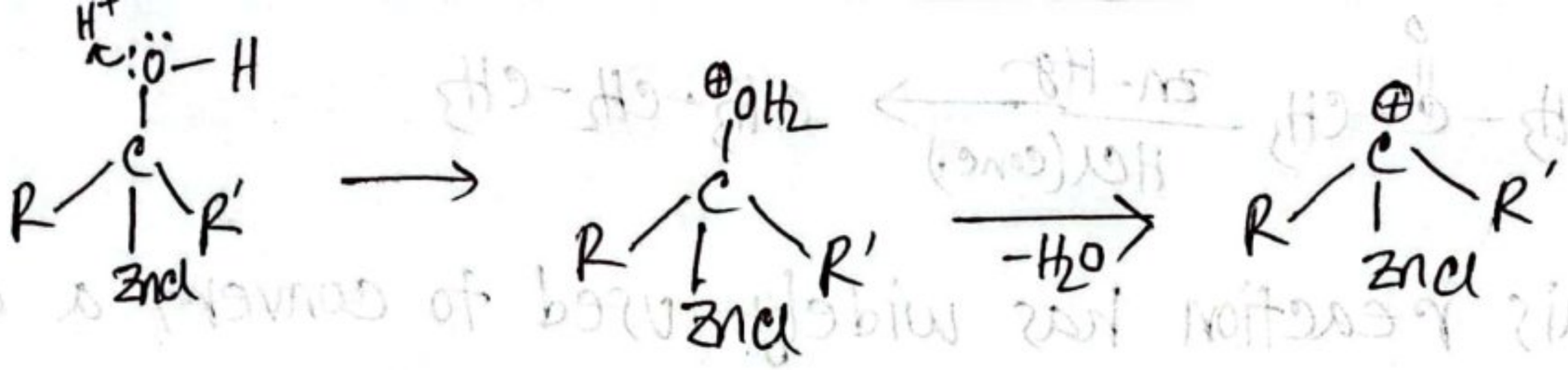
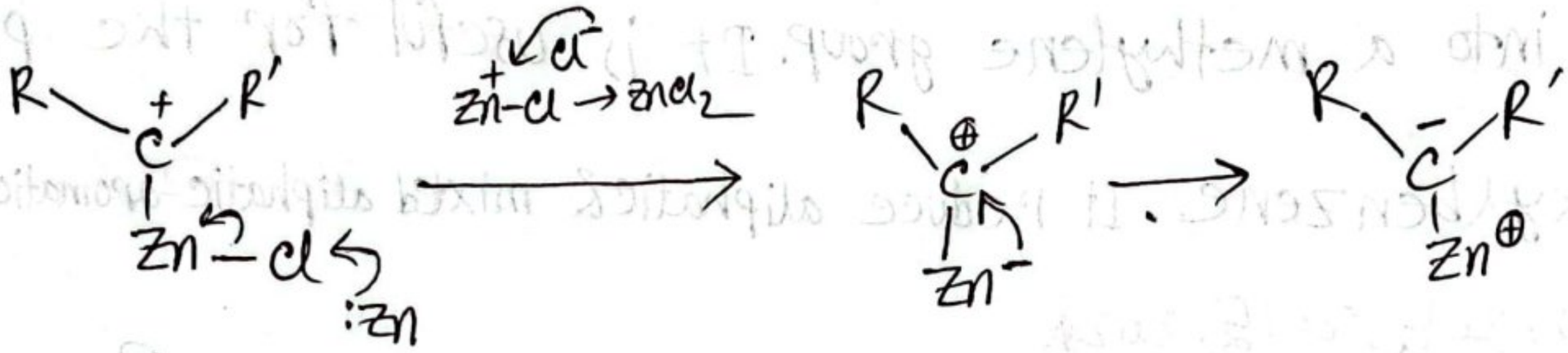
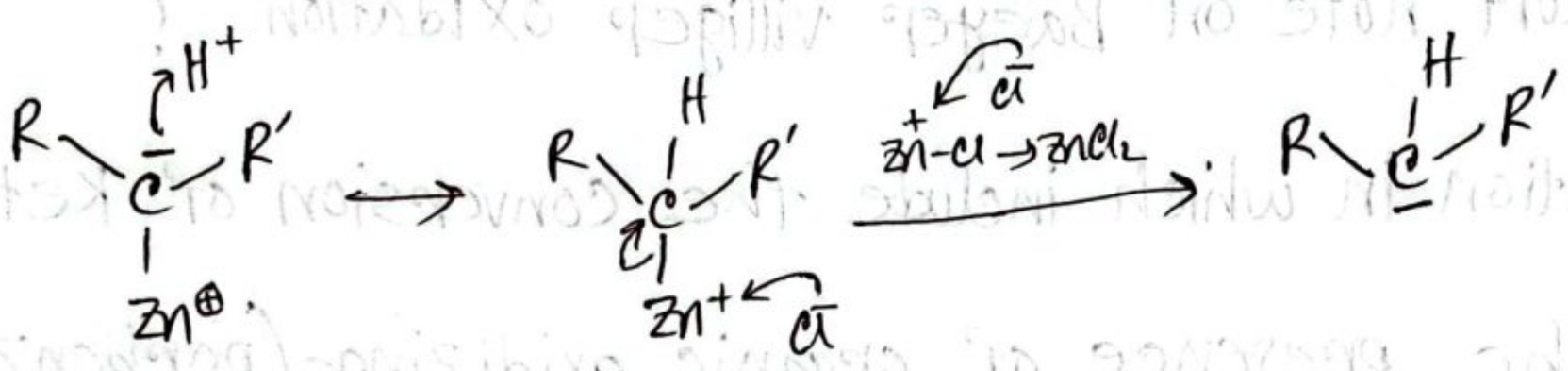
Step-3:



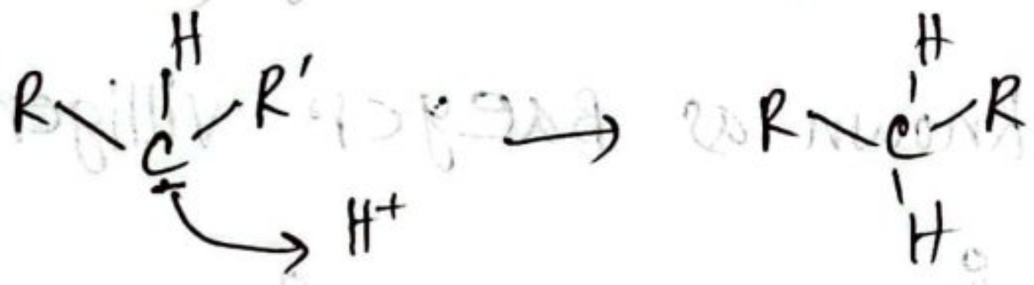
Step-4:



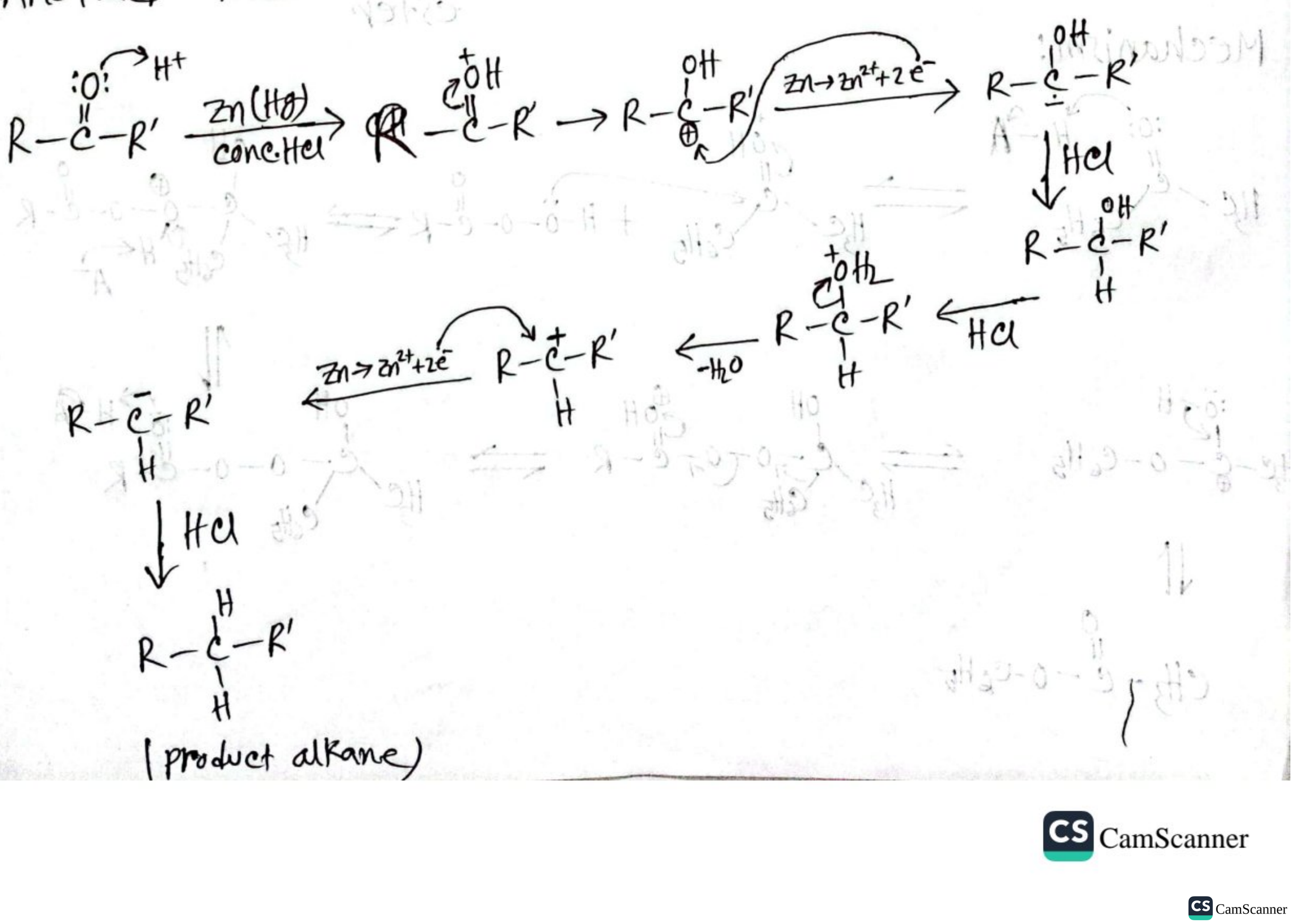
Step-5:



Step-6:



Another mechanism: (mam poraise)



(product alkane)

Example: $CH_3-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow[HCl(conc.)]{Zn-Hg} CH_3-CH_2-CH_3$

Application: This reaction has widely used to convert a carbonyl group into a methylene group. It is useful for the preparation of alkylbenzene. It reduce aliphatic & mixed aliphatic-aromatic carbonyl compound

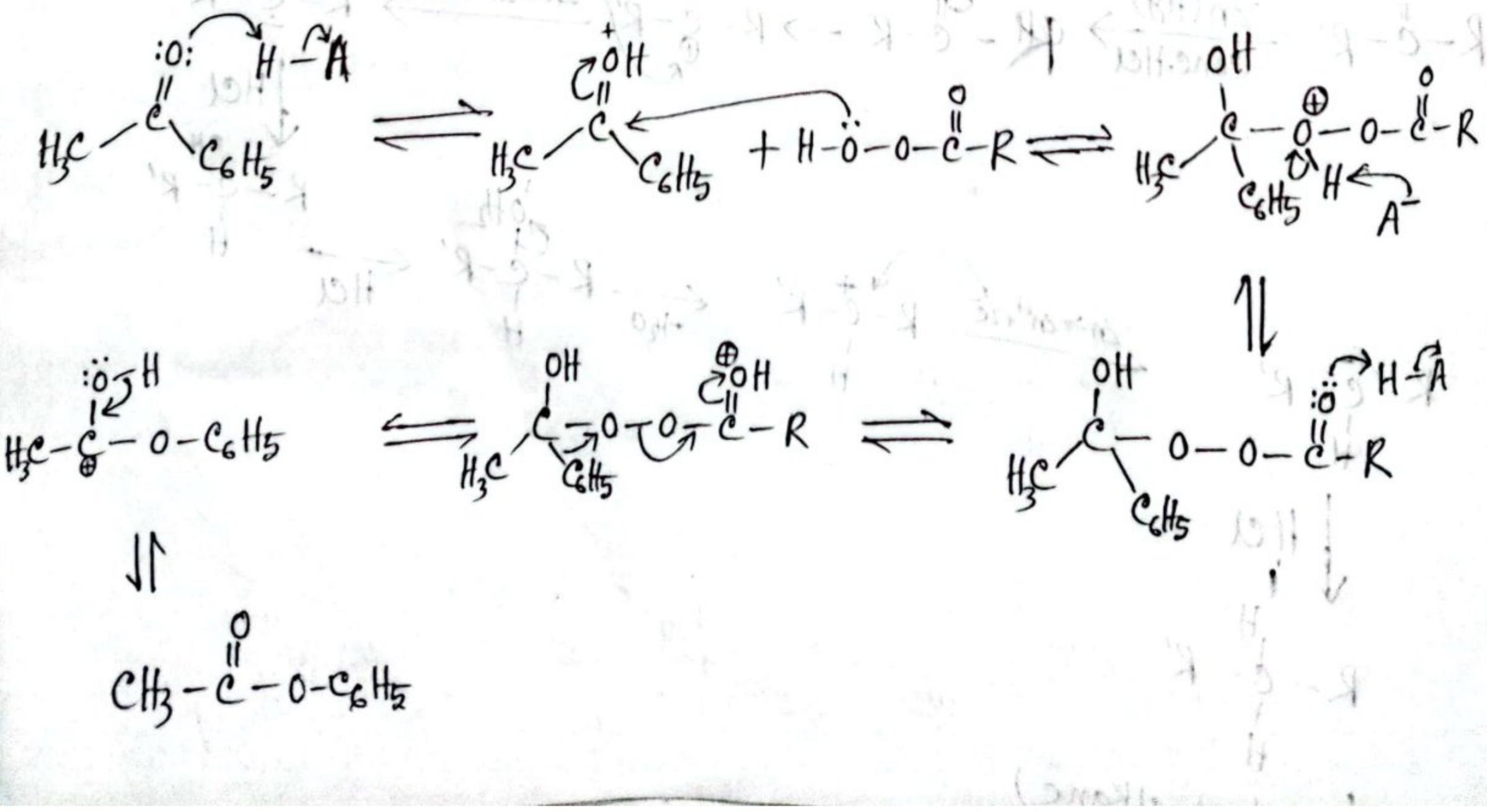
⊛ 2020, 2016, 2015, 2021 Write short note on "Baeyer villiger oxidation"?

⇒ The oxidation in which include the conversion of ketone into ester, in the presence of organic oxidizing (perbenzoic acid, peroxide, peroxy acid) is known as Baeyer villiger oxidation.

Reaction: $C_6H_5-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow{R-\overset{O}{\overset{\|}{C}}-O-OH} C_6H_5-O-\overset{O}{\overset{\|}{C}}-CH_3$

ester

Mechanism:



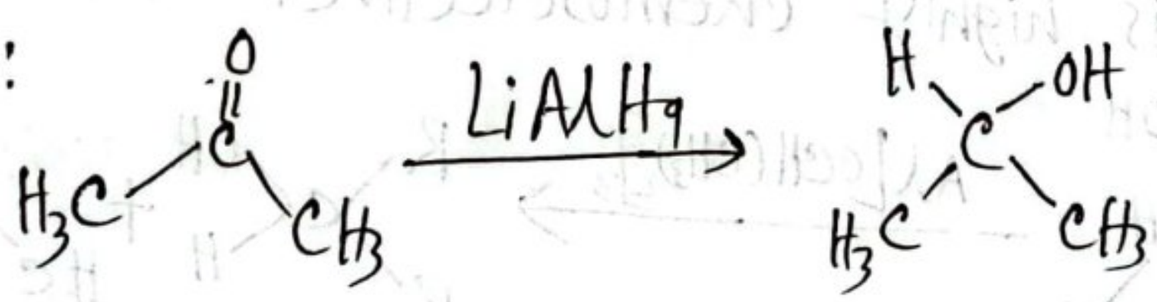
2020, 2016

Write short note on "Reduction with $LiAlH_4$" or "Reduction with metal hydride."

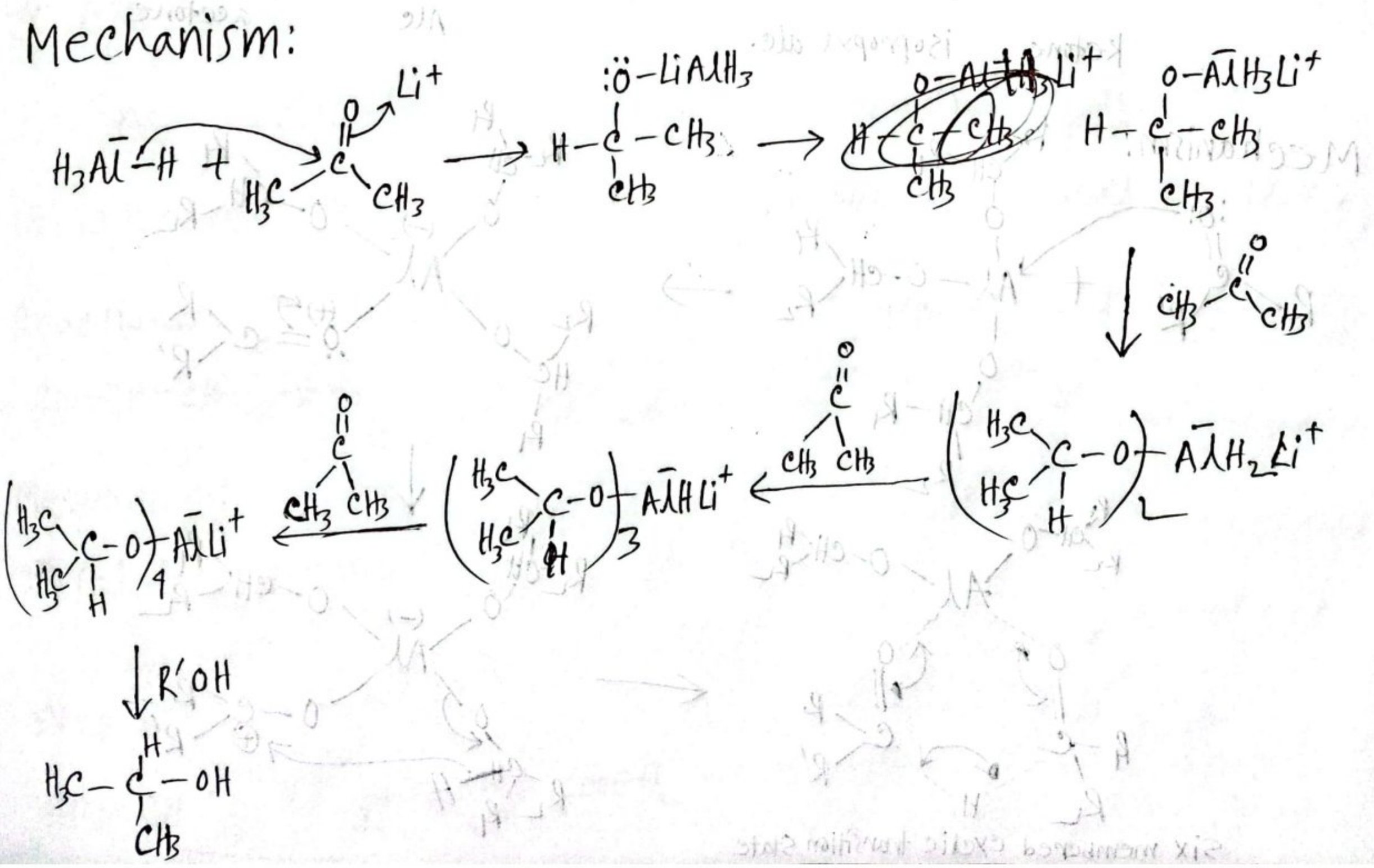
or, $NaBH_4$

⇒ $LiAlH_4$ is the most powerful metal hydride. The effective reducing agent is $\bar{A}lH_4$, which act as H^- donor. In this case reduction cannot be carried out in the protonic solvent eg H_2O, R-OH because proton abstraction would take before reduction. So non-protonic solvents e.g. ether, DME, THE can be used.

Reaction:



Mechanism:



Preparation of $LiAlH_4$:

① $4LiH + AlCl_3 \rightarrow LiAlH_4 + 3LiCl$

② $LiH + AlH_3 \rightarrow LiAlH_4$

③ $2LiH + 2Al + 3H_2 \rightarrow 2LiAlH_4$

Preparation of $NaBH_4$:

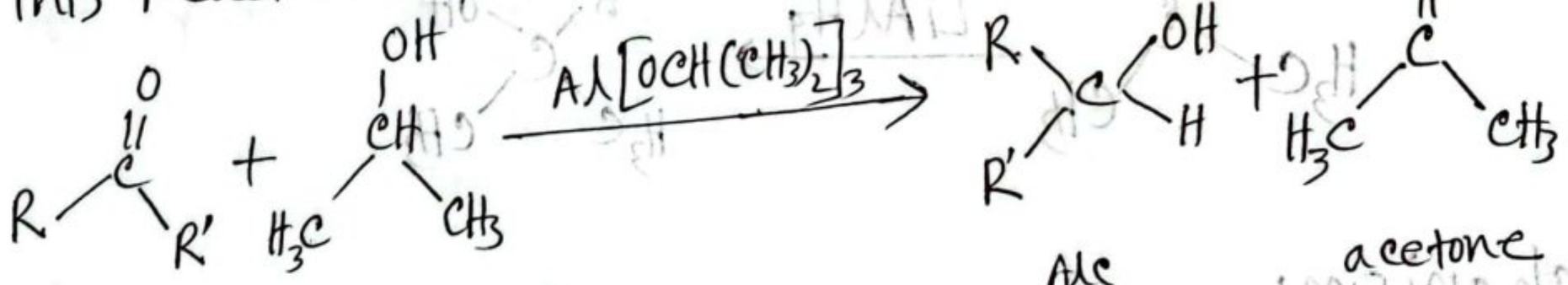
① $4NaH + (CH_3O)_3B \rightarrow NaBH_4 + 3NaOCH_3$

② $NaH + BH_3 \rightarrow NaBH_4$

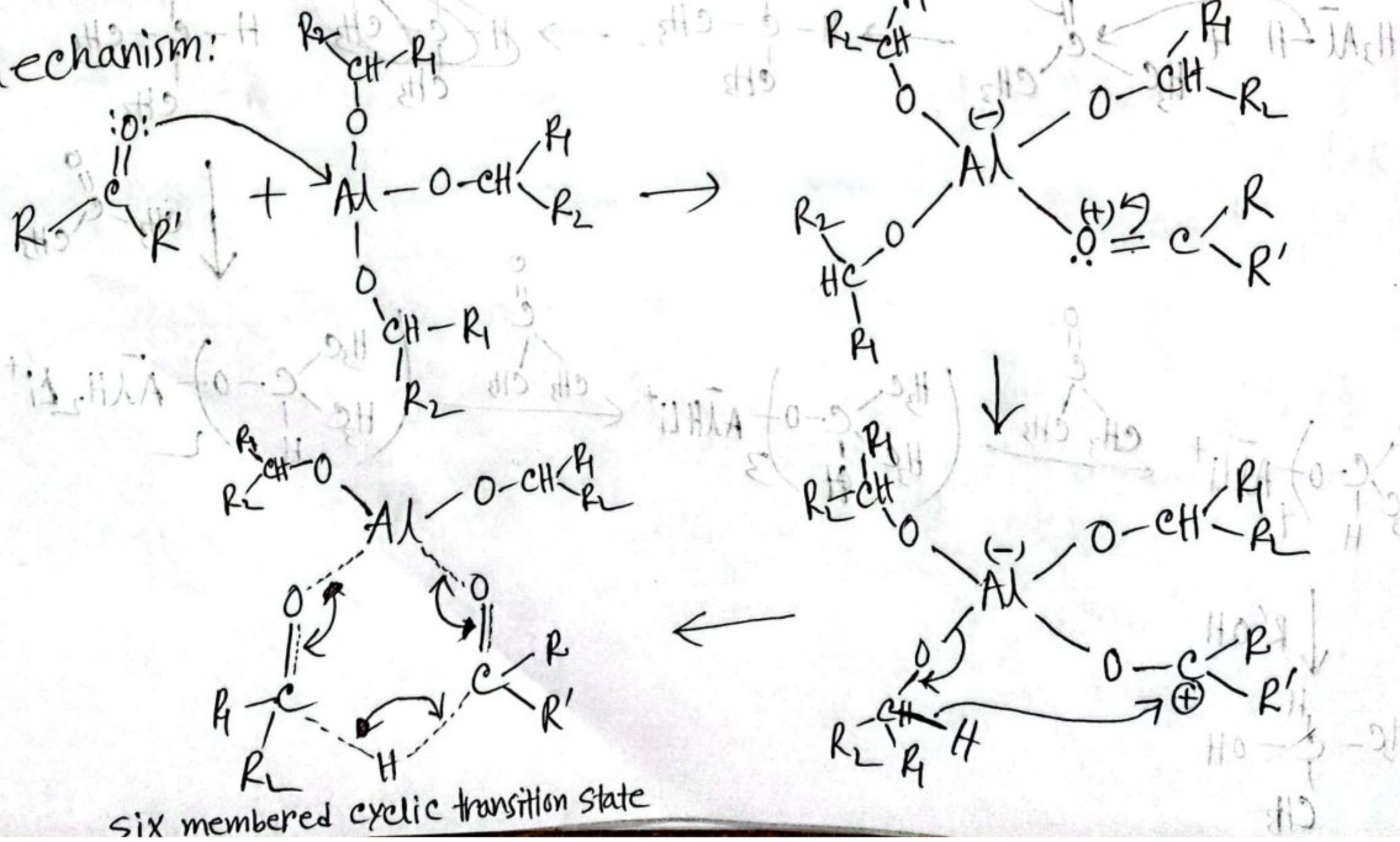
~~Q.~~ 2015, 2020 write short note on 'meerwein ponndrof verly (MPV)'?

⇒ The reduction of carbonyl group into alcohol in the presence of Aluminium Isopropoxide is known as Meerwein-ponndrof verly reduction. This reaction is highly chemoselective.

Reaction:



Mechanism:



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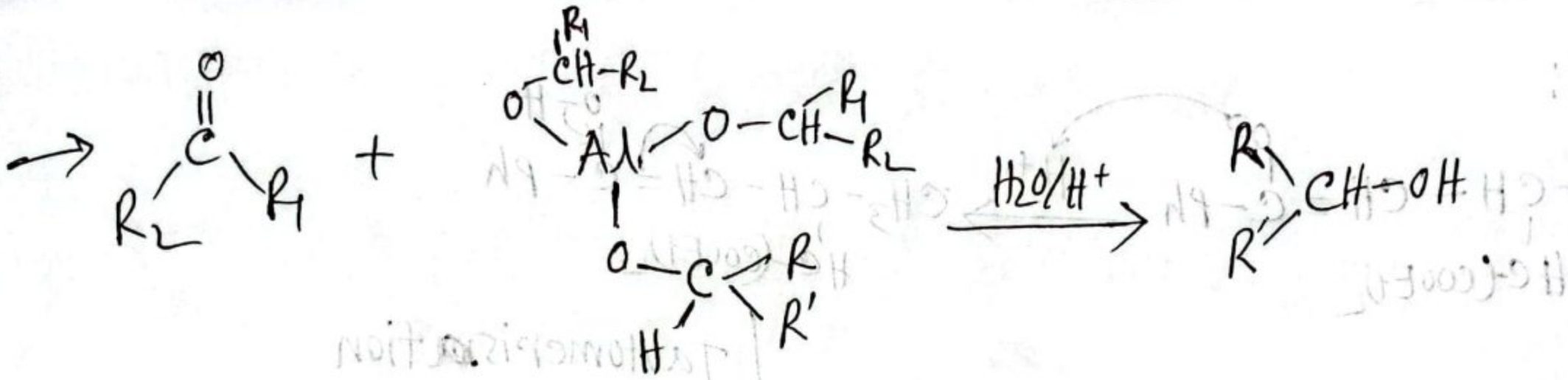


application: (i) This reaction take place under mild conditions & is highly specific for aldehyde & ketones, so that c=c & many other functional group can be present without themselves being reduced.

(ii) It is used in synthesis of Alc in fine chemical industries.

(iii) used in flavor industry for reduction of carbonyl compounds to Alc.

(iv) It is enviromental friendly.

(*) 2018 write short note on "michel reaction"?

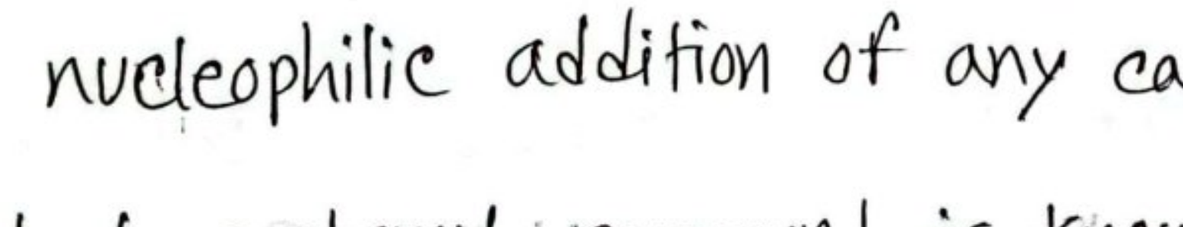
⇒ The nucleophilic addition of any carbanion to the α,β-unsaturated carbonyl compound is known as michael reaction:

Reaction:

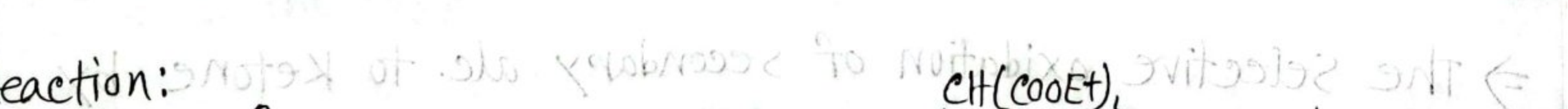
$$CH_3-CH=CH-\overset{O}{\overset{\|}{C}}-Ph + CH_2(COOEt)_2 \xrightarrow[EtOH]{NaOEt} CH_3-\underset{}{\overset{CH(COOEt)_2}{\overset{|}{CH}}}-CH_2-COPh$$

Mechanism:

Step-1:



Step-2:



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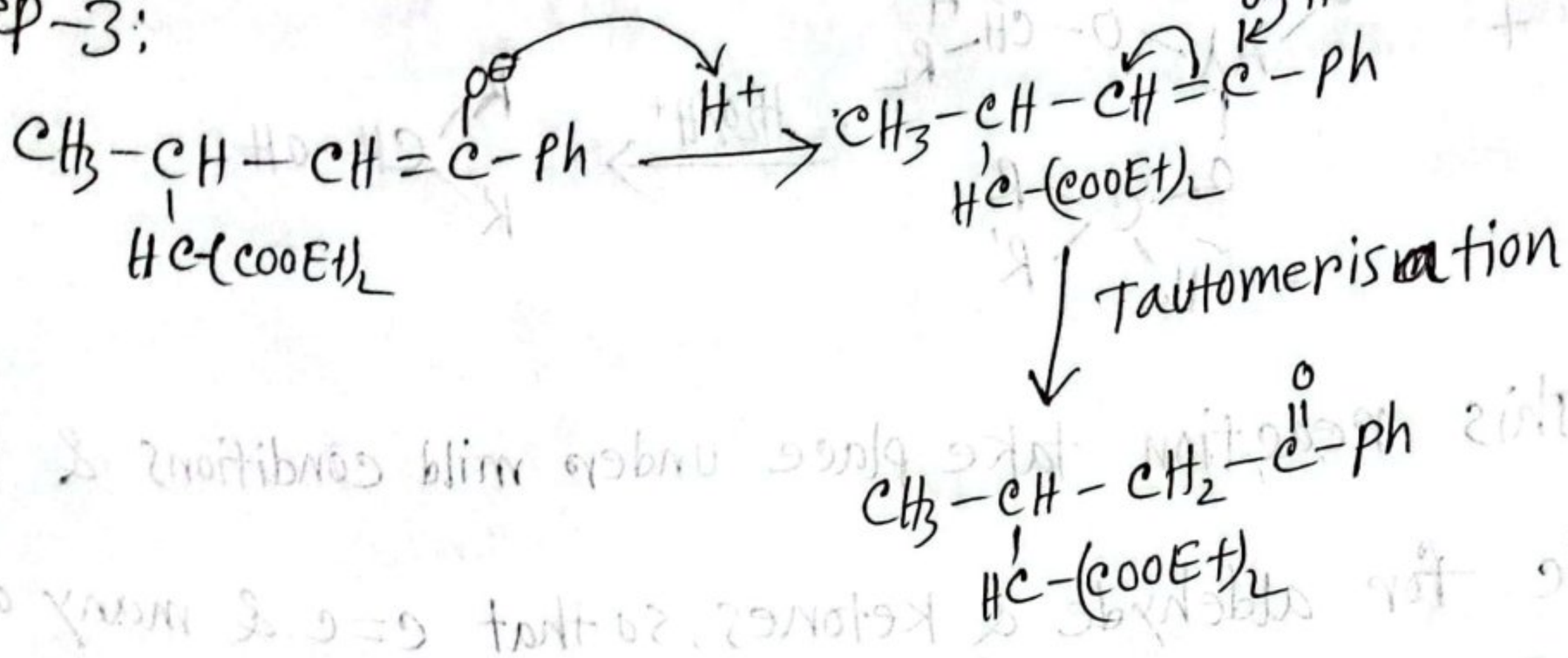
Step-3:



Application: ① Michael reaction is used in the synthesis of complex pharmaceutical compounds.

② Michael reaction facilitates the preparation of environmentally friendly polymers for pakaging applications.

③ It is widely used in synthesizing alkaloids.

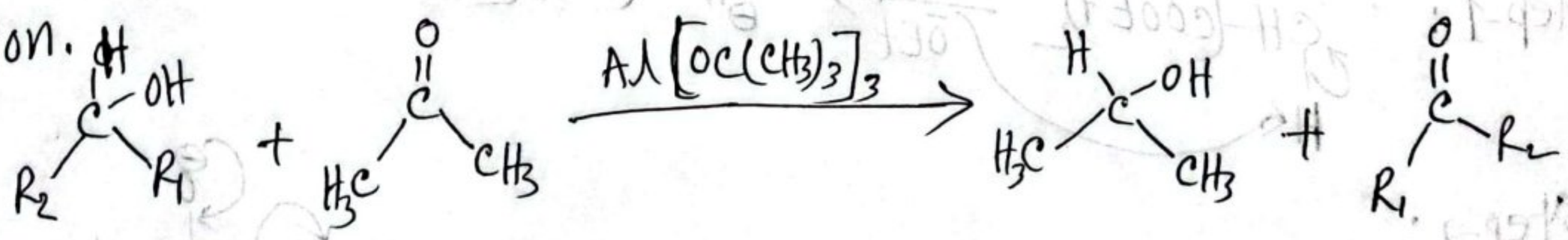
④ It can be used to modify the surface properties of materials.

2022

⇒ write short note on "oppenaver oxidation"?

⇒ The selective oxidation of secondary alc. to Ketone by treatment with aluminium tertiary-butoxide $Al[OC(CH_3)_3]_3$ in the presence of acetone is known as the oppenaver oxidation.

Reaction.



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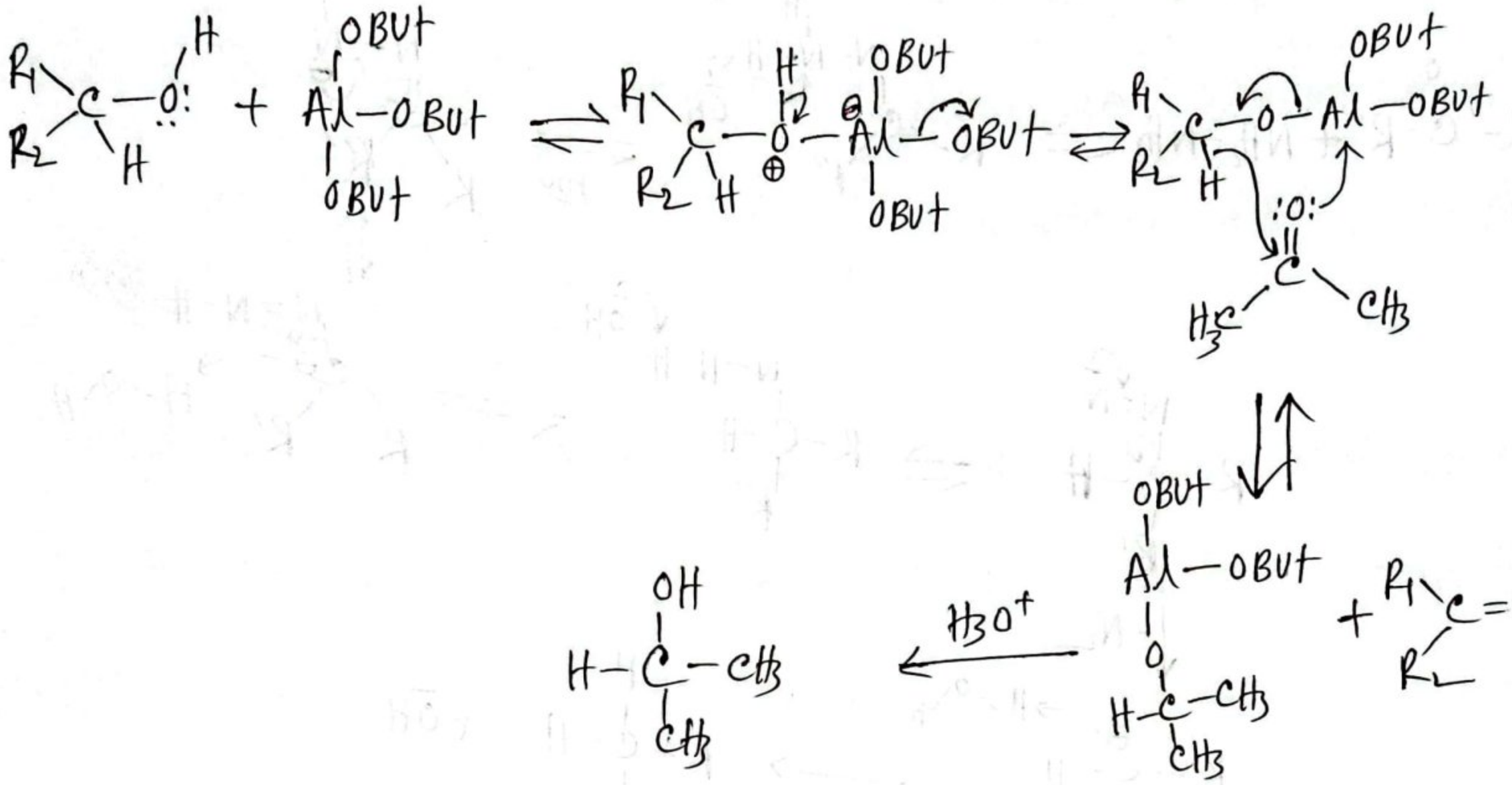
Mechanism:



Application: (I) It is used in the preparation of ketones from secondary alc. without affecting other functional group.

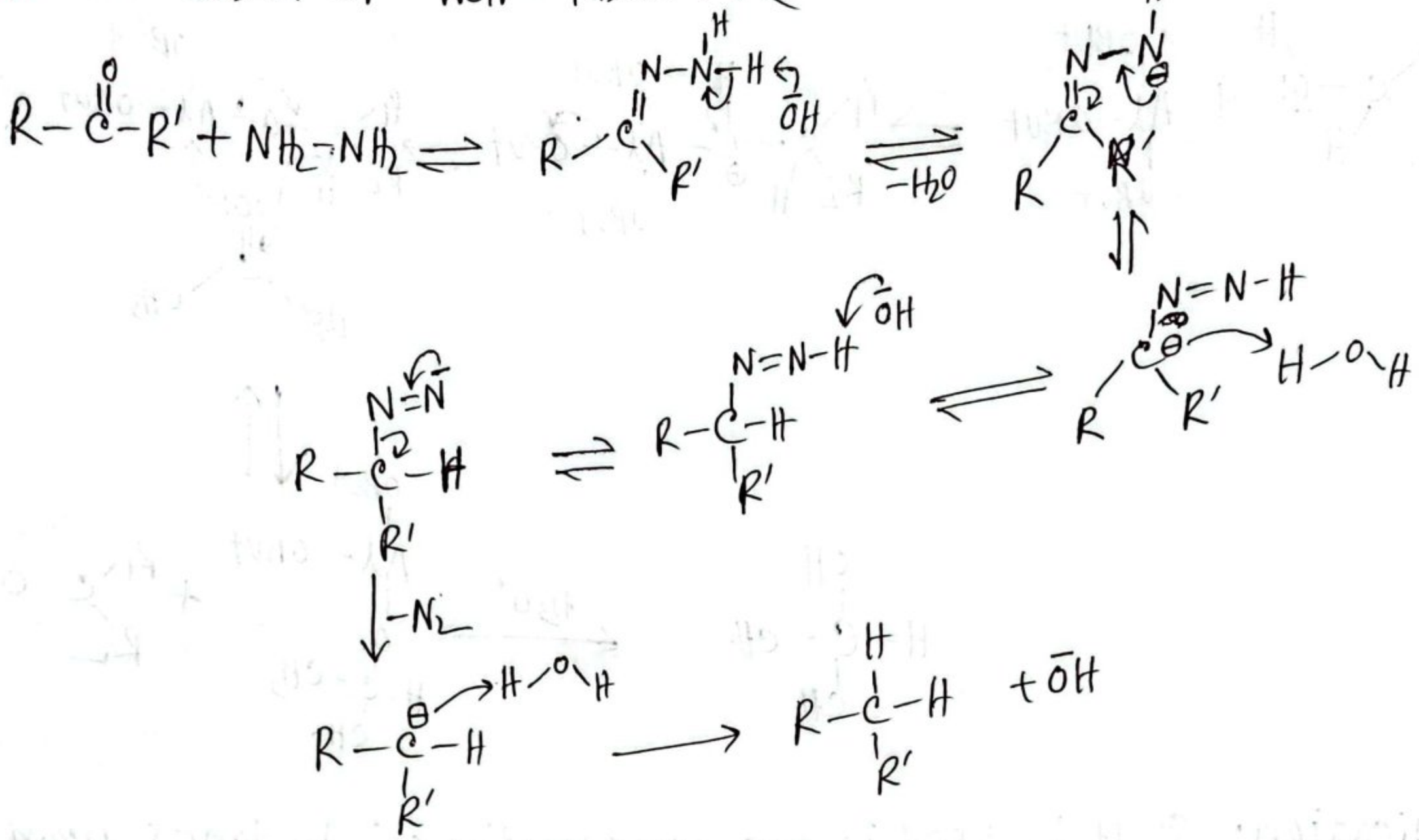
(II) used in the synthesis of steroids & hormones.

(III) used in the preparation of intermediates for drugs.

(IV) used in the synthesis of alkaloid.

N.B. ~~X~~ clemmenson থেকে wolf-kishner এ yield বেশি হয়। wolf-kishner দিয়ে alkyl & aryl দুইটাকেই reduce করা যায় but clemmenso শুধু alkyl কে Reduce করতে পারে।

Mechanism of wolf-Kishner: (mam poraise)



Application: (I) used to convert aldehydes & ketones into alkanes.

(II) used in preparation of long chain hydrocarbon which are used in fuel.

(III) useful in multistep synthesis when selective removal of the carbonyl group is desired without affecting other functional groups.

(IV) Helps stabilize molecules by removing reacting carbonyl groups.

(V) used in modifying chemical structures to improve solubility.

(VI) used for synthesis of various natural product.

(VII) It is highly selective & minimize side reactions.

Rajib
che-50

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